Università di Pisa
Department of Physics
Master's Degree in Physics

Anomalous Transport in Soaring Flights

Collective Strategies and Intermittent Search Models

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Abstract

Soaring flight couples persistent horizontal motion to the search for atmospheric lift and the management of altitude. This thesis investigates which stochastic descriptions can account for that motion using paraglider and hang-glider records from the French free-flight federation. It combines empirical observables, probabilistic flight-phase segmentation and a programme of mathematical modelling and Monte Carlo validation.

The data chapter documents recorder channels, cleaning and the limits of the retained trajectories. Global transport is then examined through displacement moments and quantiles, constant-velocity controls, signed joint distributions, regional anisotropy and velocity memory at several temporal resolutions. The analysis uses the full eligible archive over 10–10,000 seconds, with a fixed long-flight population controlling comparisons across lags. Effective exponents, changes of distribution shape and finite observation windows must be distinguished. Principal-axis analysis describes regional geometry without identifying wind or removing environmental effects. Competing stochastic models are assessed through explicit predictions, with the scope of each exclusion stated.

The segmentation chapter presents the current probabilistic decoder, readable flight examples and quantitative diagnostics. Independent manual annotation remains necessary before accuracy can be claimed. Subsequent work will measure observables conditioned on flight phase, compare solitary and group flight, construct and simulate a physically constrained stochastic model, and investigate thermal landscapes and route optimisation.

Contents

Chapter 1

Introduction

1.1 A flight is a transport process with an energy budget

A soaring pilot travels by repeatedly exchanging height for distance and recovering height in rising air. The atmosphere supplies the energy, but the pilot must locate that supply and decide how to use it. A long displacement is therefore the outcome of several coupled processes: motion through the air, encounters with lift, decisions made with incomplete information, and constraints imposed by wind and terrain. Because the recorded trajectory contains all these contributions, inferring a stochastic description requires more than fitting its geometry: the description must also preserve the physical mechanisms that allow the pilot to remain airborne.

Two elementary relations explain why neither an unconstrained random walk nor a purely geometrical description is sufficient. Writing $\mathbf{r}$ for horizontal position and z for altitude, a local kinematic description is

$$\dot{\mathbf{r}}(t) = \mathbf{v}_{\mathrm{air}}(t) + \mathbf{w}_h(\mathbf{r}(t), z(t), t), \tag{1.1}$$

$$\dot{z}(t) \simeq w_z(\mathbf{r}(t), z(t), t) - s\big(|\mathbf{v}_{\mathrm{air}}(t)|, \phi(t)\big). \tag{1.2}$$

Here $\mathbf{w}_h$ and w_z are the horizontal and vertical air velocities, $s > 0$ is the sink rate relative to the air, and ϕ is bank angle. The second relation is a quasi-steady approximation: it makes explicit the competition between lift and aerodynamic sink without attempting to model every manoeuvre. Altitude is therefore a finite resource, and a transport model must account for its use and replenishment if its trajectories are to represent sustained flight.

This physical picture suggests three descriptions of behaviour: directed *transition*, exploratory *search*, and *climb* in lift. They are hypotheses about the organisation of motion, rather than labels directly supplied by a recorder. A pilot can find lift during a transition, climb without completing a circle, or search while losing height. Determining which distinctions the data support is part of the scientific problem.

Soaring control has been studied through the problem of locating and exploiting thermals, including adaptive strategies based on local information [1]. The present work asks a complementary question: what statistical laws emerge over entire flights, and which mechanisms can account for those laws? A direct empirical precedent is the analysis of soaring transport and flight phases by Vilpellet et al. [2]. This thesis revisits that connection using the FFVL archive, explicit data-quality controls, and a separation between measured properties and assumptions used to interpret them.

1.2 The questions and the evidence needed to answer them

1.2.1 Which stochastic models survive the observations?

A first observable is the squared displacement over an interval τ,

$$M_2(\tau) = \left\langle |\mathbf{r}(t+\tau) - \mathbf{r}(t)|^2 \right\rangle . \tag{1.3}$$

Ordinary diffusion without drift has $M_2 \propto \tau$; directed motion can produce a ballistic law $M_2 \propto \tau^2$. An intermediate slope is informative, but it does not identify a mechanism, since correlated Gaussian motion, persistent motion observed during a crossover, and mixtures of flight behaviours can produce similar second moments. Random-walk frameworks offer competing explanations of anomalous transport [3]. The purpose here is to discriminate between such explanations using more than one curve.

Chapter 3 therefore measures displacement quantiles, the spectrum of moments, the shape of the increment distribution, directional anisotropy and velocity correlations alongside the MSD. Their roles are complementary: quantile ratios test whether the distribution changes shape as the lag increases, while higher moments compare the growth of ordinary and large displacements; velocity correlations resolve persistence in time. Open and closed tasks help expose the effect of the flight's intended geometry. A model is challenged where its predictions disagree with these observations under a comparable measurement protocol.

This logic also sets the limit of an exclusion. A discrepancy with a homogeneous Gaussian model for pooled flights need not exclude Gaussian fluctuations within a flight: different conditions can create a non-Gaussian mixture. Similarly, evidence against a particular constant-speed renewal Lévy walk is not a theorem against every process with finite-speed legs [4]. The chapter specifies the hypotheses being tested and leaves unsupported exclusions open.

1.2.2 How do the flight phases generate global transport?

Although an unsegmented flight supplies constraints on the whole process, it does not identify which behaviour generates a long displacement, a change of heading or a period of persistence. Chapter 4 addresses this missing link with a probabilistic segmentation into transition, search and climb. It states the mathematical model, displays its behaviour on readable trajectory excerpts, and distinguishes internal consistency from accuracy against independent annotations.

The next step will be a chapter devoted to observables conditioned on phase: episode durations, distance and height changes, speed and turning distributions, displacement scaling, and dependencies between successive episodes. A useful model must reproduce both these conditional statistics and the global ones, a stronger requirement than generating a visually plausible sequence of coloured flight segments. Windows that cross phase boundaries must also be retained when reconstructing global observables.

1.2.3 Does information from other pilots change the motion?

A nearby pilot can reveal lift before a follower reaches it. This creates a possible information channel that is absent from a model of independent searchers. A later chapter will compare solitary and group flight using both global and phase-conditioned observables. The operational definition of a group will concern simultaneous proximity and plausible visibility, not membership of a competition. A competition can contain isolated pilots, and pilots outside a competition can travel together.

The comparison will ask where differences occur: in time spent searching, climb selection, transition direction, episode persistence, or the efficiency of converting altitude into distance. Site, weather, wing and pilot characteristics must be controlled because they can produce the same differences without social interaction. Incomplete recording of nearby pilots will limit what can be called solitary flight. These are conditions for an interpretable comparison, not effects assumed in advance.

1.3 From empirical constraints to a physical model

The thesis combines empirical analysis, mathematical modelling and simulation, with a distinct role for each. Data analysis establishes the constraints that a candidate process must satisfy; mathematics then makes that process precise and derives its predictions, which Monte Carlo simulation tests under finite flight durations, heterogeneous environments and the actual observation protocol.

Model construction will follow the global analysis and the analysis conditioned on phase. Possible ingredients include finite-speed transitions, lift encounters, distributions of episode duration, and dependencies between successive directions or episodes. Renewal independence is one hypothesis to test; correlated random walks provide alternatives when that independence fails [5, 6]. The choice among these ingredients must be justified by the observations and an explicit physical mechanism, rather than fixed by the outline of the thesis. Parameters fitted on one set of flights will be checked on separate flights or days, using several observables jointly.

A reference model without wind and terrain may help isolate mechanisms, but obtaining it requires an inference about the environment: mean ground velocity alone does not determine wind, and rotating a trajectory into its principal axes does not remove orography. Artificially making a covariance matrix isotropic changes the measured object; it does not recreate a flight in still air. The analysis consequently preserves the motion in geographical coordinates and studies environmental strata before asking which properties could survive in a common reference model.

A further part of the programme concerns the spatial organisation of thermals and route choice. It will first estimate the two-dimensional distribution of *visited* lift regions, accounting for where pilots have searched. Recovering the distribution of available thermals requires an observation model: unvisited terrain is not evidence of absent lift. A mathematical description of thermal availability and strength could then support separate optimisation problems, such as minimum-time travel between two locations, a closed route, or maximum distance within a specified flight budget. Wind and, subsequently, possible geometrical constraints from terrain will be introduced in stages. A spatial density alone cannot determine an optimal path without assumptions about thermal strength, spatial dependence, altitude and the pilot's information.

1.4 Structure and current scope

Figure 1.1 shows how the measurements support the later modelling. The present document develops the data, global transport and segmentation chapters. The chapters on conditional observables, social flight, stochastic modelling and route optimisation form the research programme; their results are not claimed here. The immediate priority is to consolidate Chapters 3 and 4, then undertake the solo–group comparison with the necessary observables conditioned on phase.

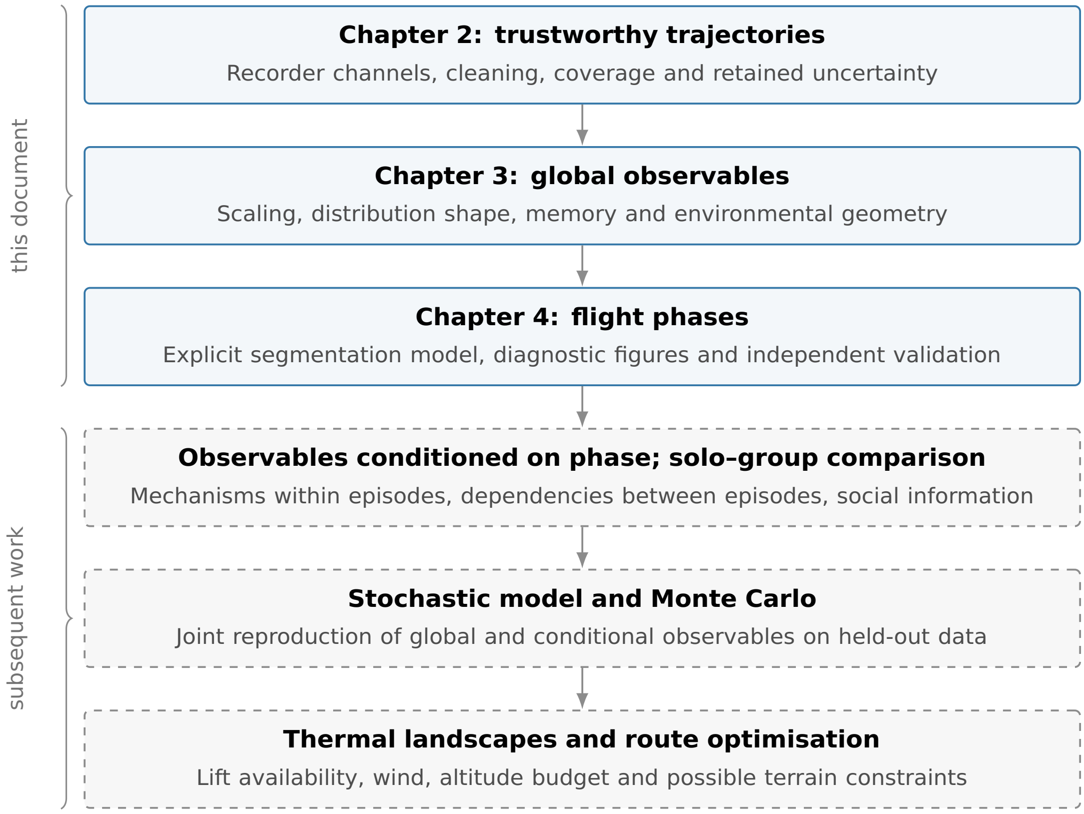


Figure 1.1. Logical structure of the thesis. Solid boxes are developed in this document; dashed boxes describe subsequent work. Model comparison must return to the measured observables, not stop at a fitted trajectory or a single exponent.

Chapter 2 follows the records from acquisition to retained trajectories, including the treatment of missing or inconsistent channels. Chapter 3 then asks what can be learned without assigning phases. Chapter 4 introduces that assignment and identifies what must be checked before its labels can support physical claims. The two appendices state the spectral conventions and the geodetic transformation used by the measurements.

1.5 Chapter summary

The thesis asks which stochastic description can account for soaring trajectories while respecting the physical roles of lift, travel and environmental constraints. The route from observations to a model begins with reliable trajectories and global transport statistics, then uses validated flight phases to identify the mechanisms mixed by those statistics. Comparisons between solitary and interacting flights, followed by simulation and thermal-landscape modelling, extend this programme. Their conclusions remain future work: the present chapters establish the empirical measurements and validation needed to make those comparisons meaningful.

Chapter 2

The dataset

The FFVL archive provides flight metadata and recorded trajectories for paragliders and hang gliders. We describe its coverage, then the cleaning, coordinate conversion and resampling that produce the trajectories used in the transport analysis.

2.1 Data acquisition

2.1.1 Source: the FFVL Coupe Fédérale de Distance

The *Coupe Fédérale de Distance* (CFD) is the cross-country competition of the French free-flight federation (FFVL). Its two archives separate paragliders (`parapente.ffvl.fr`) from hang gliders (`delta.ffvl.fr`). Each entry contains flight metadata and, usually, an IGC trajectory.

Seasons are named by their starting year: 1999 means 1999–2000. The paraglider archive begins in that season; the hang-glider archive begins in 2001–2002.

2.1.2 Collected data

For every season the pipeline retrieves two things:

- the **season XML export**, archived verbatim as provenance (`raw_xml/<year>.xml`); it is the authoritative list of flights and links;
- the **IGC tracklog files** referenced by the XML, one file per flight that has a track.

From these, a derived **catalog** is built (`catalog.csv`, one row per flight, its columns detailed in Sec. 2.3) together with a per-season index (`seasons_index.csv`). The raw data (tens of gigabytes) live on an external disk and are never committed to the repository. The catalog is itself rebuildable from the archived XMLs and likewise uncommitted (Sec. 2.3).

2.1.3 Acquisition and verification

Two command-line tools, one per discipline, share the same acquisition code. Downloads are resumable and rate-limited; each response must contain a readable IGC file with an `A` header and at least one `B` fix record (Sec. 2.2). The archived XMLs preserve the source lists, and the output location is configurable.

An independent check of the files on disk verified all 186,052 paraglider and 6,716 hang-glider tracklogs against the same content requirements. Section 2.5 compares these downloads with the full list of declared flights.

2.1.4 Further data sources

The present analysis uses the FFVL archive. WeGlide could add sailplanes, and XContest could broaden paraglider and hang-glider coverage. Both would feed the same catalog format.

Research-access requests were pending at both portals on 20 July 2026. At that date, WeGlide's default quota of roughly sixty API requests per day precluded bulk acquisition. These additional sources are not included here.

2.2 The IGC tracklog format

An *IGC file* is a plain-text flight record with fixed-position fields, defined by the FAI/IGC specification [7]. The first character identifies the record type:

Record	**Content**
`A`	Recorder identifier.
`H`	Headers: date, pilot, glider, datum, and related metadata.
`I`	Declaration of optional extension fields.
`B`	UTC time, WGS84 latitude and longitude, GNSS validity flag, pressure altitude and GNSS altitude (both in metres).

The `B`-record flag describes the GNSS solution: `A` means a three-dimensional fix; `V` means a two-dimensional fix or no GNSS data. We invalidate GNSS altitude at `V` fixes and retain their horizontal coordinates provisionally, pending the checks in Sec. 2.7.2. A pressure value alone does not establish a working barometer.

The parser rejects malformed records: wrong length, illegal times, coordinates or hemisphere letters. Physically implausible but legally encoded fixes pass to cleaning. The resulting sequence of fixes has a device-dependent cadence, typically a few seconds, which need not be uniform.

2.3 Flight metadata and the catalog

Every flight in the season XML carries a set of metadata as record attributes, which the acquisition preserves. Beyond the identifiers and links, the available fields are collected in Table 2.1.

Table 2.1. Flight metadata preserved from the season XML (raw attribute names), beyond the identifiers and links.

Field(s)	**Content / caveat**
date	flight date
pilot, club	pilot and club
take-off, landing, department	the two sites and the French department
`aile`	wing model name (raw XML field; French for “wing”)
`aile_class`	wing class, one system per discipline (Sec. 2.4)
flight type	the *scoring* category: free distance, out-and-return, flat or FAI triangle, hike-and-fly, . . . (*caveat*: a scoring label, not a glider or manoeuvre type)
distance, points	scored distance and points
duration, speed	duration and average speed, where present

The derived **catalog** (`catalog.csv`) has one row per flight: the metadata above, a `local_path` and a `downloaded` flag. The **season index** (`seasons_index.csv`) counts declared flights, track links and downloads. Both can be rebuilt from the archived XMLs.

2.4 Glider categories

Aircraft design changes speed, glide performance and turning geometry, motivating separate transport analyses [2]. Paragliders have a flexible canopy, hang gliders a framed flexible or rigid wing, and sailplanes a rigid airframe. The FFVL source identifies the first two families; sailplanes are absent from this dataset.

Within each family, `aile_class` defines the equipment groups in Table 2.2. Paraglider EN and legacy LTF codes, such as "C ou 2", are harmonized as EN A–D; CCC and tandem/non-certified entries remain distinct. Blank and placeholder entries stay unclassified.

EN labels describe safety-test responses and recovery demands, rather than measured speed or glide ratio [8]. The descriptions here use the consulted 2012 final draft; the 2013 standard and later amendments were not compared. FAI hang-glider classes distinguish design and control systems [9].

Chapter 3 uses two operational equipment proxies:

$$\text{beginners}: \ \text{EN A/B}, \qquad \text{experts}: \ \text{EN C/D/CCC}.$$

Neither measures individual experience or assumes equally spaced performance. The harmonization does not reassess certificates, and a tandem label alone does not establish EN certification.

2.5 Coverage and statistics

The statistics below describe the FFVL CFD dataset currently in hand; further sources may be added later (Sec. 2.1). It comprises 212,602 flights in total, of which 192,971 carry a GPS tracklog and 192,768 have been downloaded and verified, split across two disciplines.

The *paraglider* collection spans 27 seasons, from 1999–2000 to 2025–2026: 203,343 flights, of which 186,225 (91.6 %) carry a tracklog and 186,052 (99.9 %) are downloaded and verified (Table 2.3). The *hang-glider* collection spans 25 seasons, from 2001–2002 to 2025–2026: 9,259 flights, of which 6,746 (72.9 %) carry a tracklog and 6,716 (99.6 %) are downloaded and verified (Table 2.4). The hang-glider archive is roughly twenty times smaller, reflecting the smaller active population of the discipline.

In both disciplines the number of flights carrying a track grows by orders of magnitude over the period, reflecting the adoption of GPS loggers and online scoring: the early seasons contribute a handful of tracks each, while recent seasons contribute thousands of tracks for paragliders and several hundred for hang gliders.

Table 2.3. FFVL CFD *paraglider* flights per season (`parapente.ffvl.fr`): total declared, with a GPS track, and downloaded.

Season	Flights	With GPS	Downloaded	Coverage (%)
1999–2000	10	10	10	100.0
2000–2001	832	1	1	100.0

Season	Flights	With GPS	Downloaded	Coverage (%)
2001–2002	1,150	3	3	100.0
2002–2003	1,996	123	123	100.0
2003–2004	1,470	337	337	100.0
2004–2005	1,765	574	574	100.0
2005–2006	2,162	894	894	100.0
2006–2007	2,386	1,016	1,016	100.0
2007–2008	3,081	1,667	1,667	100.0
2008–2009	3,974	2,614	2,613	100.0
2009–2010	4,171	3,059	3,059	100.0
2010–2011	4,872	4,185	4,169	99.6
2011–2012	6,744	6,075	6,065	99.8
2012–2013	7,873	7,106	7,075	99.6
2013–2014	9,804	9,157	9,127	99.7
2014–2015	13,793	13,293	13,254	99.7
2015–2016	13,982	13,703	13,696	99.9
2016–2017	15,760	15,534	15,528	100.0
2017–2018	14,917	14,682	14,676	100.0
2018–2019	16,702	16,517	16,508	99.9
2019–2020	10,966	10,868	10,863	100.0
2020–2021	9,500	9,483	9,480	100.0
2021–2022	14,552	14,507	14,502	100.0
2022–2023	11,590	11,561	11,560	100.0
2023–2024	10,372	10,362	10,360	100.0
2024–2025	10,906	10,885	10,884	100.0
2025–2026	8,013	8,009	8,008	100.0
Total	203,343	186,225	186,052	99.9

Table 2.4. FFVL CFD *hang-glider* flights per season (`delta.ffvl.fr`): total declared, with a GPS track, and downloaded.

Season	Flights	With GPS	Downloaded	Coverage (%)
2001–2002	248	0	0	0.0
2002–2003	281	6	6	100.0
2003–2004	256	28	25	89.3
2004–2005	244	35	25	71.4
2005–2006	365	80	76	95.0
2006–2007	337	100	96	96.0
2007–2008	388	136	132	97.1
2008–2009	344	155	155	100.0
2009–2010	324	136	136	100.0
2010–2011	425	147	147	100.0
2011–2012	117	103	103	100.0
2012–2013	428	406	406	100.0
2013–2014	423	409	409	100.0
2014–2015	628	595	591	99.3
2015–2016	417	413	413	100.0
2016–2017	468	462	462	100.0

Season	Flights	With GPS	Downloaded	Coverage (%)
2017–2018	342	331	331	100.0
2018–2019	604	594	594	100.0
2019–2020	378	373	373	100.0
2020–2021	328	328	328	100.0
2021–2022	570	568	568	100.0
2022–2023	376	374	374	100.0
2023–2024	437	437	436	99.8
2024–2025	353	352	352	100.0
2025–2026	178	178	178	100.0
Total	9,259	6,746	6,716	99.6

2.6 Data quality and caveats

A few limitations are already known and will shape the analysis:

- **Flights without a track.** About 8 % of paraglider entries and about 27 % of hang-glider entries have no IGC file (older or manually declared flights); they carry metadata only. These are the flights outside the "With GPS" column of Tables 2.3 and 2.4.
- **A small residue of unrecoverable files.** A few flights advertise a track whose download fails or returns something that is not valid IGC content (broken or placeholder links); these are recorded as failures and excluded.
- **Heterogeneous logging.** Sampling rate, availability of the two altitude channels, and per-fix positioning noise depend on the device; trajectories must be resampled and quality-filtered before any ensemble statistic is computed.
- **Altitude channel.** A sizeable minority of loggers record no pressure altitude – almost a third of paraglider and roughly a sixth of hang-glider flights. GNSS altitude is used for every admitted flight, with the channel recorded per flight (Sec. 2.7.1).
- **Glider metadata.** Flights are paragliders or hang gliders, tagged by source (Section 2.4); the class field `aile_class` must be normalised before it is used to stratify.
- **Incomplete dates.** Some historical entries carry the placeholder date `0000-00-00`, written as a sentinel at acquisition. Such entries cannot support a calendar-conditioned comparison until a date is recovered independently; the relative-time preprocessing does not itself enforce this metadata exclusion. The count is small: 8 of 203,343 paraglider entries carry the sentinel (0.0039 %), 0 of 9,259 hang-glider ones, and the largest within-season fraction is only 0.17 % of its entries (2001-2002).

These caveats motivate the pre-processing described in the remainder of this chapter.

2.7 Trajectory pre-processing

Each IGC file is parsed into a sequence of fixes $(t_i, \varphi_i, \lambda_i, h_i)$ – UTC time, geodetic latitude and longitude, and altitude. The pipeline then proceeds in a fixed order:

(i) choose the altitude channel (Sec. 2.7.1);

Table 2.2. Class labels recorded in `aile_class`, one system per discipline. EN labels summarize test-based recovery characteristics; FAI classes distinguish control and design. They are used as equipment groups, without assigning pilot ability. Sources: the FAI Sporting Code, Common Section 7 [9] (§1.4.1) for the FAI classes and the sub-class ceilings; the EN class descriptions are condensed from Table 1 of the consulted 2012 final draft [8].

Class	Notes
Paragliders — harmonized EN/LTF catalogue groups	
EN A	maximum passive safety, extremely forgiving; any level of training
EN B	good passive safety, forgiving; ceiling of the FAI *Standard* sub-class
EN C	moderate passive safety, dynamic reactions; recovery needs precise input; ceiling of the FAI *Sport* sub-class
EN D	demanding, potentially violent reactions; for pilots practised in recovery
CCC	CIVL Competition Class racing wings
tandem / non-certified	two-seater and uncertified wings, pooled in this field
Hang gliders — FAI Class O (paragliders are Class 3 of the same scheme)	
Delta Class 1	weight-shift as the sole method of control
Delta Class Sport	sub-class of Class 1: type-certified production wing with a king post
Rigide Class 2	movable aerodynamic surfaces as the primary control
Rigide Class 5	as Class 2, but in the roll axis only; no fairings

(ii) clean individual fixes (Sec. 2.7.2);

(iii) trim the ground phases (Sec. 2.7.3);

(iv) filter out unfit whole flights (Sec. 2.7.4);

(v) convert to a local Cartesian frame (Sec. 2.7.5);

(vi) enforce a uniform sampling interval within each flight (Sec. 2.7.6);

(vii) smooth and differentiate the trajectory (Sec. 2.7.7).

The output consists of fix, segment and attempted-flight tables, accompanied by a table of suspect intervals retained for later sensitivity analysis. What each file holds and how the disk is organised is set out in Sec. 2.8.

Steps (i)–(iv) use raw geographic coordinates: great-circle distances give horizontal speeds, and altitude is read as logged. Step (v) places the cleaned, trimmed trajectory in a local East–North–Up frame, anchored at its first surviving fix. Resampling and differentiation then act on Cartesian components.

Table 2.8 collects the working thresholds. They are specified independently of the transport exponents; Table 2.7 records their effect on retention.

Status of the numerical snapshot. The census, transport measurements and phase analysis use the archive regenerated with pipeline version 2.3.0. It applies the local paired-barometer requirement, the $10\,\mathrm{m\,s^{-1}}$ vertical-speed threshold and the propagation of reconstructed altitude through derivative-estimation windows. Both inter-fix gaps and intervals between finite altitudes obey the same bridge-or-split bound; longer altitude gaps split the complete trajectory. The checks establish agreement with these numerical rules, not the absence of every recorder error. The witness audit below concerns the isolated version-2.0.1 correction. The targeted comparison reran the legacy and corrected rules on 2,388 candidates: 189 flights changed. Frozen-run counts changed in 188 cases; interior-ground counts changed in 2 cases (the categories can overlap). The corrected witness retained 22,468 additional horizontal fixes; 0 flight admission decisions changed. These counts describe the targeted set, not detector error rates. They do not replace regeneration of the archive and its observables.

2.7.1 Choice of altitude channel

The analysis uses *logged GNSS altitude* throughout: $z = h_{\mathrm{GNSS}}$. This keeps one channel across flights but leaves the recorder's vertical datum unresolved:

Recorder specification	GNSS height convention
IGC-approved	Ellipsoidal height [7].
CIVL hang-/paragliding	Geoid-referenced height in the consulted document, which contains draft annotations [10].

Compliance has not been established across this historical archive, and the parser preserves the numerical field. Absolute-height comparisons therefore need a datum audit.

A constant height offset cancels from increments; spatially varying geoid corrections and changing sensor biases do not. Pressure altitude also depends on atmospheric pressure and temperature. Using GNSS consistently does not make it uniformly more accurate.

Comparing the barometric and GNSS channels. From 3,000 sampled raw flights per discipline, we retain those with at least 95% pressure-altitude presence and median cadence within 0.25 s of the pooled mode (median intervals rounded to seconds).

Each pair of spectra uses the same longest block of finite, nonzero readings with valid GNSS flags. Duplicate/backward timestamps and gaps above 1.5 modal intervals break the block; ties in fix count select the earliest block. Linear resampling at the modal interval requires at least 256 raw and 256 grid fixes.

Figure 2.1 shows a time-varying channel difference: a constant offset does not align the 30 min traces. The high-frequency median spectra nearly coincide, with broad variation between flights. These comparisons exclude missing-channel runs and do not represent every sampled recorder. Appendix 2.A defines the spectral estimator.

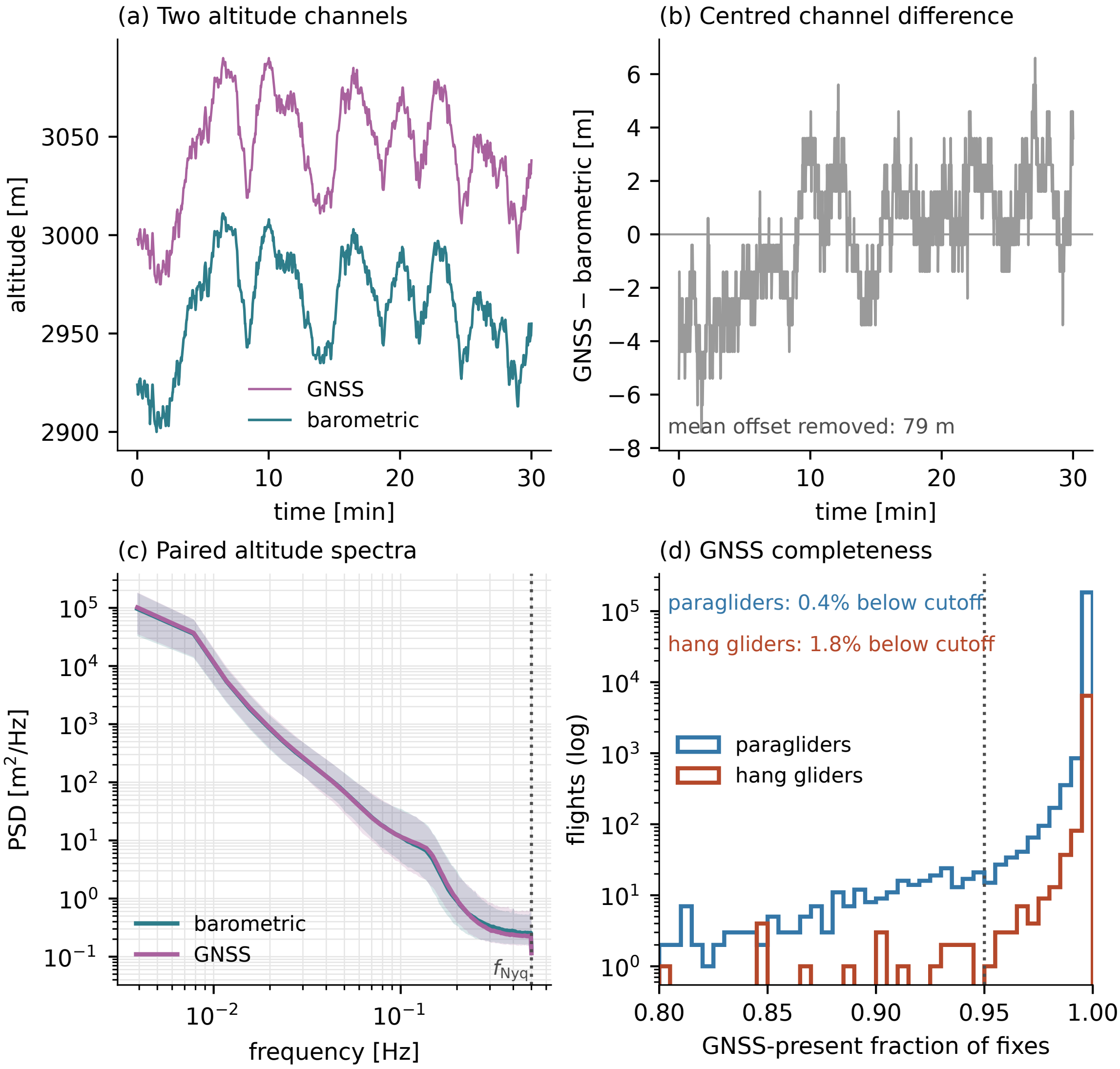


Figure 2.1. Barometric and GNSS altitude. (a) Paired raw channels over 30 min; (b) their demeaned difference. (c) Per-frequency median Welch PSD with the 10–90% range across paired flights, pooled over both disciplines (Appendix 2.A). The band describes flight variability; the dotted line marks $f_{\mathrm{Nyq}} = 1/(2\Delta t)$. (d) GNSS-presence fractions by discipline, shown over 0.80–1.00 with logarithmic counts. The dotted line is the 95 % admission threshold.

Variation between flights. Similar medians can conceal different upper tails. We therefore measure each flight's median PSD in the 0.35 Hz to 0.50 Hz band and compare it with the median of these values across barometric records. Of the 2,008 paired flights contributing to the spectrum, 6.5 % carry a GNSS floor above three times the barometric channel's typical one, and 2.1 % above ten times it. Multipath is one possible GNSS error source [11]; this comparison does not attribute the measured excess to a unique physical mechanism.

Sensitivity to the observation interval. A same-flight comparison separates the selection of valid intervals from subsequent trajectory smoothing. Using the same decoder and Welch estimator, we compare a whole-record rule, which can retain missing-value codes and interpolate across recording gaps, with the paired valid-block rule above. On the same 2,008 flights, the 90th percentile of the per-flight GNSS spectral floor decreases from 3.758 to 0.590 $\mathrm{m}^2\,\mathrm{Hz}^{-1}$; the corresponding barometric values change from 0.545 to 0.508 $\mathrm{m}^2\,\mathrm{Hz}^{-1}$. Here the floor is each flight's median PSD over 0.35 Hz to 0.50 Hz; its percentile is distinct from a per-frequency percentile in Fig. 2.1.

Neither comparison uses the pipeline's Savitzky–Golay output: both use logged altitudes, linear resampling and Welch detrending. The smaller GNSS upper tail therefore reflects the changed observation interval in this comparison. It does not demonstrate lower physical sensor noise. The experiment changes the combined validity, missing-value and gap-selection rule, without identifying the contribution of each criterion separately. These spectra remain conditional on recorders and intervals with adequate paired-channel support.

Availability. A flight needs finite, nonzero GNSS altitude at 95 % of its fixes and an altitude range of at least 30 m. These criteria reject incomplete or nearly constant channels, including genuine flights with too little vertical excursion. No pressure altitude is substituted.

The presence criterion excludes 0.4 % of paraglider and 1.8 % of hang-glider flights. Both channels fall below their presence thresholds in 0.14 % and 1.13 %, respectively. Within admitted flights, missing GNSS altitude initially leaves the horizontal fix intact. At resampling, short altitude holes are reconstructed and flagged; long holes split the full trajectory (Sec. 2.7.6). Missing readings affect 6.1 % of paraglider and 10 % of hang-glider flights, with a median of 5 missing altitudes among affected paraglider flights.

Local barometric support. The pressure channel has a separate role: it can corroborate a suspected frozen GNSS position and an interior ground interval (Secs. 2.7.2 and 2.7.3). Its flight-level eligibility requires at least 95 % nonzero finite readings and a range of at least 30 m. The corresponding archive fractions, 72.1 % and 84.7 %, describe eligibility, not coverage of every candidate interval.

At fix i, define $b_i = 1$ when a finite, nonzero pressure altitude accompanies the horizontal position in the same decoded record, and $b_i = 0$ otherwise. For a candidate interval I, the barometric test is available only if

$$B_I = B_{\text{flight}} \prod_{i \in I} b_i = 1. \tag{2.1}$$

Zero pressure values are treated as missing before duplicate timestamps are merged: the pair $\{0, 1000\}$ therefore supplies one 1,000 m reading. No temporal interpolation supplies a witness. Equation (2.1) requires pressure and position in every record of I, even when GNSS altitude is invalid.

An eligible barometer with incomplete local coverage causes the barometric test to abstain. Flights without an eligible barometer remain in the analysis, subject to the weaker recorder-declaration rule below. Restricting a sensitivity analysis to $B_I = 1$ would measure the effect of this difference in evidence.

2.7.2 Fix-level cleaning

Fix-level cleaning distinguishes three cases (Fig. 2.2):

- An *off-the-trend defect*, such as a horizontal spike, may receive a Hampel flag. The separate impossibility gate decides deletion.
- An *on-the-trend defect*, such as a duplicate timestamp or frozen position, requires timestamp or recorder checks.

- A *vertical defect* invalidates altitude while provisionally retaining the horizontal fix.

These local tests leave slowly varying and plausible-looking errors unresolved. Deletions, boundaries and reconstruction flags record what the pipeline changed.

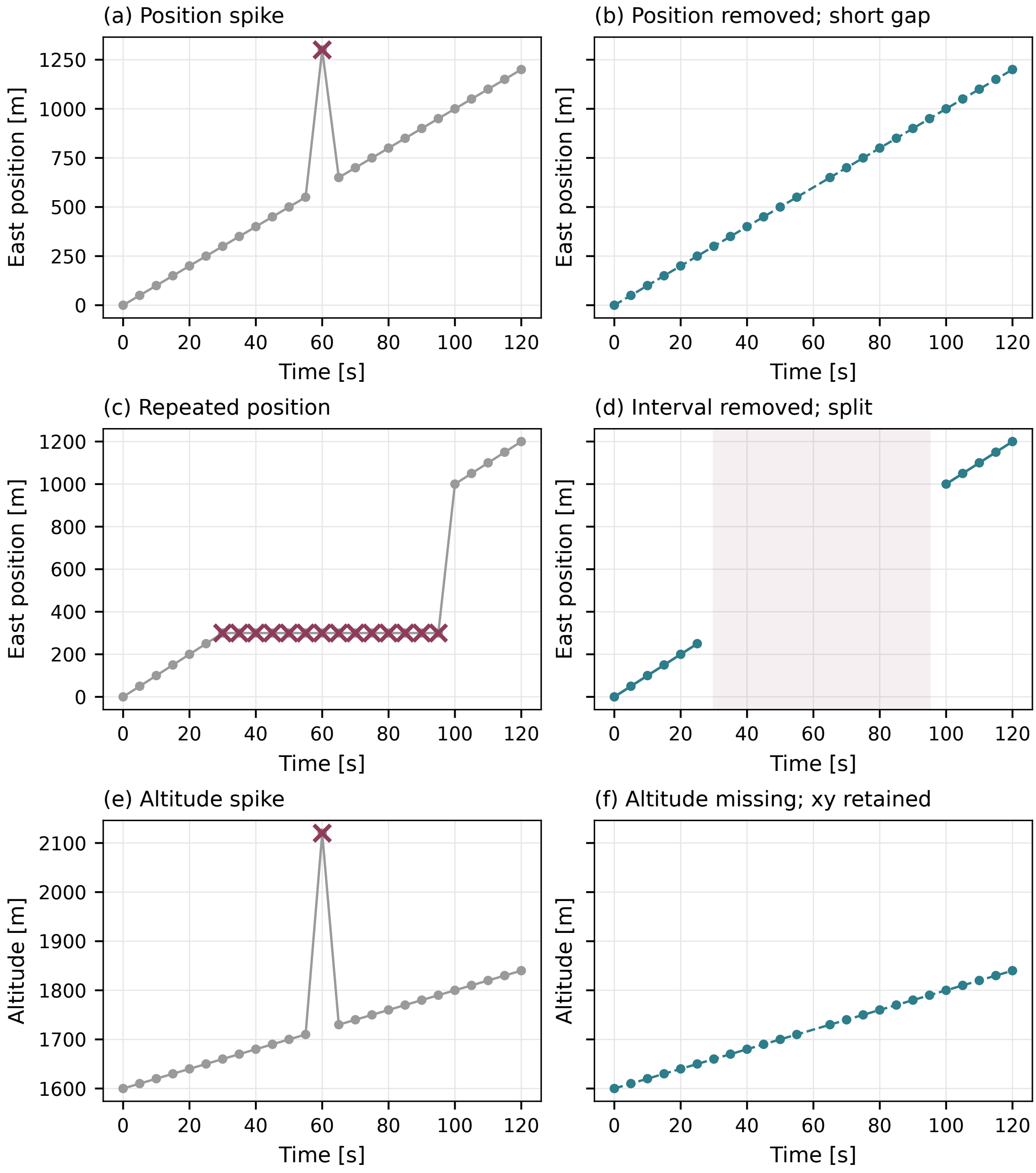


Figure 2.2. Synthetic cleaning examples. Crosses mark suspect readings and dots retained information. A horizontal spike is deleted after the speed/rejoin test; a frozen-position run is removed and split when duration and witness conditions hold; an altitude spike leaves horizontal fixes intact. Dashed paths indicate possible reconstruction at resampling.

Geometry on the raw coordinates. The cleaning runs on the raw geodetic coordinates (φ_i, λ_i), before the Cartesian conversion (Sec. 2.7.5), so distances between fixes are great-circle (haversine) distances [12] on a sphere of mean radius $R_\oplus = 6{,}371\,\mathrm{km}$:

$$\Delta s_i = 2R_\oplus \arcsin \sqrt{\sin^2 \frac{\varphi_{i+1} - \varphi_i}{2} + \cos \varphi_i \cos \varphi_{i+1} \, \sin^2 \frac{\lambda_{i+1} - \lambda_i}{2}}\,, \tag{2.2}$$

The same distances give step velocities, total path length and extent:

$$\begin{aligned} v_{xy,i} &= \frac{\Delta s_i}{t_{i+1} - t_i}, \quad & v_{z,i} &= \frac{h_{i+1} - h_i}{t_{i+1} - t_i}, \\ L &= \sum_i \Delta s_i, \quad & D &= \max_i \Delta s(\mathrm{fix}_0, \mathrm{fix}_i). \end{aligned} \tag{2.3}$$

Near 45° latitude, the spherical distance error is at most about 0.3%: 3 cm on a 10 m step, below metre-scale GNSS noise. Appendix 2.B derives the ellipsoidal correction and its latitude dependence.

The Hampel identifier. Let the window of fix k be the fixes within $\pm w$ of t_k, and $\tilde{\varphi}_k, \tilde{\lambda}_k$ the component-wise window medians. The residual is the great-circle distance from that median point,

$$r_k = \Delta s\big((\varphi_k, \lambda_k), (\tilde{\varphi}_k, \tilde{\lambda}_k)\big), \tag{2.4}$$

with local scale $\mathrm{MAD}_k = \mathrm{med}\{r_j : |t_j - t_k| \le w\}$, and the fix is flagged when

$$r_k > \max\big(k\,\sigma_k,\ \varepsilon_{\min}\big), \qquad \sigma_k = 1.4826\,\mathrm{MAD}_k, \tag{2.5}$$

σ_k a robust local scale [13]. The factor 1.4826 is the usual one-dimensional Gaussian MAD normalization; for these two-dimensional radial residuals it is a scale convention, not an exact standard deviation or a calibrated false-alarm probability.

The working values are $k = 5$, $w = 20\,\mathrm{s}$ and $\varepsilon_{\min} = 15\,\mathrm{m}$, with at least 5 fixes per window.

Flags contribute to the anomaly count; they never gate deletion. A thermalling circle can depart from its local median while remaining reachable at the absolute speed bound.

Reach, rejoin, or cut. Write a for the last fix the scan accepted, the *anchor*, and measure every candidate against the chord from it,

$$V(a, i) = \frac{\Delta s\big((\varphi_a, \lambda_a), (\varphi_i, \lambda_i)\big)}{t_i - t_a}. \tag{2.6}$$

An unreachable fix opens a pending block. Its anchor a stays fixed while elapsed time grows. The scan then applies three actions:

Condition	**Action**
Reachable fix, no pending block	Keep it and advance the anchor.
First rejoin within w	Delete the pending predecessors; keep the rejoin as the new anchor.
No rejoin within w	Keep the block, split at its first fix and re-anchor on the current fix.

The cap starts at the pending block's first fix. It reuses the Hampel half-window without a shared calibration. Gaps left by deletion are bridged or split at resampling (Sec. 2.7.6).

Figure 2.3 compares a spike, a moderate offset and a large offset. The moderate offset becomes reachable as time grows and can cost genuine fixes; the large offset remains unreachable and is split. This gate cannot identify which branch contains the true positions.

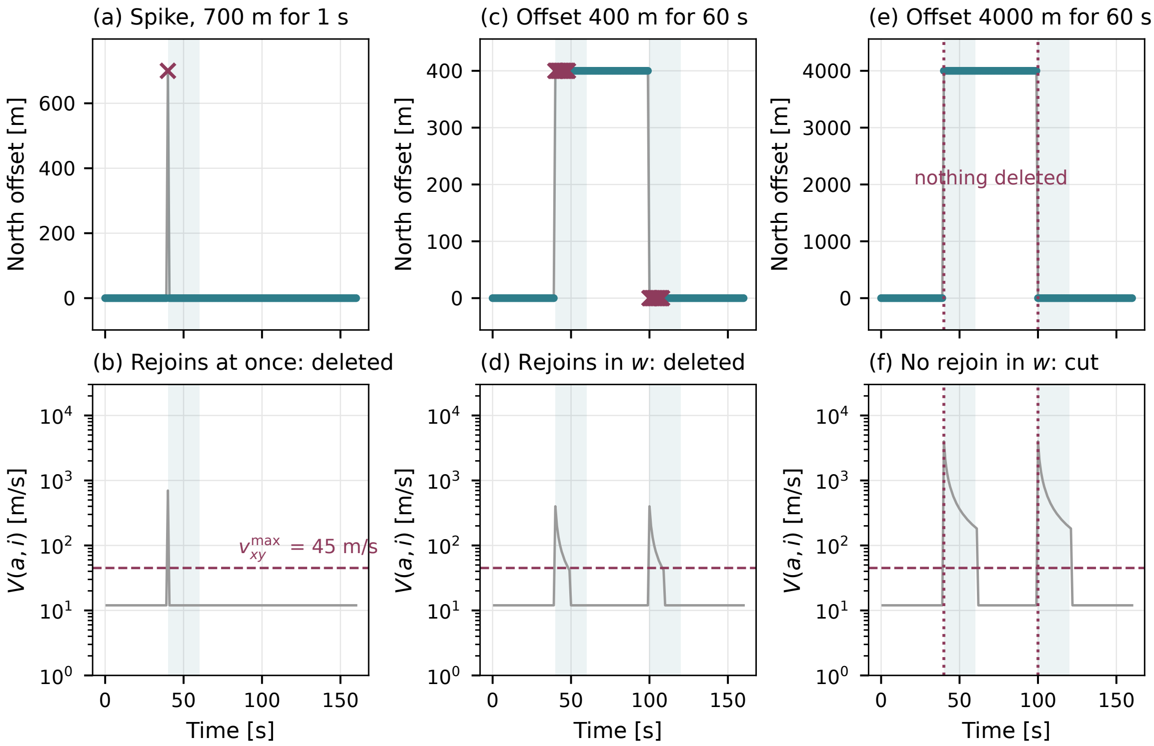


Figure 2.3. Implemented gate on synthetic tracks cruising east at $12\,\mathrm{m\,s^{-1}}$, sampled every second. Northward offsets are 700 m for one fix (a,b), 400 m for 60 s (c,d), and 4,000 m for 60 s (e,f). Crosses mark deletions, dotted lines boundaries, and shading the block cap w. Lower panels show chord speed from the current anchor and the bound $v_{xy}^{\max}$. The moderate offset removes nine fixes at each end, including genuine ones; the large offset is kept as a separate segment.

Vertical-speed checks. Write the observed altitude as $h_i = z_i + \epsilon_i$. Even a smooth true trajectory produces noisy finite differences,

$$v_{z,i}^{\mathrm{obs}} = \frac{h_{i+1} - h_i}{t_{i+1} - t_i} = \overline{v}_{z,i}^{\mathrm{true}} + \frac{\epsilon_{i+1} - \epsilon_i}{t_{i+1} - t_i}. \tag{2.7}$$

For independent errors of variance σ_z^2, the differenced noise has variance $2\sigma_z^2/\Delta t^2$. Short cadences therefore amplify altitude error. This calculation illustrates the effect; real GNSS errors may be correlated.

The sustained-excess statistic is

$$\tilde{v}_j = \mathrm{med}\{|v_{z,i}^{\mathrm{obs}}| : |m_i - m_j| \le w_v,\ v_{z,i}^{\mathrm{obs}} \text{ finite}\}, \qquad m_i = \tfrac{1}{2}(t_i + t_{i+1}), \tag{2.8}$$

with $w_v = 5\,\mathrm{s}$. The statistic measures local speed magnitude. Each of the $N-1$ steps is assigned to its midpoint m_i. Figure 2.4 illustrates window selection and the median: for $2r+1$ finite values, at least $r+1$ must exceed the bound for the median to do so. For an even count, the two central values are averaged.

With fewer than 5 finite steps, the implementation does not use the median. Writing n_j for that count, the quantity used by the cleaning is

$$u_j = \begin{cases} \tilde{v}_j, & n_j \ge 5, \\ |v_{z,j}^{\mathrm{obs}}|, & n_j < 5. \end{cases}$$

The second branch is the sparse-window fallback. It can occur at a slow recording cadence, near missing altitudes, or at the ends of a record. The sustained-excess rule marks the altitude

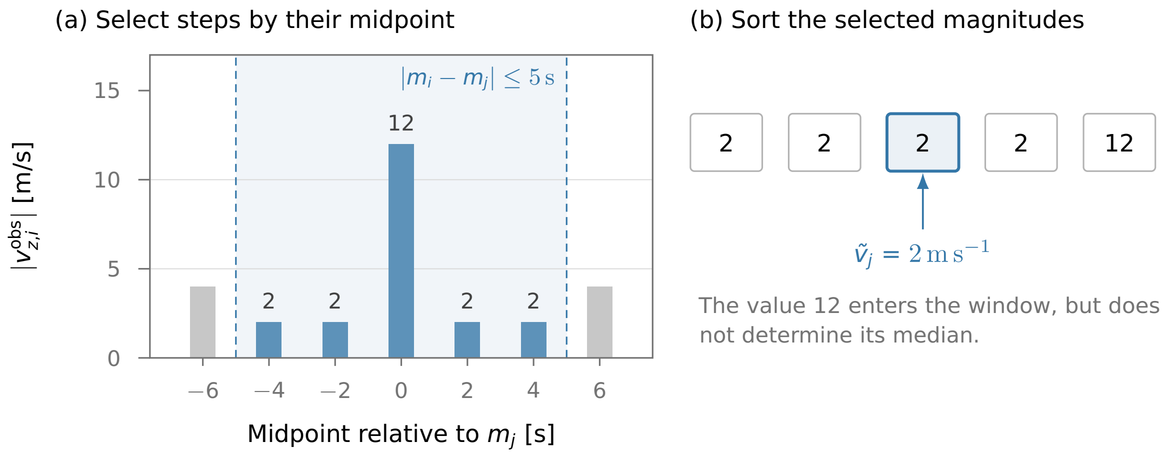
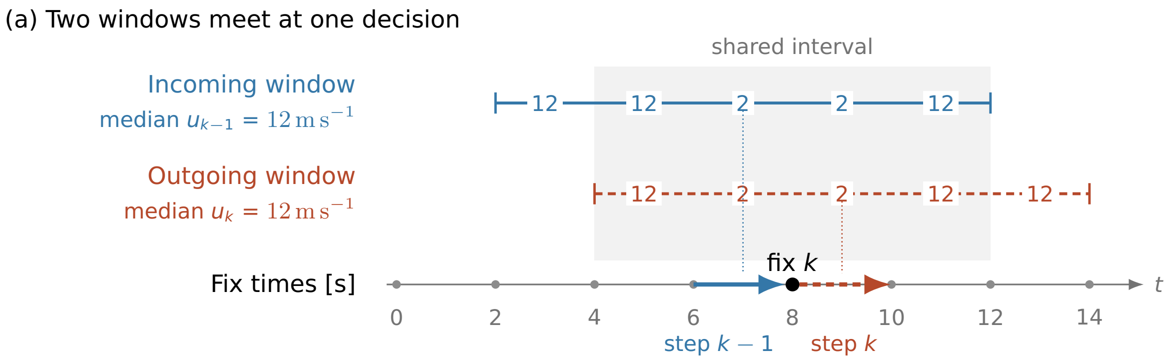


Figure 2.4. How Eq. (2.8) is evaluated. This schematic uses fixes every 2 s and a half-window of 5 s: five step midpoints lie inside it. Grey bars are outside the window and are excluded, regardless of their height. Missing or non-finite speeds are also excluded. Taking absolute values makes an ascent and a descent of equal magnitude contribute equally. These are illustrative values, not flight data.

of an interior fix k as missing when

$$u_{k-1} > v_z^{\max} \quad \text{and} \quad u_k > v_z^{\max}, \qquad v_z^{\max} = 10\,\mathrm{m\,s^{-1}}. \tag{2.9}$$

Both the incoming and outgoing window statistics must exceed the bound strictly. With populated windows, the rule can invalidate altitude even when the two steps touching fix k are below the bound (Fig. 2.5). The overlapping windows locate a neighbourhood of excessive speeds; they do not identify its erroneous altitude independently.



(b) Apply the AND condition

u_{k-1} [m/s]	u_k [m/s]	Both $> 10\,\mathrm{m\,s^{-1}}$?	Result of this rule
2	2	No	Keep altitude
2	12	No	Keep altitude
12	2	No	Keep altitude
12	12	Yes	Mark altitude missing

Figure 2.5. Two overlapping windows determine the altitude decision at fix k. Fixes are 2 s apart and the half-window is 5 s; numbers along the bars are speed magnitudes in m/s. Both medians equal $12\,\mathrm{m\,s^{-1}}$, although the steps touching k have magnitude $2\,\mathrm{m\,s^{-1}}$. Equation (2.9) therefore invalidates its altitude at the $10\,\mathrm{m\,s^{-1}}$ bound. The table illustrates the AND condition. Timestamp and horizontal position remain intact.

An isolated spike on a low-speed background leaves a well-supported median low (Fig. 2.6).

A separate out-and-back rule invalidates it when both adjacent steps exceed $v_z^{\max}$ with opposite signs. Sustained excess instead raises the median, whether caused by recorder error or exceptionally strong physical motion.

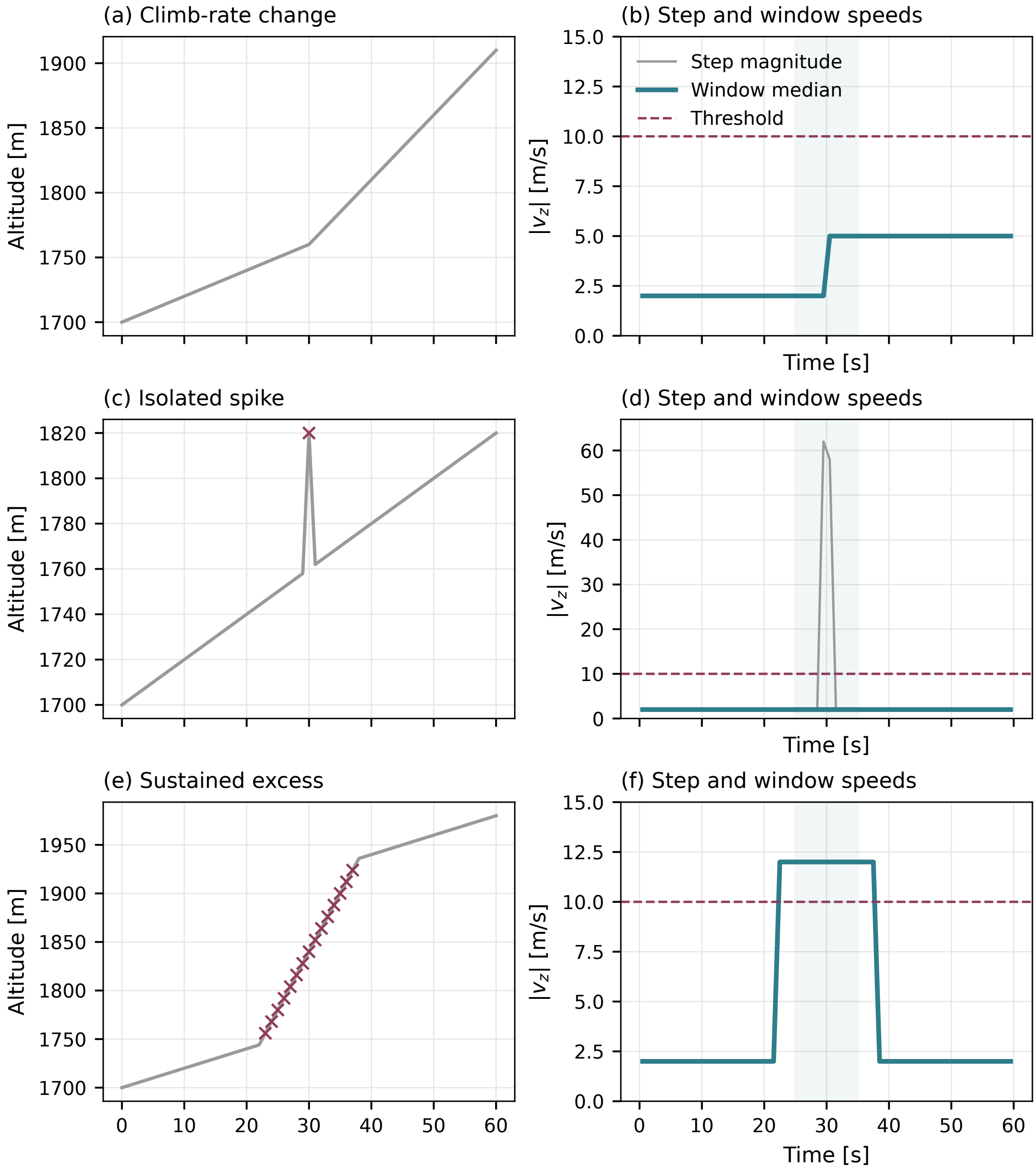


Figure 2.6. The median test at a one-second cadence, using the configured window and speed threshold. Left: synthetic observed altitude. Right: absolute per-step speed (grey), window median (solid), and the bound (dashed). A change between plausible climb rates remains below the bound; an isolated spike barely affects the median and is handled by the separate spike rule; a sustained excessive interval raises the median. Crosses mark altitudes invalidated by the actual implementation.

Unreturned vertical steps. An isolated jump without a prompt return can escape both checks. Define

$$e_i = 1\{|v_{z,i}^{\mathrm{obs}}| > v_z^{\max}\}, \qquad J_i = |h_{i+1} - h_i|, \tag{2.10}$$

$$d_i^{\mathrm{return}} = \min_{\substack{j=i+1,\ldots,i+5 \\ h_j \text{ finite}}} |h_j - h_i|. \tag{2.11}$$

The minimum uses the next five fix positions, fewer at the record end, and equals $+\infty$ if none has finite altitude. Set $e_i = 0$ outside the observed steps. The counter `n_alt_level_shift` records candidates satisfying

$$e_i = 1, \qquad e_{i-1} = e_{i+1} = 0, \qquad d_i^{\mathrm{return}} > 0.3 J_i. \tag{2.12}$$

This short return check does not establish a permanent offset. It runs before trimming and segment selection, and can overlap other altitude rules.

Among retained-flight metadata, 47,440 paraglider and 1,350 hang-glider records contain at least one candidate. Reprocessing eight previously selected examples with the current pipeline reproduces their counters. In each discipline these include two high-count records and two seeded positive-count records. All eight still have candidate times near usable phase-feature centres. This is temporal association, not proof that an uncorrected sensor defect enters a feature. These counts describe candidates before later cuts, rather than surviving confirmed defects. Automatic repair is not applied: a level shift and a real large increment can have similar signatures. Their effect on phase features still requires a sensitivity analysis.

Figure 2.7b motivates the working bound $10\,\mathrm{m\,s^{-1}}$: around $9\,\mathrm{m\,s^{-1}}$ to $10\,\mathrm{m\,s^{-1}}$, the density begins to decay more slowly and the paraglider curve crosses the hang-glider curve. The transition is gradual. The bound is an empirical cleaning choice, so extreme genuine descents can be invalidated too. In the full diagnostic sample, 0.069 % of window medians exceed it. This exceedance fraction differs from the removal rate, which also depends on neighbouring windows and the spike rule.

Deletion and mandatory boundaries. Altitude rules retain timestamp and horizontal coordinates; position and timestamp rules may remove the whole fix. Reconstruction is deferred to resampling. After all cleaning checks, every surviving step above the horizontal speed bound becomes a mandatory segment boundary, counted as `n_boundaried`. This also catches an impossible return from a block that the forward scan kept.

The forward scan does not re-evaluate its initial anchor, so an erroneous first fix can survive this local procedure and displace the origin of the later Cartesian frame. A separate flight-level test therefore rejects a track containing a fix beyond the distance reachable from its first retained position (Sec. 2.7.4). The reach test acts on spherical distances before projection and smoothing, with a 50 m tolerance. The written Cartesian trajectories undergo separate diagnostic checks (Sec. 2.9); those operations do not preserve the reach inequality as an exact invariant.

Frozen-position runs and the evidence for GNSS lock loss. A logger may repeat its last position after losing GNSS lock while timestamps continue. An unremoved run then resembles a period of spatial confinement. The detector uses three joint conditions,

$$D_I < \delta_{xy}, \qquad t_{\max I} - t_{\min I} \geq \tau_{\mathrm{freeze}}, \qquad W_I = 1, \tag{2.13}$$

where D_I is the great-circle diagonal of the latitude–longitude bounding box of candidate run I, evaluated after longitude unwrapping. This is the implemented extent proxy, not an exact all-pairs diameter. The working values are $\delta_{xy} = 15\,\mathrm{m}$ and $\tau_{\mathrm{freeze}} = 60\,\mathrm{s}$.

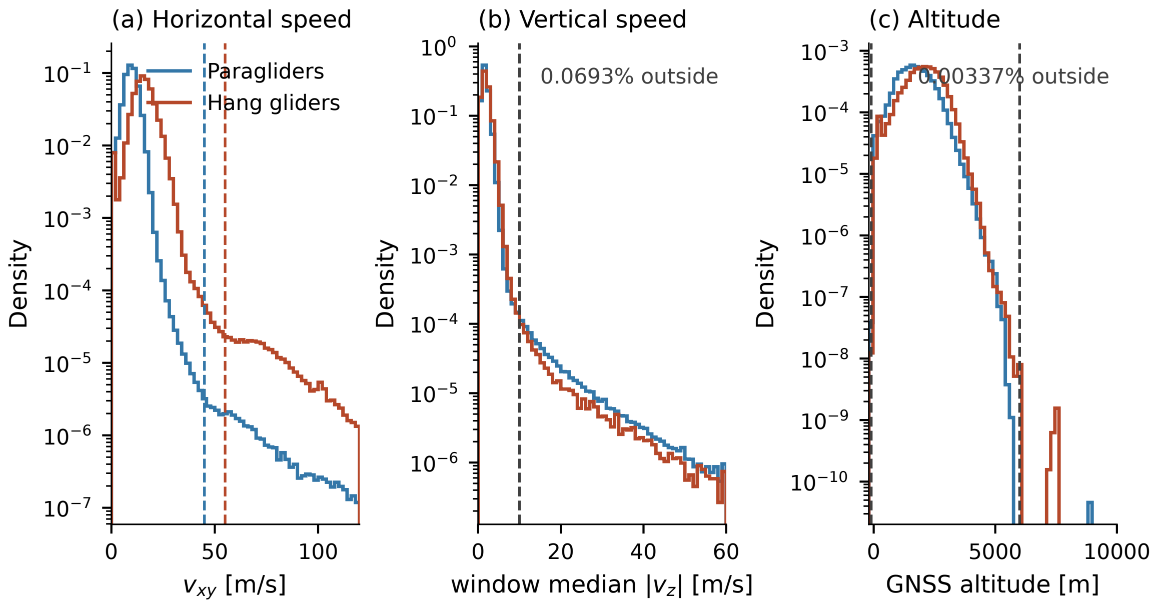


Figure 2.7. Raw diagnostics for the absolute bounds: (a) horizontal step speed, (b) windowed GNSS med $|v_z|$, and (c) GNSS altitude. Dashed lines mark the working thresholds. Densities pool a large paraglider subsample and the complete hang-glider population, using the full finite-sample denominator even where the displayed range cuts off a tail. Logarithmic ordinates expose rare values; abscissae are linear. Annotations count bound exceedances, while actual deletion and invalidation also depend on the rejoin and neighbouring-window rules.

Let E_I denote exact equality of both coordinate fields throughout the run, d_I the fraction of its fixes with a `V` flag or missing GNSS altitude, and Δb_I the median pressure altitude of the final quarter minus that of the first quarter. With B_I defined in Eq. (2.1), the witness is

$$W_I = E_I \;\vee\; \begin{cases} |\Delta b_I| < \delta_z, & B_I = 1, \\ \text{false}, & B_{\text{flight}} = 1,\; B_I = 0, \\ d_I > 1/2, & B_{\text{flight}} = 0, \end{cases} \qquad \delta_z = 5\,\text{m}. \tag{2.14}$$

Removed frozen-lock runs always split the trajectory. Exact coordinate repetition is a separate witness; otherwise incomplete local pressure coverage forces abstention.

The signatures have limits. Quantized coordinates can repeat for a stationary receiver, and sparse fixes do not establish continuous stillness between them. Small net pressure change can hide an excursion; a barometer can also continue climbing during a horizontal GNSS freeze. Shared logger failures can affect both channels.

For scale, a coordinated turn in still air has

$$R = \frac{v_{\text{air}}^2}{g\tan\phi}, \qquad T_{\text{turn}} = \frac{2\pi R}{v_{\text{air}}}. \tag{2.15}$$

At $v_{\text{air}} = 10\,\text{m}\,\text{s}^{-1}$ and bank angle $\phi = 35°$, $2R \simeq 29\,\text{m}$ exceeds δ_{xy}, while a sustained $1\,\text{m}\,\text{s}^{-1}$ climb changes pressure altitude far more than δ_z. These are reference scales, not universal minimum diameters or guarantees for every wind, bank angle and recorder cadence.

A genuine small manoeuvre can satisfy all signatures and be cut; a jittering freeze without a witness can survive. Runs shorter than τ_{freeze} are retained. Independent inspection or labelled injections are needed to estimate these error rates.

Defect	Detected by	Action
Off the trend — *horizontal position defects*		
Position spike, out-and-back jump	Unreachable block with a rejoin within w (Eq. 2.6). The Hampel flag [14] only records local disagreement.	delete block; apply gap rule at resampling
Position discontinuity (re-acquisition offset)	No reachable rejoin within the block cap w	keep fixes; split at the discontinuity
Impossible step still standing	any surviving step past the v_{xy} bound, whatever the scan resolved (postcondition)	make it a segment boundary
Corrupt block closing the record	a block still open when the track ends: nothing follows it to rejoin, and by construction it spans at most w	delete the block, instead of splitting
On the trend — *record structurally wrong, horizontal position agrees with its neighbours*		
Duplicate timestamp	two or more fixes in one UTC second	merge to one (centroid)
Backward timestamp	time decreases (not a midnight roll-over)	delete the fix
Frozen-lock run	Spatial collapse, minimum duration and witness (Eqs. 2.13–2.14)	remove run; split trajectory
Out of range — *horizontal fix intact; altitude unavailable or suspect*		
Missing / out-of-band altitude, sustained $\lvert v_z \rvert$ excess, or $\lvert v_z \rvert$ spike	Absent reading; outside $[-100, 6000]$ m; both adjacent window statistics above $v_z^{\max}$ (Eq. 2.9); or an opposite-sign spike pair above that bound	invalidate altitude; keep horizontal fix
Policy: altitude-only invalidation retains the timestamp and horizontal coordinates. Position and time rules may remove complete fixes. These are operational decisions, not known sensor truth. At resampling, short altitude holes are reconstructed and long ones exclude the full interval and split the trajectory (Sec. 2.7.6).		

Table 2.5. Fix-level decisions. Thresholds appear in Table 2.8; Fig. 2.7 shows the absolute-bound diagnostics. Horizontal flags, deletions and altitude invalidations remain separate operations.

Integrity and validation. The integrity gate rejects flights whose cleaning burden exceeds 10 %. Counts use only the airborne window selected by trimming (Sec. 2.7.3), so discarded ground recordings do not affect this decision. Let N_{del} count deleted fixes, N_{keep} surviving fixes and $N_{z\text{bad}}$ survivors whose altitude was invalidated. The burden is

$$f_{\text{clean}} = \frac{N_{\text{del}} + N_{z\text{bad}}}{N_{\text{keep}} + N_{\text{del}}}. \tag{2.16}$$

Altitude invalidation includes out-of-band values, sustained excess, spikes and `V` declarations. Duplicate-second merges and later reconstruction are excluded. This gate removes 550 paraglider and 53 hang-glider flights. The larger loss occurs later when no segment survives resampling: 20,798 and 193 flights, respectively.

Removal fractions measure the amount changed; threshold sweeps measure sensitivity. Classification error rates require independent labels. Synthetic regressions and the real examples in Fig. 2.8 check specific failure modes, while a representative labelled archive audit remains to be done.

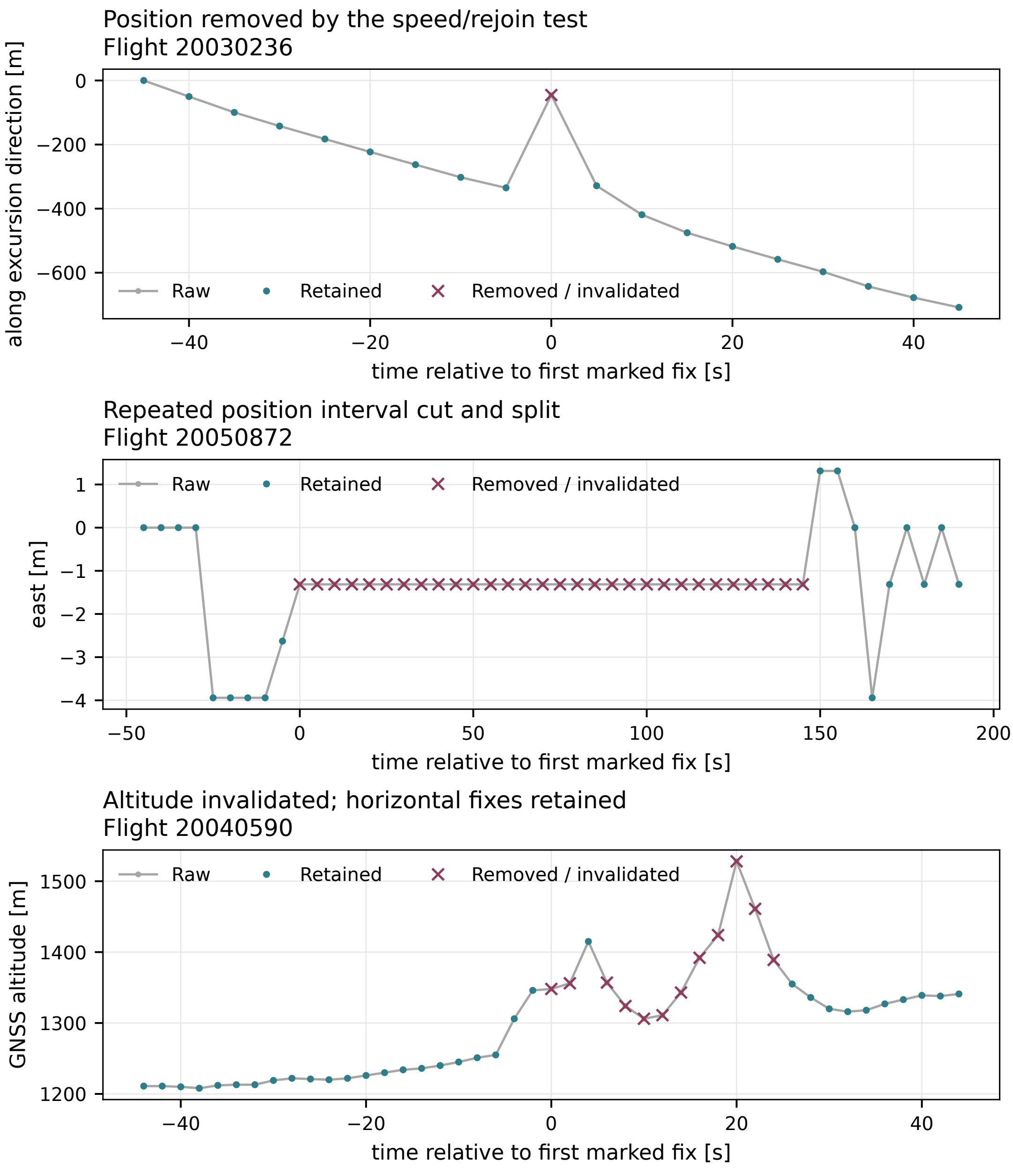


Figure 2.8. Short examples extracted from the interior of raw flight records and reprocessed with the current cleaning implementation. Top: the horizontal coordinate projected along the detected excursion direction, making the removed spike visible. Middle: a collapsed-position interval; crossed marks are removed and the surviving sides split. Bottom: GNSS altitude invalidation, with all horizontal fixes retained. Flight identifiers and plotted raw values are saved with the figure. The examples document detector decisions; no independent sensor truth is available for these intervals.

2.7.3 Trimming of ground phases

Outer ground phases (take-off and landing). The logger starts recording before take-off and stops after landing; these ground phases must be stripped before any kinematic analysis, using the horizontal speed v_{xy} alone. With $v_0 = 5\,\mathrm{m\,s^{-1}}$ and $T_0 = 30\,\mathrm{s}$ (Table 2.8), the retained airborne segment is $[t_{\mathrm{on}}, t_{\mathrm{off}}]$ with

$$t_{\mathrm{on}} = \min\{t: \ v_{xy} > v_0 \text{ on } [t,\, t+T_0]\}, \qquad t_{\mathrm{off}} = \max\{t: \ v_{xy} > v_0 \text{ on } [t-T_0,\, t]\}, \tag{2.17}$$

the same sustained-speed rule read forward and time-reversed. Here v_{xy} is the speed inferred on each observed step, not a continuously measured velocity. Interior slow periods do not define these outer boundaries, although slow flight at either end can be clipped. A flight with no sustained stretch at all is discarded.

The persistence time T_0 reduces sensitivity to brief speed excursions. It does not bound take-off or landing error by T_0: a sustained fast ground movement can be retained, while a long airborne interval below v_0 at either end can be removed. The trimmed fraction is recorded per flight, and such cases require direct inspection.

The median time removed before onset and after landing is 33 s and 71 s for paragliders, and 19 s and 62 s for hang gliders (Fig. 2.9). Counts cover 185,131 and 6,580 flights reaching a defined window. The stored `trimmed_fraction` combines the two ends. Longer post-landing intervals are consistent with continued logging after touchdown.

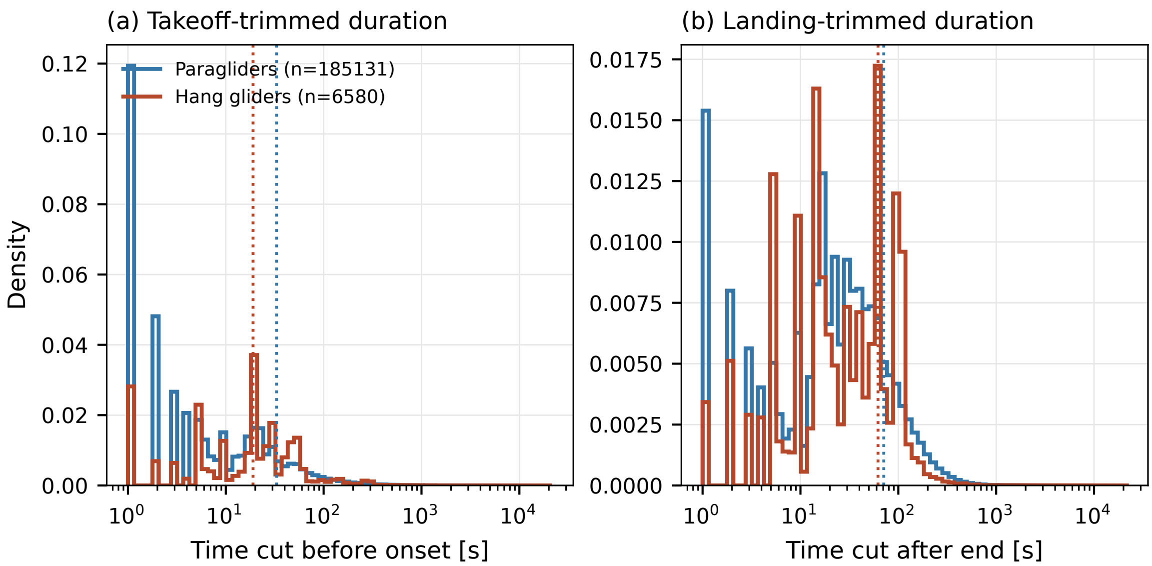


Figure 2.9. Time removed (a) before estimated onset and (b) after estimated landing, with one observation per flight per panel. Dotted lines mark discipline medians. The sustained-speed interval T_0 does not bound the time removed; the tails extend to hours.

Interior ground intervals. A landing and relaunch in one log can imitate a long waiting episode. The detector requires sustained low horizontal speed on an interval I containing at least three fixes, all with paired barometry. Fit $h_i^{\mathrm{baro}} = \widehat{\alpha} + \widehat{\beta} t_i + r_i$ by least squares. The joint conditions are

$$\begin{aligned} &v_{xy} < v_0 \text{ on } I, \qquad |I| \geq T_{\mathrm{ground}}, \qquad B_I = 1, \\ &Q_{.95}(r) - Q_{.05}(r) \leq \delta_z^{\mathrm{gr}}, \qquad |\widehat{\beta}| \leq \delta_z/\tau_{\mathrm{freeze}}, \end{aligned} \tag{2.18}$$

where $|I|$ denotes duration, $T_{\mathrm{ground}} = 10\,\mathrm{min}$ and $\delta_z^{\mathrm{gr}} = 5\,\mathrm{m}$. The residual span tolerates isolated extremes; the slope bound prevents detrending from erasing a steady climb. Without complete eligible barometry the detector abstains, since slow ground-relative flight can otherwise resemble a landing.

Excised intervals split the trajectory. Subsequent flight-level cuts apply to the parent flight, summing duration and path length over its remaining blocks; they are not reapplied separately to each segment.

Interior excisions occur in 231 paraglider flights (0.12 %) and 3 hang-glider flights (0.05 %). Most affected flights have one excision; the maxima are 5 and 2 (Fig. 2.10).

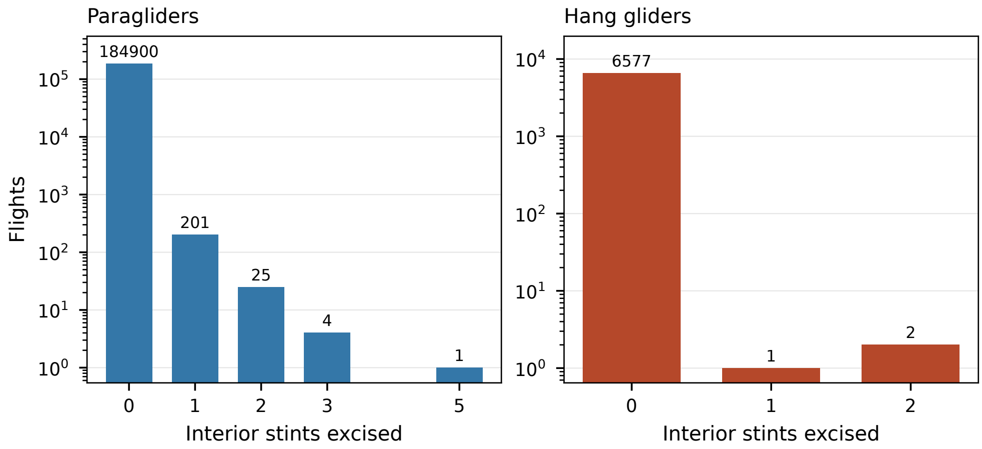


Figure 2.10. Interior intervals excised per flight among all flights reaching this stage. Logarithmic counts expose the rare nonzero cases. Empty bars denote counts absent from the archive.

Shorter intervals satisfying the same speed and barometric conditions are *flagged* when their duration reaches 60 s, the configured minimum passed to this stage. They remain in the trajectories. The stored flags support a future sensitivity analysis with and without these observations; that comparison has not yet been carried out.

2.7.4 Flight-level filtering

The cuts below define the analysed population. They act on the trimmed parent flight, with duration T_{rec} and path L accumulated only within contiguous blocks. Forced boundaries and gaps longer than $g_{\max}$ contribute to neither sum; elapsed span T_{span} includes them.

Quantity	**Condition**	**Purpose**
Recorded duration	$T_{\mathrm{rec}} \geq 40\,\mathrm{min}$	Exclude short records.
Elapsed span	$T_{\mathrm{span}} \leq 16\,\mathrm{h}$	Screen long recording sessions.
Path length	$L \geq 20\,\mathrm{km}$	Require horizontal travel.
GNSS altitude range	$\max z - \min z \geq 600\,\mathrm{m}$	Require vertical excursion.
Mean speed	$L/T_{\mathrm{rec}} \leq v_{xy}^{\max}$	Check consistency with step bounds.

These are sample restrictions, not a soaring classifier: a genuine flight with little vertical excursion can fail. Mean speed adds no independent evidence when all included steps already satisfy the bound. Figure 2.11 shows sample-size sensitivity; a plateau does not establish unbiased selection.

Raw records near the cuts. The raw census contains 9 paraglider records spanning 16 h to 166 h, and no overlong hang-glider records. Among raw records above the duration floor, the path cut excludes

	Below path floor	Above duration floor
Paragliders	236	184,541
Hang gliders	7	6,638

These raw counts are distinct from the sequential post-trimming removal counts.

Reach from the first fix. Before projection and smoothing, every fix must satisfy

$$\Delta s(\mathrm{fix}_0, \mathrm{fix}_i) \le v_{xy}^{\max} t_i + \varepsilon_{\mathrm{reach}}, \qquad \varepsilon_{\mathrm{reach}} = 50\,\mathrm{m}, \qquad \forall i, \tag{2.19}$$

with t_i measured from the first trimmed fix. Testing each elapsed time prevents a later long duration from excusing an early impossible displacement. It also catches a corrupt initial anchor whose first step is no longer summed into L.

This rejects 217 paraglider and 5 hang-glider flights in the stored snapshot. The written Cartesian trajectories receive separate checks with operational tolerances (Sec. 2.9).

Unclassified equipment remains in pooled analyses but is excluded from class-resolved comparisons (Sec. 2.4). Sampling regularity is tested after resampling (Sec. 2.7.6).

2.7.5 From geographic to local Cartesian coordinates

The cleaned, trimmed track is next mapped from geographic to local Cartesian coordinates: resampling and smoothing operate on the coordinates *componentwise* (Sec. 2.7), and latitudes and longitudes cannot play that role, being angles on a curved surface whose componentwise differences are not displacements. The mapping proceeds in two steps.

Step 1: geodetic to ECEF. Each fix is converted to Earth-Centred, Earth-Fixed (ECEF) coordinates on the WGS84 ellipsoid (semi-major axis $a = 6{,}378{,}137\,\mathrm{m}$, flattening $f = 1/298.257223563$, $e^2 = f(2-f)$) [15, 16],

$$N(\varphi) = \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi}}, \qquad \begin{aligned} X &= (N + h) \cos\varphi \cos\lambda, \\ Y &= (N + h) \cos\varphi \sin\lambda, \\ Z &= \big(N(1 - e^2) + h\big) \sin\varphi, \end{aligned} \tag{2.20}$$

$N(\varphi)$ the prime-vertical radius of curvature and h the ellipsoidal height along the normal, derived step by step in Appendix 2.B.

Geodetic versus geocentric latitude. WGS84 reports geodetic latitude φ, measured along the ellipsoid normal. Geocentric latitude ψ follows the radius from the Earth's centre (Fig. 2.12). Their difference is approximately

$$\varphi - \psi \simeq \frac{e^2}{2} \sin 2\varphi, \qquad \max|\varphi - \psi| \simeq 0.19^\circ.$$

Confusing them can shift a position by about 20 km (Appendix 2.B). Equation (2.20) already takes the geodetic latitude recorded in IGC; no latitude conversion is needed.

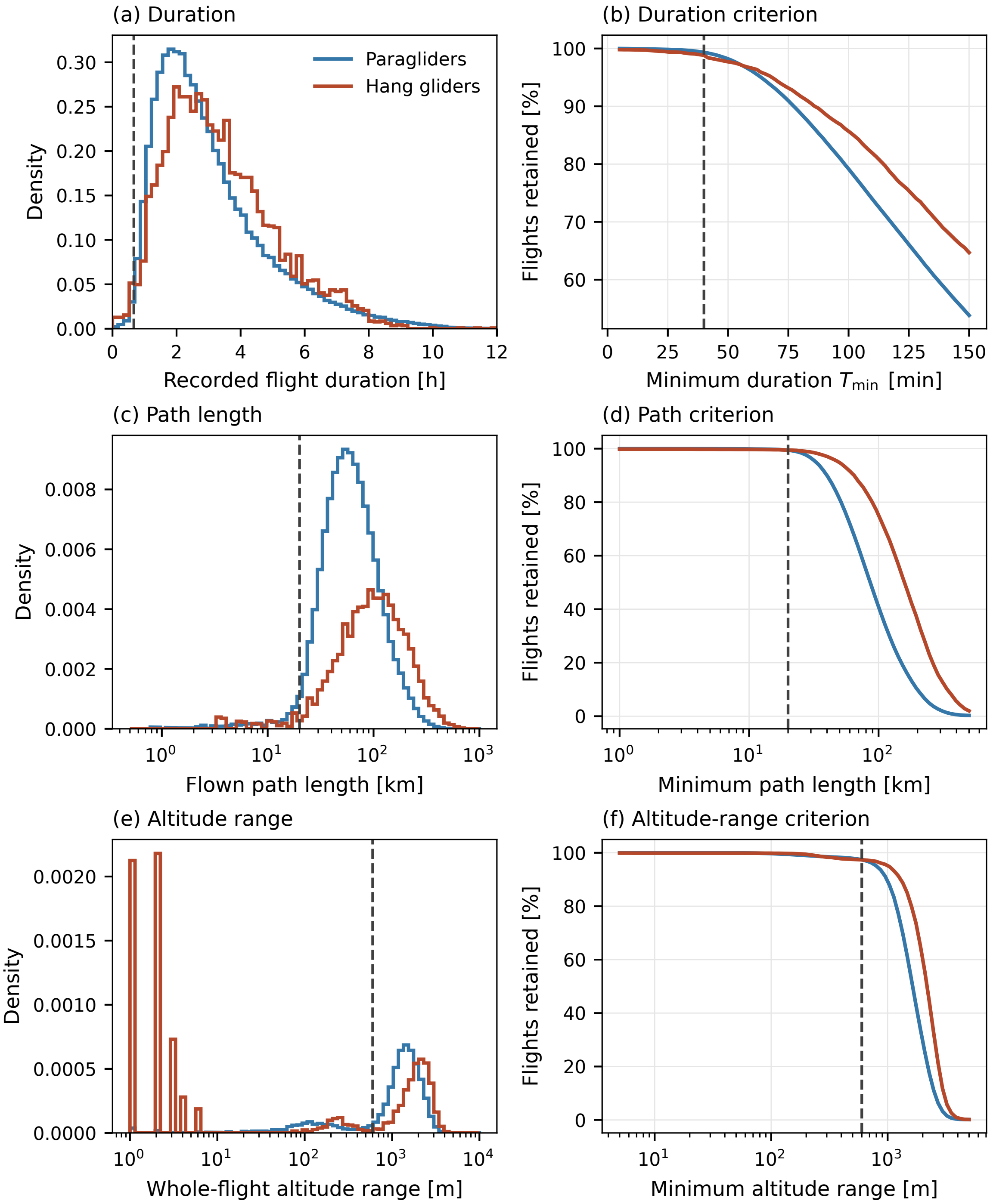


Figure 2.11. Raw flight-level diagnostics by discipline. Left column: (a) recorded duration, (c) flown path length and (e) GNSS altitude range. Right column: retention under the corresponding single criterion, evaluated separately on these raw summaries. Dashed lines mark the working thresholds ($T \geq 40$ min, $L \geq 20$ km and altitude range ≥ 600 m). The altitude panels require the configured minimum GNSS coverage. Histogram densities retain their full finite-sample denominators when tails extend outside the displayed range. These marginal raw diagnostics are not the retention fractions of the sequential cleaned-flight cuts.

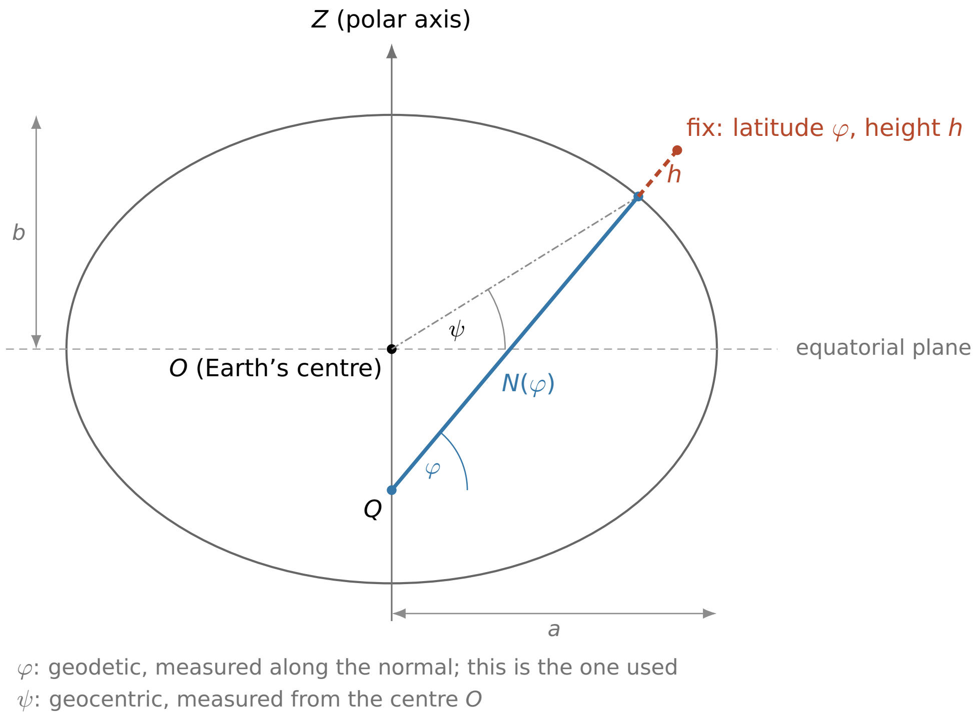


Figure 2.12. Geodetic latitude and height on the reference ellipsoid, in meridian cross-section (flattening exaggerated; the true WGS84 $f \approx 1/298$ would be indistinguishable from a circle): the semi-axes a and b; the geodetic latitude φ, the angle between the ellipsoid normal and the equatorial plane; the height h along that normal; and $N(\varphi)$, the distance along the normal from the surface to the point Q on the polar axis. The normal does not pass through the centre O, so φ differs from the geocentric latitude ψ (dash-dotted). Longitude λ is out of the page here; it is shown in Figure 2.13.

Step 2: ECEF to local ENU. Taking the first fix $(\varphi_0, \lambda_0, h_0)$ of the *trimmed* track as the origin — close to but not the take-off point, since the ground phase has been removed (Sec. 2.7.3) — the ECEF displacement $\Delta\mathbf{X} = \mathbf{X} - \mathbf{X}_0$ is rotated into the local East–North–Up (ENU) frame tangent to the ellipsoid at that point,

$$\begin{pmatrix} E \\ N \\ U \end{pmatrix} = \begin{pmatrix} -\sin\lambda_0 & \cos\lambda_0 & 0 \\ -\sin\varphi_0\cos\lambda_0 & -\sin\varphi_0\sin\lambda_0 & \cos\varphi_0 \\ \cos\varphi_0\cos\lambda_0 & \cos\varphi_0\sin\lambda_0 & \sin\varphi_0 \end{pmatrix} \Delta\mathbf{X}, \tag{2.21}$$

The rotation is orthogonal with unit determinant: it preserves the three-dimensional ECEF chord and angles (Appendix 2.B). Keeping only (E, N) then projects the chord onto the tangent plane.

For a point at surface distance d from the origin on a sphere,

$$r_{EN} = R_\oplus \sin(d/R_\oplus), \qquad d - r_{EN} = \frac{d^3}{6R_\oplus^2} + O(d^5/R_\oplus^4). \tag{2.22}$$

The horizontal shortening is about 0.03 m at 20 km and 0.51 m at 50 km; it reaches about 33 m at 200 km. The often-used $d^3/(24R_\oplus^2)$ expression instead compares an arc with its full three-dimensional chord. Neither estimate is a uniform error bound for every pair of fixes in a fixed frame.

The height input is the logged GNSS altitude, whose datum has not been harmonized. Its use as ellipsoidal height introduces a horizontal error of order

$$|\delta\mathbf{r}_{EN}| \sim |\delta h| \frac{d}{R_\oplus}.$$

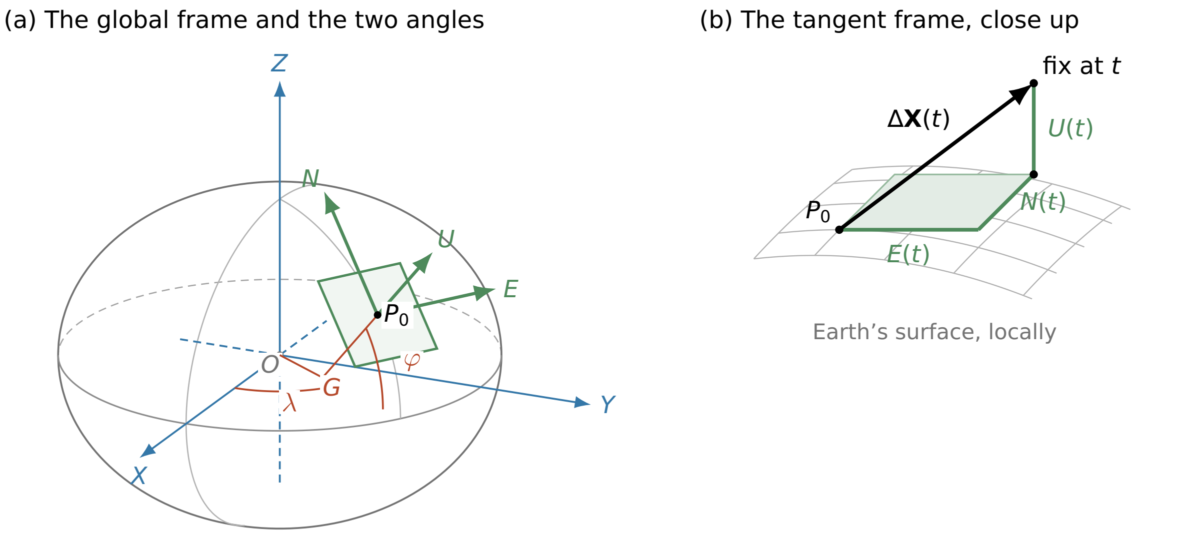


Figure 2.13. The local ENU frame. (a) The origin P_0 is the first fix of the trimmed track. Longitude λ and geodetic latitude φ orient its green unit vectors within the blue ECEF frame. Up follows the ellipsoid normal through G, with flattening exaggerated. (b) A later fix's displacement is resolved into E, N and U: the box edges project its space diagonal.

For example, a 100 m height error at 50 km contributes about 0.8 m. This illustrates the scale; it is not an archive-wide bound.

The vertical coordinate is $z(t) = h_{\mathrm{GNSS}}(t)$, without subtracting the origin altitude. The rotated component U also contains Earth curvature and is not used for vertical dynamics. Velocities depend on altitude increments; absolute heights retain the datum limitation of Sec. 2.7.1.

2.7.6 Uniform sampling within a flight

The smoothing and derivative estimates in the next section require equally spaced observations. Each flight keeps its native interval, $\Delta t = \mathrm{median}_i(t_{i+1} - t_i)$: a flight recorded every second stays at one second, and a flight recorded every five seconds stays at five seconds. Figure 2.14 shows these intervals in the archive.

Fill a short gap; split at a long one. For a gap $g = t_{i+1} - t_i$ between surviving fixes, interpolation is allowed when $g \le g_{\max}$, with

$$g_{\max} = \min(10\,\Delta t,\ \max(20\,\mathrm{s},\ 2\Delta t))\,. \tag{2.23}$$

At fast cadences the relative limit binds; at slower cadences the absolute cap binds, with a $2\Delta t$ floor allowing one missed fix. Thus the configured 20 s value is not an absolute ceiling for every logger: at $\Delta t = 15\,\mathrm{s}$, the bound is 30 s. For example,

$$\Delta t = 1\,\mathrm{s} \ \Rightarrow\ g_{\max} = 10\,\mathrm{s}, \qquad \Delta t = 5\,\mathrm{s} \ \Rightarrow\ g_{\max} = 20\,\mathrm{s}.$$

A longer gap starts a new segment. The same bound applies between consecutive finite altitude readings, even with horizontal fixes in between. Cleaning boundaries always force a split. The cap limits reconstruction but cannot ensure that manoeuvres inside short gaps are resolved.

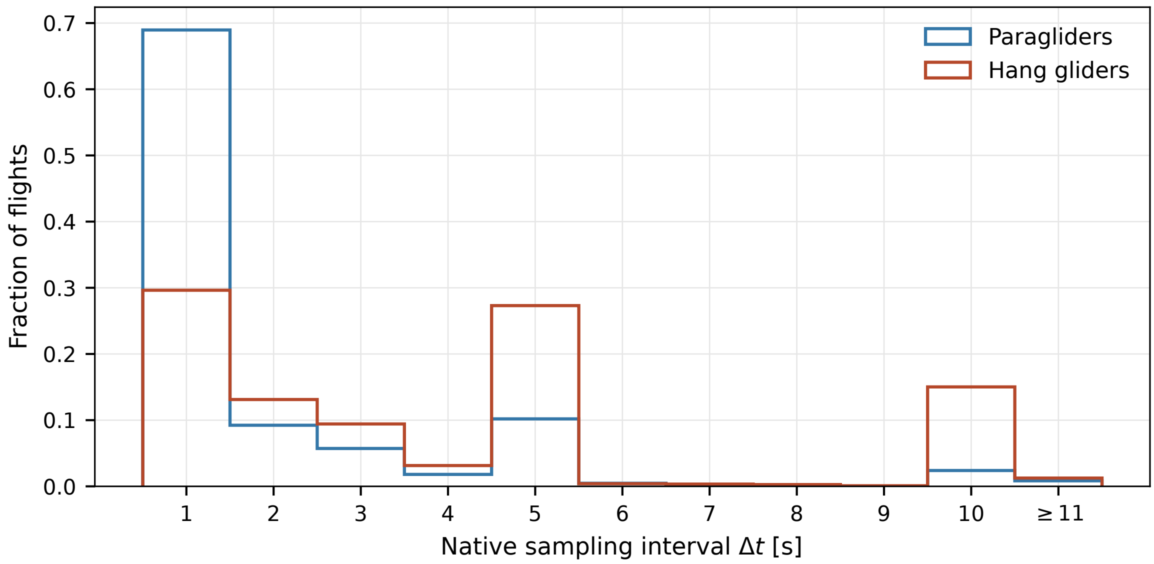


Figure 2.14. Native sampling interval per flight, measured as the median inter-fix time in the raw archive. Bars give the fraction of flights in one-second bins; the last bin also includes intervals beyond 11 s. Paragliders concentrate at 1 s, while hang gliders are spread across several cadences. Resampling preserves each flight's native interval.

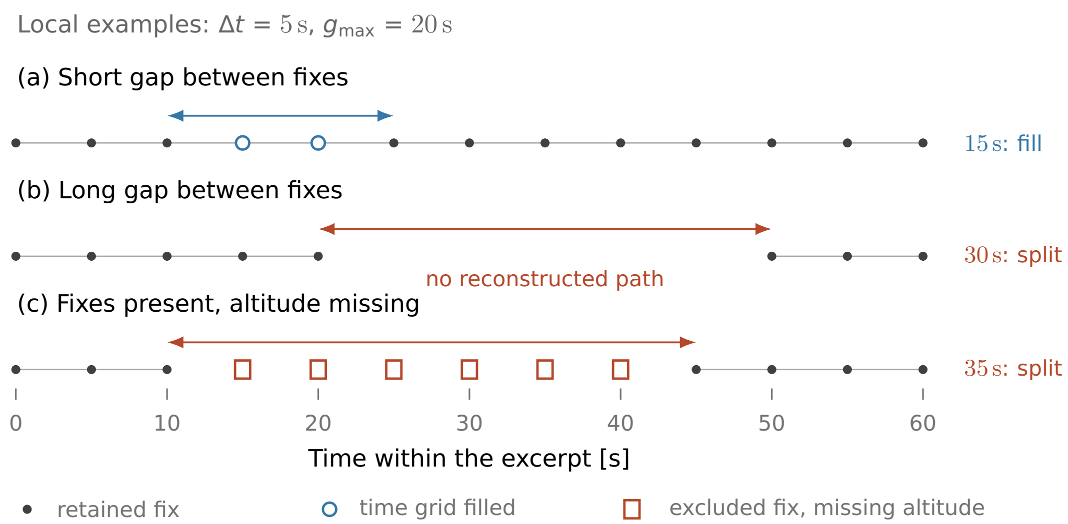


Figure 2.15. Three local examples of resampling. Circles mark new grid times in a short gap; a long gap leaves separate segments. In (c), squares mark existing horizontal fixes without valid altitude. The separation between valid altitude endpoints exceeds $g_{\max}$, so those fixes are excluded and the full trajectory is split.

Reconstructing the coordinates. Within segment s, the grid is $t_k^{(s)} = t_{\text{start}}^{(s)} + k\Delta t$, ending at or before its last fix. An already uniform, complete series is unchanged. Otherwise, E and N are evaluated with a shape-preserving cubic interpolant (PCHIP) [17], and z with linear interpolation. Linear filling avoids overshoot in altitude but can smooth away a climb–glide transition. Missing-altitude prefixes and suffixes are excluded before the grid is built; altitude is never extended beyond finite support.

For altitude missing at existing fixes, let g_z separate the surrounding finite altitude readings. The rule is

$$\begin{cases} g_z \le g_{\max}: & \text{fill altitude linearly,} \\ g_z > g_{\max}: & \text{exclude intervening fixes and split the full trajectory.} \end{cases}$$

It applies to both missing raw readings and values invalidated by cleaning (Fig. 2.15c). Unsupported intervals remain recorded as rejected segment candidates. Sacrificing their horizontal coverage avoids constructing long unsupported three-dimensional paths.

Three flags record reconstruction for the phase analysis (Chapter 4):

Flag	**Marked grid times**
`interpolated`	No surviving fix within $\Delta t/2$.
`z_reconstructed`	The above, plus times whose nearest fix has no altitude.
`z_derivative_reconstructed`	Derivative windows touched by altitude reconstruction.

The flags identify grid times or derivative windows touched by reconstruction; numerical completeness alone does not establish accuracy. The altitude-gap bound concerns the separation of finite source readings, not the duration of a consecutive run of flags on the new grid. With irregular source timing, nearest-fix flagging can merge adjacent reconstructed portions into a longer flagged run even when every finite-altitude bracket satisfies the bound.

Which segments are retained. A segment must span at least 90 s, have at most 10 % of its grid times flagged as `interpolated`, and yield finite values in all three coordinates. The temporal-coverage fraction and the altitude-gap bound are separate criteria; the latter is applied before interpolation. A rejected segment leaves the other segments available; the parent's flight-level cuts are not reapplied.

Segments retain the flight's spatial origin and elapsed time. Time-lagged displacement pairs and analysis windows are formed within segments. Splitting therefore changes the available coverage and can truncate flight phases; it must be accounted for when comparing observables. Figure 2.16 describes gaps in the raw archive; observables that depend on the recording interval, such as fix-to-fix turning angles, require a common temporal scale or a comparison within cadence groups.

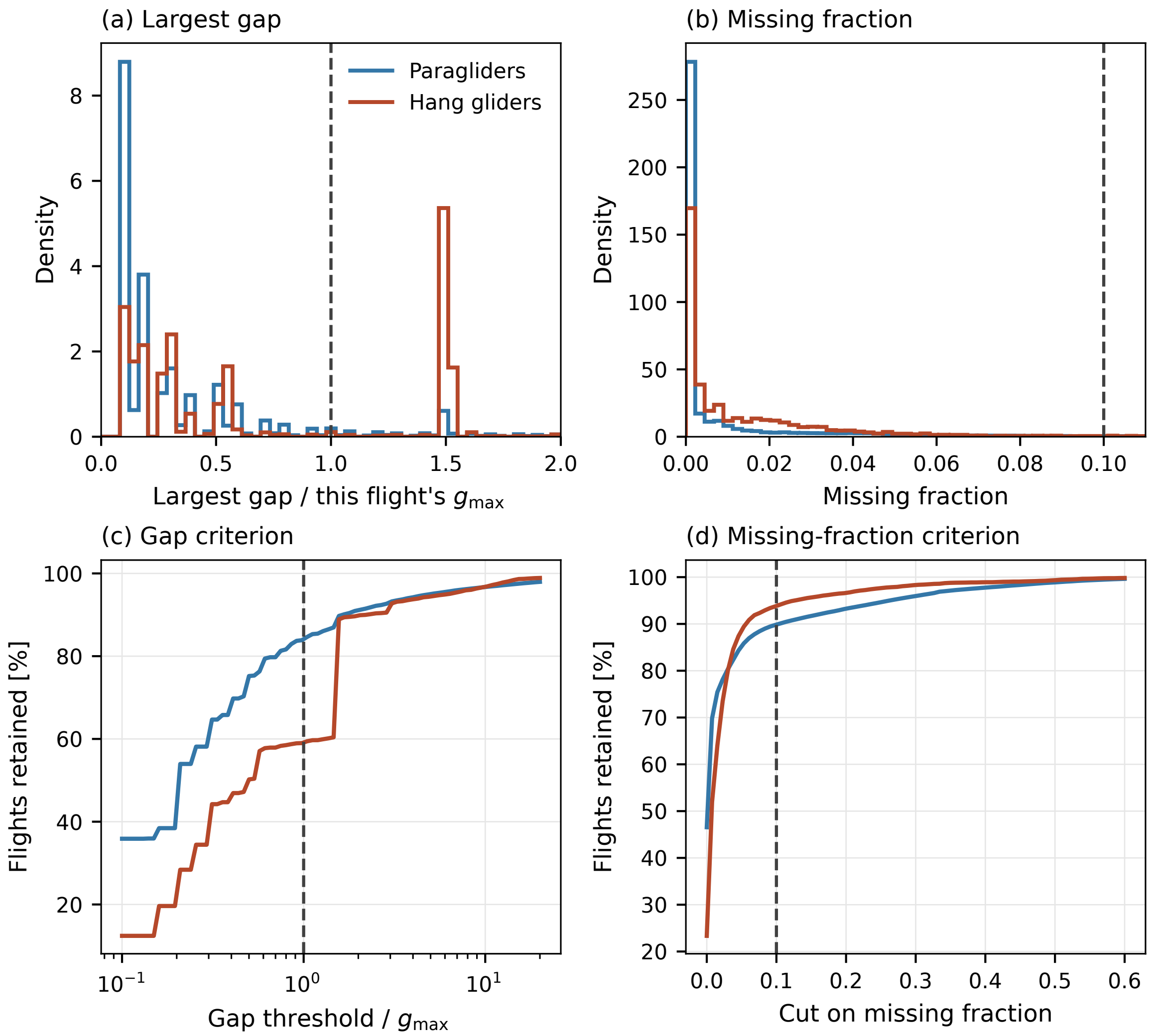


Figure 2.16. Sampling gaps in the raw archive, before cleaning and splitting. (a)/(b): distributions of the largest gap relative to $g_{\max}$ and of the missing fraction on a native-rate grid. (c)/(d): fractions of flights satisfying each criterion separately as its threshold varies. These are diagnostics of whole raw flights; the operational missing-fraction gate acts on segments after splitting, so these curves are not final flight-retention rates.

2.7.7 Smoothing and differentiation

Position errors are amplified by differentiation. We estimate position, velocity and acceleration from the same local polynomial, fitted independently to E, N and z within each retained segment. For $w = 2m + 1$ equally spaced samples around fix i, the Savitzky–Golay fit [18] is

$$P_i(u) = \sum_{k=0}^{p} a_k(i) u^k, \qquad u = -m, \dots, m, \tag{2.24}$$

$$\mathbf{a}(i) = \arg\min_{\mathbf{a}} \sum_{u=-m}^{m} [x_{i+u} - P_i(u)]^2. \tag{2.25}$$

Here u is a sample offset. The physical quantities at the window centre are

$$\widehat{x}_i = a_0(i), \qquad \widehat{v}_i = \frac{a_1(i)}{\Delta t}, \qquad \widehat{a}_i = \frac{2a_2(i)}{\Delta t^2}. \tag{2.26}$$

The weights. With $V_{u,k} = u^k$ and $\mathbf{x}_i = (x_{i-m}, \ldots, x_{i+m})^\top$, the mathematical least-squares solution is

$$\mathbf{a}(i) = (V^\top V)^{-1} V^\top \mathbf{x}_i. \tag{2.27}$$

The inverse is a formula for the weights, not a recommendation to form a matrix inverse numerically. The implementation uses SciPy's Savitzky–Golay coefficients. For fixed (w, p), row k defines weights $c_k(u)$ with

$$a_k(i) = \sum_{u=-m}^{m} c_k(u) x_{i+u}. \tag{2.28}$$

This forward-offset convention is a correlation; convolution uses the reversed weights. For an odd derivative the reversal changes their signs. Table 2.6 gives the forward-offset weights, before conversion to seconds.

Table 2.6. Interior Savitzky–Golay weights for $w = 5$, $p = 3$. Velocity and acceleration weights must be divided by Δt and Δt^2, respectively.

	sample offset u				
	-2	-1	0	1	2
c_0	$-3/35$	$12/35$	$17/35$	$12/35$	$-3/35$
c_1	$1/12$	$-8/12$	0	$8/12$	$-1/12$
$2c_2$	$2/7$	$-1/7$	$-2/7$	$-1/7$	$2/7$

The order $p = 3$ allows nonzero acceleration and locally varying acceleration. A quadratic already estimates velocity and acceleration; a cubic is a modelling choice, not a mathematical necessity. On a symmetric window, adding a cubic term changes the first-derivative weights but leaves the value and second-derivative weights of the quadratic fit unchanged.

Window length and cadence. A raw spectral diagnostic suggests a change in slope around 0.2 Hz (Fig. 2.18). It motivates the working timescale

$$\tau_c = 1/f_c = 5\,\mathrm{s}, \qquad w = \max\left(\left\lceil \frac{\tau_c}{\Delta t} \right\rceil_{\mathrm{odd}}, 5\right). \tag{2.29}$$

Horizontal and vertical timescales are separate configuration entries, currently equal. The code converts this timescale to a *number of samples*: the distance in time between the first and last sample of a window is $(w-1)\Delta t$. Thus $w = 5$ spans 4 s at a 1 s cadence, and 20 s at a 5 s cadence. Slower recording therefore changes the physical smoothing scale and can remove motion that faster loggers resolve. Comparisons of short-time observables must control for cadence.

The spectral knee is neither a fitted receiver-noise model nor the cutoff frequency of this filter. Rounding a field to steps of size q has variance $q^2/12$ only under a uniform-error approximation; a flat quantization spectrum additionally requires temporal independence. Similar high-frequency spectra do not establish those assumptions or identify the source of the observed knee.

Frequency response. In the segment interior, a Fourier mode $x_i = e^{2\pi \mathrm{i} f i \Delta t}$ is multiplied by

$$G(f) = \sum_{u=-m}^{m} c_0(u) e^{2\pi \mathrm{i} f u \Delta t}, \qquad S_{\hat{x}}(f) = |G(f)|^2 S_x(f). \tag{2.30}$$

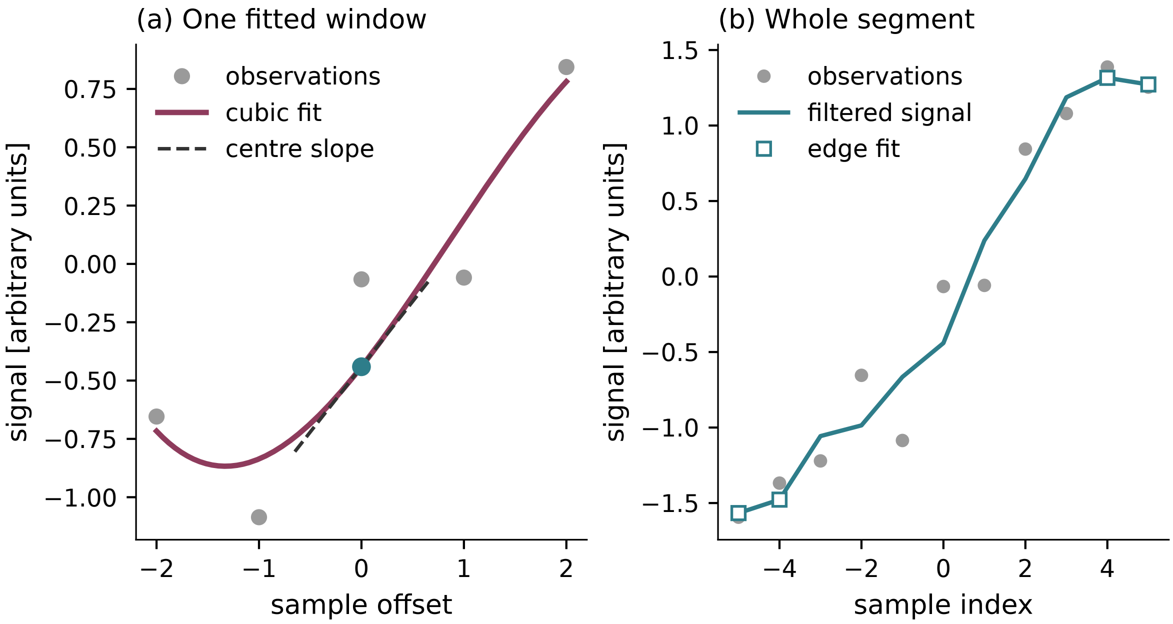


Figure 2.17. A cubic fit to a synthetic signal. Left: one five-sample window, its fit and the fitted value and slope at the centre. Right: the output of the implemented filter over the same series; squares mark the two edge samples at each end. Synthetic values are $\sin(0.28j) + \varepsilon_j$, with independent $\varepsilon_j \sim \mathcal{N}(0, 0.35^2)$.

The response depends on frequency and cadence through

$$\nu = f\Delta t \in [0, \tfrac{1}{2}], \tag{2.31}$$

in cycles per sample, with Nyquist at $\nu = 1/2$. Thus the same $G(\nu)$ applies to every cadence: $\nu = 0.2$ corresponds to $0.2\,\mathrm{Hz}$ at $\Delta t = 1\,\mathrm{s}$ and $0.04\,\mathrm{Hz}$ at $\Delta t = 5\,\mathrm{s}$.

Symmetric weights give a real response, $G(\nu) = c_0(0) + 2\sum_{u=1}^{m} c_0(u)\cos(2\pi\nu u)$. The phase is unchanged where $G > 0$ and reversed by π where $G < 0$. For the five-sample cubic fit (Table 2.6),

$$G(\nu) = \frac{17 + 24\cos(2\pi\nu) - 6\cos(4\pi\nu)}{35}. \tag{2.32}$$

Figure 2.19 shows the resulting power transmission. The filter preserves constants, $G(0) = 1$, and transmits half the power near $\nu = 0.24$. It vanishes at $\nu_0 = \arccos(1 - \sqrt{35/12})/(2\pi) \approx 0.375$, then rises to $|G(1/2)|^2 = (13/35)^2 \approx 0.14$. High frequencies are therefore attenuated unevenly by this short kernel.

Effect on the MSD. Write one recorded coordinate as $\widetilde{x}_i = x_i + \varepsilon_i$. For zero-mean errors independent of the true motion and of one another, with $\mathbb{E}[\varepsilon_i\varepsilon_j] = \sigma^2\delta_{ij}$, the one-coordinate MSD at lag $\ell \geq 1$ is

$$\mathbb{E}\big[(\widetilde{x}_{i+\ell} - \widetilde{x}_i)^2\big] = \mathbb{E}\big[(x_{i+\ell} - x_i)^2\big] + \underbrace{\mathbb{E}\big[(\varepsilon_{i+\ell} - \varepsilon_i)^2\big]}_{2\sigma^2}. \tag{2.33}$$

The constant noise contribution matters most at short lags, where the true displacement is small.

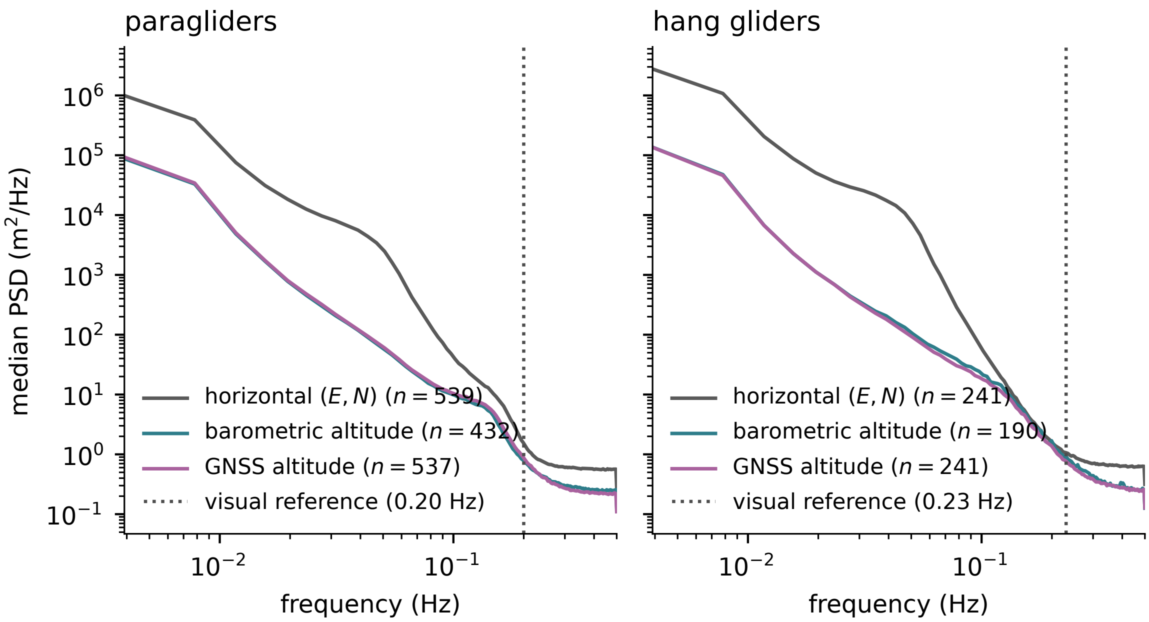


Figure 2.18. Median raw Welch PSDs, by discipline and coordinate channel, on a seeded sample of up to 900 flights per discipline. Spectra use recording intervals near 1 s; legend counts give the usable flights. Horizontal coordinates use a local equirectangular approximation before cleaning, and the curve pools both component spectra from each admitted flight. Channel eligibility is assessed separately, so the three medians need not represent identical flights or intervals. The vertical dotted lines are visual reference frequencies, 0.20 Hz and 0.23 Hz, not estimates with an uncertainty interval. Recording precision, sensor error, interpolation and real motion can all affect these spectra.

Smoothing replaces ε_i by the weighted average $\eta_i = \sum_u c_0(u)\varepsilon_{i+u}$, with c_0 extended by zero outside $|u| \leq m$. Independence of the raw errors leaves only the terms that read the same fix, so with the weight autocorrelation $R(\ell) = \sum_u c_0(u)c_0(u-\ell)$,

$$\mathbb{E}[\eta_i\eta_{i+\ell}] = \sigma^2 R(\ell), \tag{2.34}$$

$$B(\ell) = \mathbb{E}[(\eta_{i+\ell} - \eta_i)^2] = 2\sigma^2\left[R(0) - R(\ell)\right], \tag{2.35}$$

For $|\ell| \geq w$, the two windows share no raw errors and $R(\ell) = 0$.

For $w = 5$, $p = 3$ the weights of Table 2.6 give

$$35^2 R(\ell) = 595,\ 336,\ 42,\ -72,\ 9,\ 0 \qquad (\ell = 0, \ldots, 5), \tag{2.36}$$

Thus $B(\ell)/\sigma^2 \approx 0.42, 0.90, 1.09, 0.96$ at $\ell = 1, \ldots, 4$, settling at $34/35 \approx 0.97$ for $\ell \geq 5$. The filter roughly halves the noise contribution beyond its window but makes it lag dependent inside, with an overshoot at $\ell = 3$ (Fig. 2.19). A horizontal MSD adds both coordinate contributions.

This calculation assumes independent errors; it does not calibrate GNSS noise. The filter also changes the true motion unless each window follows a polynomial of degree at most p. A Hurst exponent alone cannot bound that change.

Edges and validation. At either end of a segment, the filter fits the nearest full window and evaluates that polynomial inside its fitted interval. These are off-centre evaluations, not extrapolation beyond the observations. The first and last m samples carry an `edge` flag because their weights and error variance differ from the interior. The filter can overshoot despite earlier cleaning, so speed bounds are not guaranteed by smoothing.

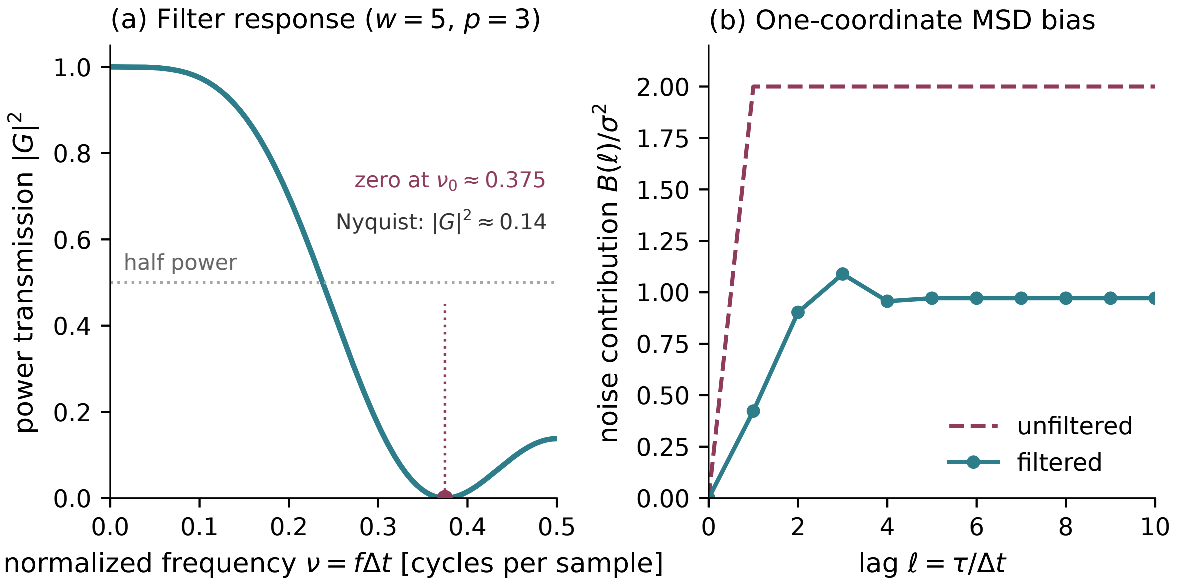


Figure 2.19. Exact interior response of the five-sample cubic filter. Left: power transmission $|G(\nu)|^2$, with $\nu = f\Delta t$ in cycles per sample and Nyquist at 1/2. Right: additive one-coordinate MSD bias $B(\ell)/\sigma^2$ for independent measurement errors, compared with the unfiltered value 2. Overlapping windows produce the short-lag variation.

The kernel identities and derivative units are checked in automated tests. Archive-level validation remains separate: after rebuilding the cleaned archive, compare the MSD and phase features at neighbouring odd window lengths, within each cadence class, and with edge and reconstructed samples excluded. A flat residual spectrum is not a sufficient criterion: even white input noise has residual PSD $|1 - G(f)|^2 S_\varepsilon(f)$. Until these comparisons are reported, the adopted window is a working choice rather than a certified optimum.

2.7.8 Notation

The coordinates and their derivatives are

$$\begin{aligned} \mathbf{r}(t) &= (E(t), N(t)), & z(t) &= h_{\mathrm{GNSS}}(t), \\ \mathbf{v}(t) &= \dot{\mathbf{r}}(t), & v_z(t) &= \dot{z}(t), \\ v_{xy}(t) &= |\mathbf{v}(t)|, & \Delta\mathbf{r}(t_0, \tau) &= \mathbf{r}(t_0 + \tau) - \mathbf{r}(t_0). \end{aligned}$$

The first trimmed fix sets $t = 0$ and the horizontal origin; smoothing can shift its position slightly. Altitude keeps its logged value. Every segment retains the parent flight's origin and clock. A bare *position* denotes the horizontal vector $\mathbf{r}$; three-dimensional position is stated explicitly.

2.8 Pipeline summary

Figure 2.20 gathers the processing steps of Sec. 2.7 in the order they are applied, from raw IGC tracks to the retained trajectories. Each stage identifies the records it acts on, its effect and the section that explains it. Section 2.9 then describes the resulting dataset.

Each flight is processed independently using its own cadence and geometry, with the shared thresholds in Table 2.8. Adding flights to the archive therefore does not change how an existing record is cleaned.

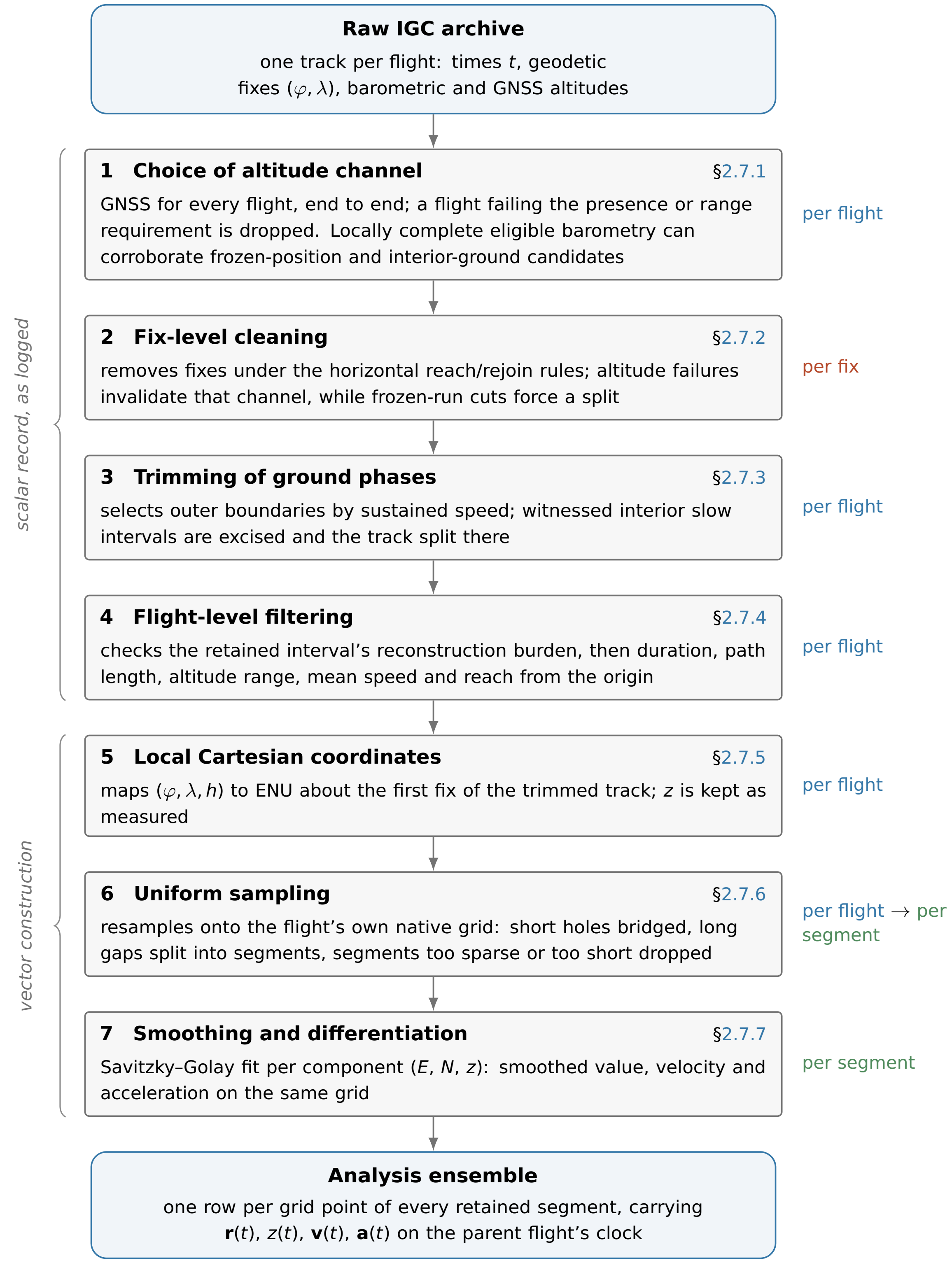


Figure 2.20. The processing chain, from the raw archive to the analysis ensemble. Each stage carries its effect and the section that argues the choice; the badge to the right is the level it acts at (per fix, per flight, per segment). Every check on the scalar record (stages 1–4) precedes the construction of vector quantities (stages 5–7), so no coordinate conversion is computed for a record about to be repaired or dropped. The thresholds each stage applies are listed, with their symbols, in Table 2.8.

What the chain retains. Run over the whole archive, the cascade retains 156,406 of 186,052 paraglider flights (84.1 %) and 6,094 of 6,716 hang-glider flights (90.7 %). Table 2.7 says where the rest went. Counts form a cascade: each rejected flight reached that stage after passing earlier criteria. They differ from the single-criterion raw diagnostics in Secs. 2.7.4 and 2.7.6.

Within the retained flights, cleaning deletes 1.83 % of paraglider fixes and 0.72 % of hang-glider ones, and invalidates an altitude on a further 0.056 % and 0.036 %. The local outlier test flags far more than is deleted, and by design: it never gates, so 0.68 % and 0.96 % of fixes are flagged and *kept*. The flag records local disagreement, while the impossibility gate controls horizontal deletion (Sec. 2.7.2). Resampling reconstructs 0.58 % and 0.30 % of grid points. A split is common but not the rule: 9.9 % of retained paraglider flights and 20.5 % of hang-glider ones carry at least one, the difference following the cadences of Fig. 2.14 as Sec. 2.7.6 anticipated.

Table 2.7. The pre-processing cascade over the full archive, in the order the pipeline applies it. *Standing* counts flights that survived earlier criteria; *share* is the fraction removed from that remaining population. Resampling leaves no retained segment in 20,798 paraglider flights (11.7 % of those reaching that step) and 193 hang-glider flights (3.1 %). These conditional removal fractions differ from the separate raw-population diagnostics in Secs. 2.7.4 and 2.7.6.

	paragliders			
Why the flight was dropped	**standing**	**removed**	**share (%)**	**remaining**
unreadable track	186,052	27	0.01	186,025
no usable GNSS altitude	186,025	697	0.37	185,328
never airborne	185,328	197	0.11	185,131
integrity gate	185,131	550	0.30	184,581
shorter than the duration cut	184,581	2,084	1.13	182,497
longer than the duration cut	182,497	3	0.00	182,494
path below the floor	182,494	173	0.09	182,321
altitude activity below the floor	182,321	4,899	2.69	177,422
out of reach of its own first fix	177,422	217	0.12	177,205
no segment survived resampling	177,205	20,798	11.74	156,407
no segment carries the smoothing window	156,407	1	0.00	156,406
retained				**156,406**

	hang gliders			
Why the flight was dropped	**standing**	**removed**	**share (%)**	**remaining**
no usable GNSS altitude	6,716	135	2.01	6,581
never airborne	6,581	1	0.02	6,580
integrity gate	6,580	53	0.81	6,527
shorter than the duration cut	6,527	79	1.21	6,448
path below the floor	6,448	1	0.02	6,447
altitude activity below the floor	6,447	155	2.40	6,292
out of reach of its own first fix	6,292	5	0.08	6,287
no segment survived resampling	6,287	193	3.07	6,094
retained				**6,094**

Table 2.8 lists the working thresholds and their symbols by stage.

Table 2.8. Working values of the pre-processing pipeline, by stage. Symbols are those used in the text; configured values are read from `configs/preprocessing.yaml`. The minimum smoothing window is fixed by the polynomial-order rule in Sec. 2.7.7. The listed sections explain each criterion.

Symbol	Value	What it controls
Altitude admission (Sec. 2.7.1)		
—	95 %	minimum fraction of finite, nonzero GNSS readings; a flight below this completeness threshold is excluded
—	30 m	minimum GNSS range among finite, nonzero readings; a flight below it is excluded, without establishing a sensor failure
—	95 %	minimum barometric presence for witness eligibility; each candidate also requires complete local support
—	30 m	minimum barometric range for the same witness eligibility
Fix-level cleaning (Sec. 2.7.2)		
k	5	robust-σ multiplier of the Hampel test, Eq. (2.5)
w	20 s	half-width of the Hampel window; the implementation reuses it as the longest a pending block may stay open before the track is cut (Sec. 2.7.2)
$\varepsilon_{\min}$	15 m	absolute floor of the Hampel test: no flag below this residual, whatever the local scale
—	5 fixes	smallest window population for which the test is evaluated
$v_{xy}^{\max}$	$45\,\mathrm{m\,s^{-1}}$ (para), $55\,\mathrm{m\,s^{-1}}$ (hang)	absolute inter-fix speed bound; the impossibility gate, on its own (Sec. 2.7.2)
$\lvert v_z\rvert^{\max}$	$10\,\mathrm{m\,s^{-1}}$	cutoff for the two adjacent window statistics in Eq. (2.9); also used by the out-and-back spike rule
w_v	5 s	half-width of that window
—	5 finite steps	smallest window population for which the median is estimable; below it the per-step value stands in
$[h_{\min}, h_{\max}]$	$[-100, 6000]$ m	sensor-plausibility window on the altitude
δ_{xy}	15 m	diameter within which a frozen-lock run must stay
δ_z	5 m	altitude flatness tolerance of the same rule
τ_{freeze}	60 s	minimum duration before a run can be cut by the frozen-position rule
—	10 %	share of fixes above which the cleaning is declared to have failed and the flight is dropped
Ground trimming (Sec. 2.7.3)		
v_0	$5\,\mathrm{m\,s^{-1}}$	ground-speed threshold used to identify take-off and landing candidates
T_0	30 s	how long $v_{xy} > v_0$ must persist for take-off (and, time-reversed, landing)
T_{ground}	10 min	minimum duration of an interior ground stint before it is excised
—	5 m	altitude flatness tolerance over such a stint
Flight-level filtering (Sec. 2.7.4) — discards whole flights that cannot support the statistics		

continued on the next page

Table 2.8 (continued)

Symbol	**Value**	**What it controls**
—	40 min	minimum recorded airborne duration, accumulated within blocks
—	20 km	minimum flown path length
—	16 h	maximum elapsed span, including gaps; records above it are excluded
—	600 m	minimum whole-flight altitude range
—	$v_{xy}^{\max}$ (above)	plausibility bound on the mean ground speed of the cleaned track; reuses the fix-level bound, read over a whole flight (Sec. 2.7.4)
—	$v_{xy}^{\max}$ (above)	reach bound applied at every fix: $d_i \le v_{xy}^{\max}(t_i - t_0) + 50\,\mathrm{m}$, where d_i is its distance from the first trimmed fix
Resampling (Sec. 2.7.6) — one uniform grid per flight, holes bridged or split		
$g_{\max}$	$\min\bigl(10\,\Delta t,\ \max(20\,\mathrm{s},\ 2\Delta t)\bigr)$	longest gap that is interpolated instead of splitting the flight
—	10 %	largest share of grid points a retained *segment* may have had to interpolate
—	90 s	minimum retained segment duration; smoothing additionally requires enough grid samples for its window
Smoothing (Sec. 2.7.7) — Savitzky–Golay filter and its derivatives		
p	3	polynomial order of the filter
τ_c	5 s (horizontal), 5 s (vertical)	working timescale setting the window through Eq. (2.29); equal values are adopted for the two channels, without identifying a physical correlation time
—	5 samples	minimum odd window for the adopted cubic fit. The temporal span is $(w-1)\Delta t$, with w given by Eq. (2.29)

The analysis ensemble. The analysis ensemble comprises the retained segments written to `fixes.parquet`, with one row per grid time carrying (E, N, z) and their first two derivatives. A flight must pass stages 1–4 and retain at least one segment after resampling and smoothing. Segment origins and clocks remain those of the first trimmed fix of the parent flight.

Transport and phase estimators impose their own cadence, lag-support and quality-mask requirements, so their contributing populations can differ. The file schema and estimator inputs are documented in `docs/guide/data-on-disk.md`.

2.9 Preliminary characterization

This section describes the retained flights: their duration, path length, recording interval and geographical coverage. These measurements require no phase segmentation. Directional differences and their relation to geography are examined with the transport observables in Sec. 3.4.

Trajectory statistics. Figure 2.21 compares the retained trajectories. Hang-glider flights are typically longer and logged less frequently:

	Paragliders	**Hang gliders**
Retained flights	156,406	6,094
Median retained span [h]	2.63	3.07
Median path [km]	87	151
Median cadence [s]	1	3
Recorded at 1 Hz [%]	69.9	28.0

Paraglider interquartile ranges are 1.77 h to 3.93 h for duration and 57 km to 134 km for path length. The median record contains 7,132 fixes. The native cadence sets each flight's smoothing window (Sec. 2.7.7).

The dataset audit counts 0 non-finite coordinates and 0 non-monotone flight clocks in the paraglider audit. Its largest filtered interior step speed is $49\,\mathrm{m\,s^{-1}}$, with 26 interior steps above the working bound but below the verifier's gross-anomaly threshold. Across segment boundaries, 30 paraglider and 7 hang-glider flights exceed its separate speed-envelope diagnostic, which compares displacement with elapsed time and the declared coordinate tolerance. These boundary anomalies do not establish a unique sensor-error mechanism. Splitting prevents increment windows from bridging them; it does not independently validate absolute positions after reacquisition.

These checks identify structural failures and large kinematic anomalies. Finite coordinates and bounded speeds do not exclude plausible-looking recorder errors.

Panels (a) and (b) describe retained flights. Here duration is the elapsed span from the first to the last retained fix, including intervening gaps; path length sums only within-segment steps. These quantities can be smaller than those of the trimmed parent flight tested in Sec. 2.7.4, because short, sparse or unreconstructable segments can be discarded. The resulting distributions include 609 paraglider flights (0.39 %) and 6 hang-glider flights below the original 40 min or 20 km threshold. The parent-flight quantities passed those cuts; the cuts are not reapplied to the shorter retained trajectory.

Retention across recording intervals and seasons. The cascade in Table 2.7 retains most attempted flights. Loss of every segment at resampling is the largest conditional cut in both archives:

	Paragliders	**Hang gliders**
End-to-end retention [%]	84.1	90.7
No segment survives [%]	11.7	3.1
Altitude-range cut [%]	2.7	2.4
Ratio of the two cut rates	4.4	1.3

Each cut rate uses the flights reaching that stage. The 20.798 paraglider losses at resampling combine segments that are too short, too sparse or unreconstructable; they cannot all be attributed to the missing-fraction bound.

Across the six most common known cadences, the paraglider share with no surviving segment ranges from 0.0 % to 22.4 %. This variation calls for sensitivity checks of the missing-fraction and minimum-duration thresholds.

Across the 20 seasons with at least a thousand flights, overall retention ranges from 69.8 % to 91.9 % (Fig. 2.22b). Similar rates can conceal changes in cadence, region and pilot composition.

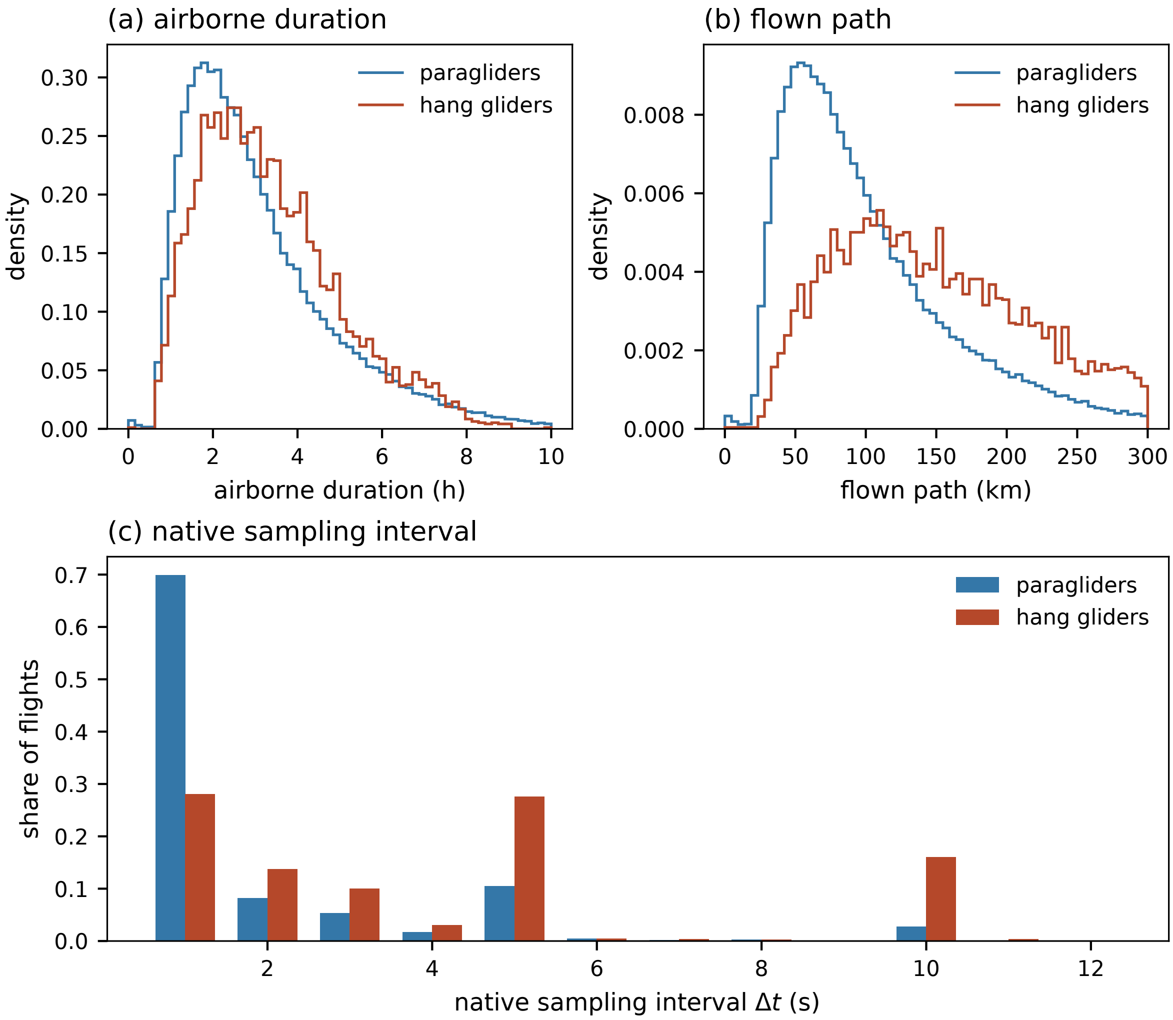


Figure 2.21. The records the pipeline retains. (a) Airborne duration, (b) flown path and (c) native sampling interval Δt, per discipline, over the retained flights rather than over every parsed flight—which is what distinguishes these from Figs. 2.16 and 2.14. The duration labelled in panel (a) is the first-to-last retained-fix span, including gaps, rather than observed airborne time. Path length in (b) sums only within-segment steps. These retained-record quantities differ from the trimmed-parent quantities tested by Sec. 2.7.4; no parent-flight threshold is reapplied after resampling.

Where the flights are. The distribution of take-off sites is presented across three separate panels to account for the geographic coverage of the CFD: Figure 2.23 displays these sites against coastline and national borders.

- **Metropolitan France.** 94.7 % of retained paraglider flights launch here, against 96.7 % of hang-glider ones — the hang gliders are slightly more domestic.
- **La Réunion.** 3.2 % (4,946 flights), the one overseas department with a substantial share.
- **Elsewhere.** 2.2 % (3,379 flights): the Canaries, Andalusia, the Italian Alps, and further out Colombia, Nepal, India, South Africa and Brazil.

Within metropolitan France the ensemble does not sample the country evenly. It occupies 5,341 distinct take-off sites, where distinct sites are launches rounded to a hundredth of a degree (∼1 km); by share of paraglider launches:

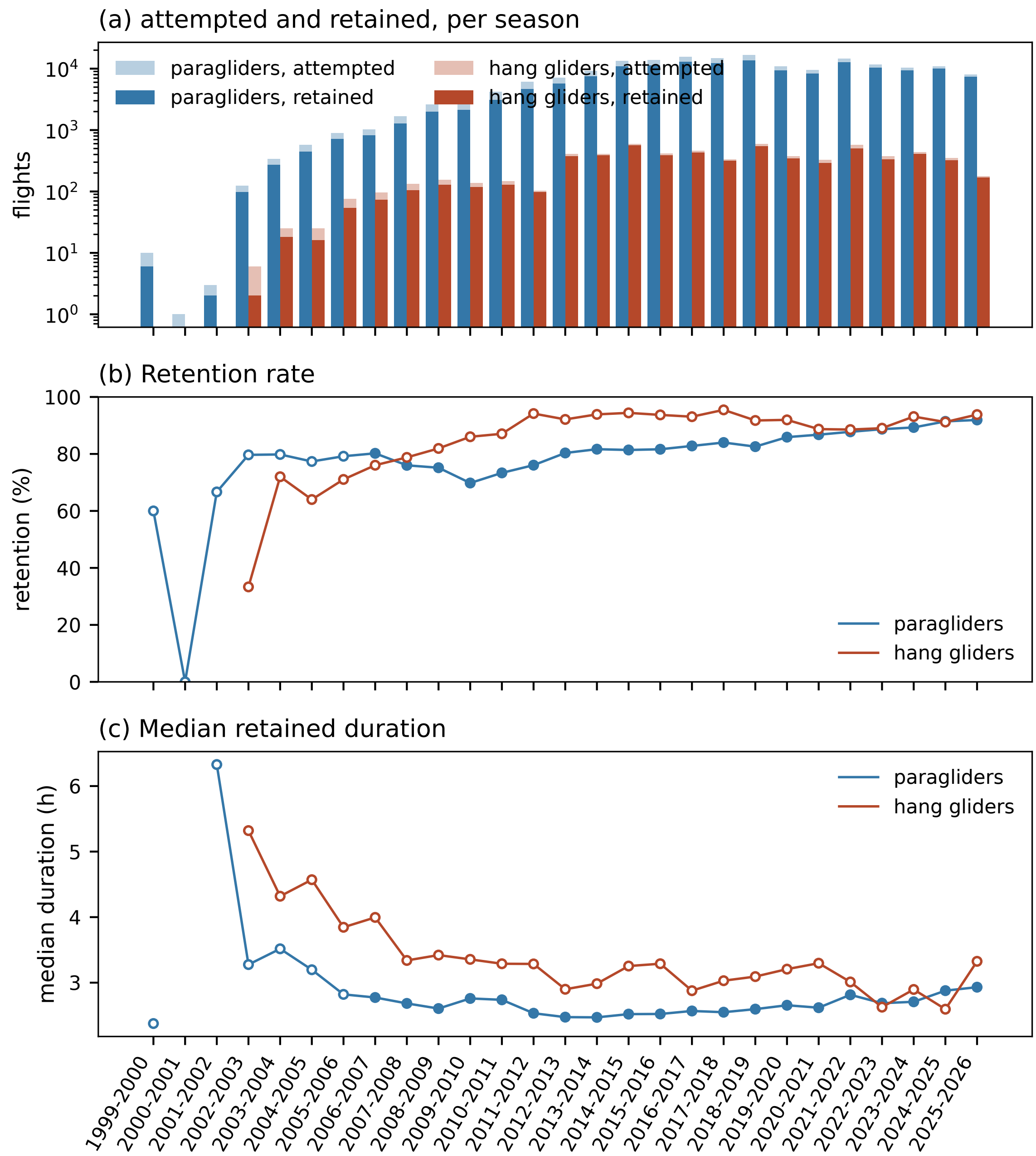


Figure 2.22. The archive season by season. (a) Flights attempted and retained, on a logarithmic scale: the ladder grows by three orders of magnitude over its life. (b) Retention rate and (c) median airborne duration of the survivors; open markers are seasons of fewer than 1,000 flights. The aggregate retention rate is approximately flat across the larger seasons; this does not establish equal selection within cadence or region groups.

- **Alps.** 58.8 %.
- **Pyrenees.** 7.9 %.
- **Massif Central.** 8.2 %.
- **Channel Coast.** 3.2 %: a box over low-relief terrain (Normandy, Picardy, the coastal Nord), not one carved out by excluding the other three.
- **Outside the massifs.** 16.7 %: everywhere else in metropolitan France, a residual defined by exclusion and not a terrain claim.

These groups are longitude and latitude boxes, drawn on Fig. 2.23a; membership is assigned from the first recorded position.

Altitude at the beginning of the record. Figure 2.24 groups flights by their first raw altitude, before trimming: GNSS where available, pressure altitude otherwise. This classifies 99.9 % of retained paraglider flights. Their shares are

Legend label	Logged altitude [m]	Share [%]
Plains	< 300	7.6
Hills	300–800	11.2
Low mountains	800–1500	50.7
High mountains	≥ 1500	30.5

The bands follow Hernández-Aguayo, Cristelli and Benzaquen [19]. They simplify the full classification of Kapos et al. [20]; their historical names do not measure local relief or slope.

The first fix can be away from launch, already airborne, or affected by sensor settling. Datum and atmospheric offsets can move it between bands. This exploratory map therefore describes recorded starting altitude, with pressure fallback confined to this map.

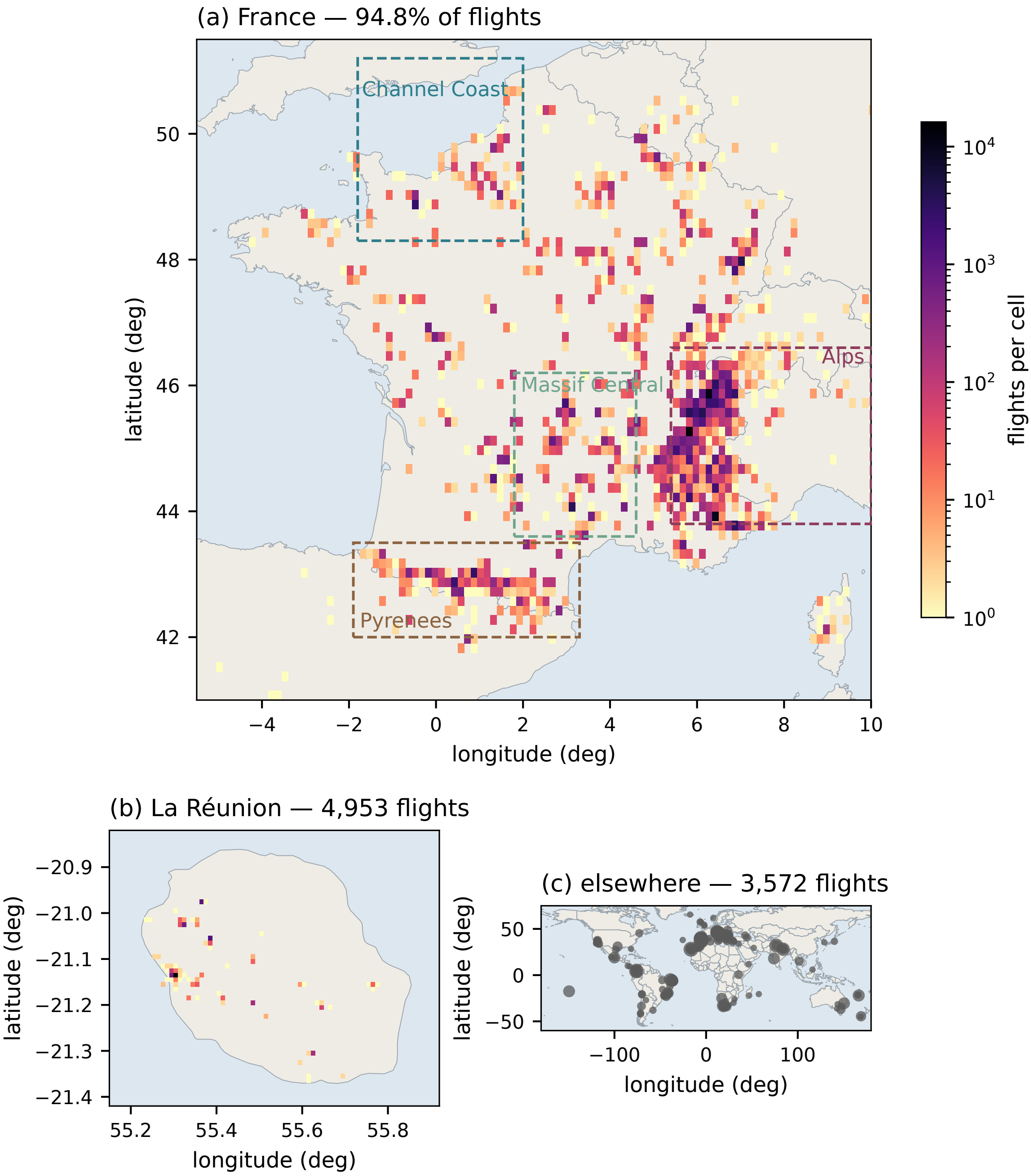


Figure 2.23. Take-off sites of the retained ensemble, on three frames, pooling both disciplines throughout. (a) Metropolitan France, density on a 0.15° cell; the dashed boxes are the orographic groups used as strata in Fig. 3.15 (Sec. 3.4). (b) La Réunion, on a 0.01° cell, the one overseas department with a substantial share. (c) Everything else, one marker per 0.5° cell with the area set by its count.

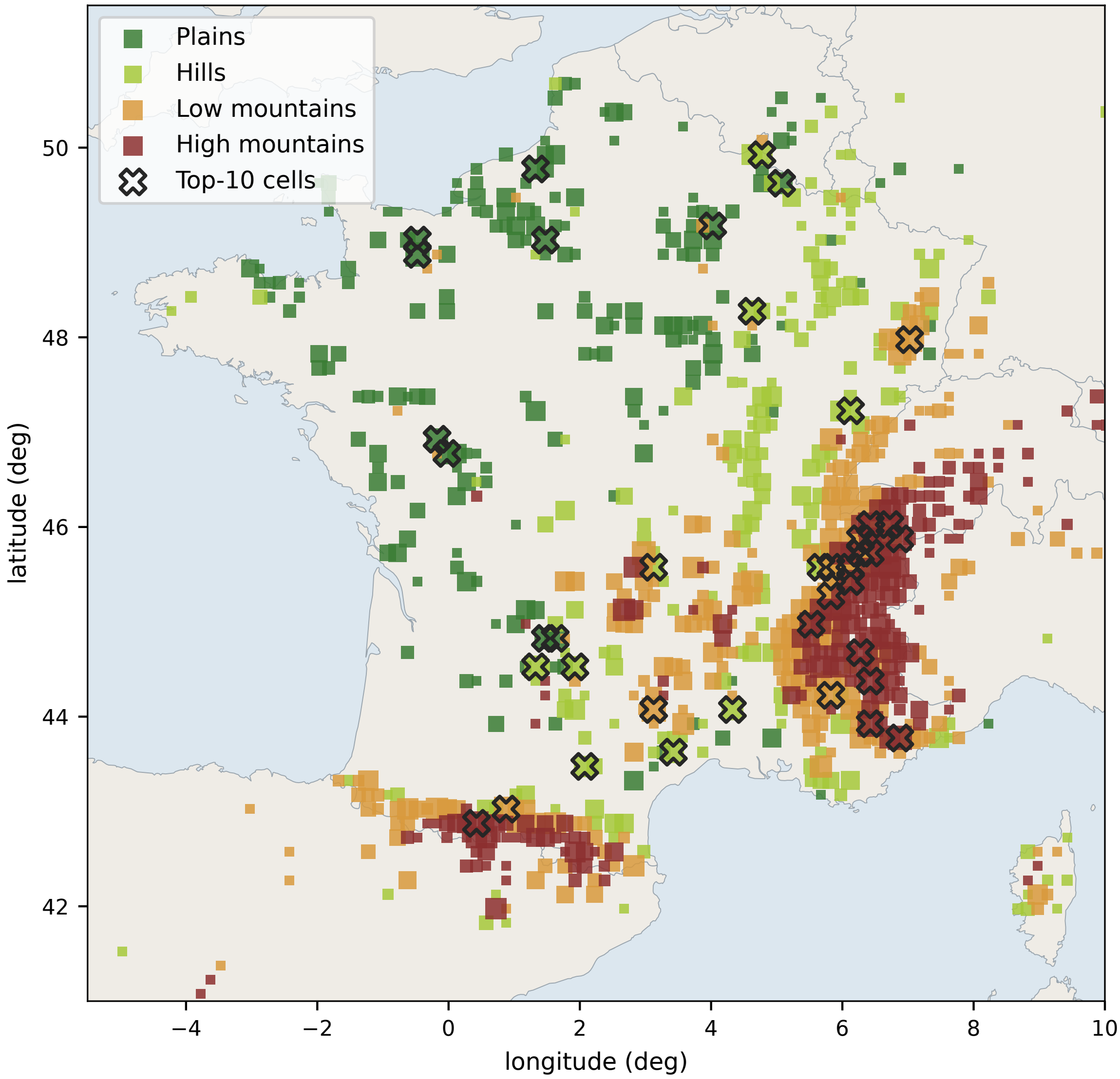


Figure 2.24. First recorded positions of the retained ensemble (both disciplines pooled), grouped by logged altitude using the elevation bands of [19]. These are approximate launch-altitude groups, not measured terrain classes. One marker per occupied 0.15° cell, coloured by its majority class and sized by the logarithm of its flight count; crosses mark the 10 busiest cells of each class.

2.10 Trajectories for the transport analysis

The retained trajectories have a uniform grid within each segment, local horizontal coordinates and explicit reconstruction flags. Phase features can exclude flagged vertical support, although plausible unflagged defects remain possible. Geography, cadence and flight duration also vary across the archive.

Chapter 3 defines each transport observable together with its weights, lag support and contributing population.

Chapter 3

Global transport

Circling, gliding and searching for lift combine into the displacement of a whole flight. We measure its growth, variability, preferred directions and velocity memory, then compare these observations with stochastic models. The analysis precedes the phase labels introduced in Chapter 4.

The archive contains 156,406 paraglider and 6,094 hang-glider flights in the retained transport snapshot. A complete scan of the cleaned archive supplies measurements spanning 10 s to 10,000 s, including regional principal axes and velocity correlations at several resolutions. All retained segments meeting the common-grid cadence and lag-support requirements enter; there is no cap on the number of flights.

Table 3.1. Population support in this chapter. The common-grid analysis includes every eligible flight. The fixed long-flight population supplies an additional control with identical flights and origins across lags (Sec. 3.2.2).

Population	Paragliders	Hang gliders
Retained cleaned archive	156,406	6,094
Eligible on the 10-s grid	155,085	6,060
Fixed long-flight control	14,360	563

The chapter first defines displacement growth and drift-removing variations. Quantiles then test scalar rescaling, with a fixed-population control, before the component and joint laws address vector scaling and dependence. A separate section examines regional geometry and equipment contrasts. Duration and equipment composition then check how the observed growth depends on the selected flights. With those population effects established, moment spectra and velocity memory provide further constraints for the model comparison.

3.1 Displacement and time scales

Let $\mathbf{r}_f(t) = (E_f(t), N_f(t))$ be the horizontal position of flight f. Its retained segments form the set $\mathcal{S}_f$, with fs denoting segment s of flight f.

A displacement is

$$\mathbf{X}_f(t,\tau) = \mathbf{r}_f(t+\tau) - \mathbf{r}_f(t), \qquad R_f(t,\tau) = |\mathbf{X}_f(t,\tau)|. \tag{3.1}$$

For segment fs, of duration T_{fs}, the time-averaged MSD is

$$\overline{\delta^2_{fs}}(\tau) = \frac{1}{T_{fs}-\tau}\int_0^{T_{fs}-\tau} R_f(t,\tau)^2 \,\mathrm{d}t, \qquad 0 < \tau < T_{fs}, \tag{3.2}$$

with t measured from the start of that segment. Only windows lying entirely within a retained segment are admitted, so an increment never crosses a data gap. Let $n_{fs}(\tau)$ count the admissible windows of segment fs and $n_f(\tau) = \sum_{s\in\mathcal{S}_f} n_{fs}(\tau)$ their total over the flight. Combining a flight's segment sums with those counts gives the flight-level mean

$$\overline{\delta_f^2}(\tau) = \frac{\sum_{s\in\mathcal{S}_f} n_{fs}(\tau)\,\overline{\delta_{fs}^2}(\tau)}{n_f(\tau)}, \tag{3.3}$$

a window-count-weighted average of the segment TA-MSDs contributing to that flight. Averaging further over the population requires a choice of weights:

$$M_2^{\text{flight}}(\tau) = \frac{1}{N_{\text{fl}}(\tau)} \sum_f \overline{\delta_f^2}(\tau), \qquad M_2^{\text{window}}(\tau) = \frac{\sum_f n_f(\tau)\overline{\delta_f^2}(\tau)}{\sum_f n_f(\tau)}, \tag{3.4}$$

where $N_{\text{fl}}(\tau)$ is the number of flights with at least one admissible window. The first gives each contributing flight equal weight, whereas the second gives each admissible window equal weight and consequently weights longer flights more strongly. Both averages still combine flights made under different meteorological and geographical conditions.

Figure 3.1 uses a third convention, equal weight per segment:

$$M_2^{\text{segment}}(\tau) = \frac{1}{N_{\text{seg}}(\tau)} \sum_f \sum_{s\in\mathcal{S}_f} \overline{\delta_{fs}^2}(\tau). \tag{3.5}$$

Only segments with at least eight fixes and support at τ enter. A split flight can contribute several segments.

In discrete time, a segment fs of N_{fs} fixes $\mathbf{r}_0, \ldots, \mathbf{r}_{N_{fs}-1}$ at interval Δt_{fs} contributes

$$\widehat{\delta_{fs}^2}(k\Delta t_{fs}) = \frac{1}{N_{fs}-k} \sum_{j=0}^{N_{fs}-k-1} |\mathbf{r}_{j+k} - \mathbf{r}_j|^2, \qquad 1 \le k < N_{fs}. \tag{3.6}$$

The common-grid diagnostics below use a 10 s grid, excluding segments with slower native cadence or fewer than two grid points. Starting times are $0, \tau, 2\tau, \ldots$ within each segment, giving non-overlapping first differences. Segment sums and origin counts are combined within flights. The duration analysis in Sec. 3.5 instead uses the native-grid segment TA-MSDs, with its own support and weighting stated there.

The second moments V_1 and V_2 then receive equal flight weights; baseline quantiles and higher moments pool windows. Section 3.2.2 fixes flights, weights and origins to control their change with lag.

Figure 3.1a also gives the launch-synchronised average, $\text{MSD}_{\text{launch}}(t) = \langle|\mathbf{r}_f(t) - \mathbf{r}_f(0)|^2\rangle_f$. Flights occur in different places, seasons and years; this curve describes the archive without assigning it a common process exponent.

3.1.1 Effective and local growth exponents

The comparison interval is 10 s to 10,000 s. For a positive observable $Y(\tau)$ we distinguish its fitted slope and its local slope,

$$\log Y(\tau) \simeq a + b\log(\tau/\tau_0), \qquad b_{\text{loc}}(\tau) = \frac{\mathrm{d}\log Y}{\mathrm{d}\log\tau}, \tag{3.7}$$

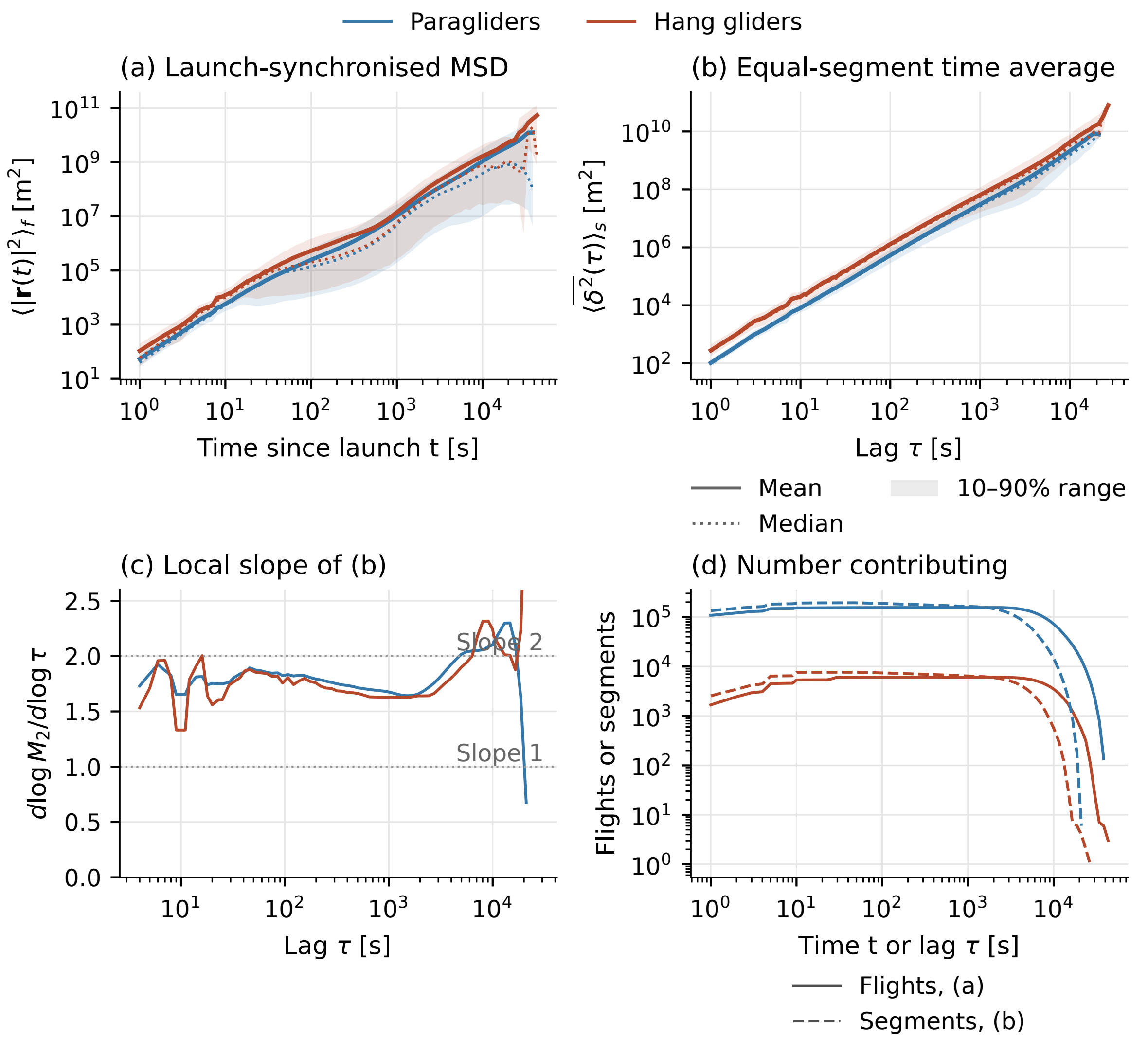


Figure 3.1. Archive displacement summaries. (a) Launch-synchronised average, retained as a description of the heterogeneous collection. (b) Time-averaged MSD, averaged equally over retained segments. Solid lines are means, dotted lines medians, and shading the 10–90% range across contributing flights in (a), or segments in (b); shading is not an uncertainty band on the mean. (c) Local slope of (b). (d) Contributing flights for (a) and segments for (b). Panel (a) uses time since launch; panels (b,c) use within-segment lag.

Here logarithms use the stated units, equivalently $\log(Y/Y_0)$, and $\tau_0 = 1\,\mathrm{s}$. The common-grid and duration reports fit ordinary least squares in $\log_{10} Y$, with equal weight per admitted lag. A changing local slope makes b an effective summary of that interval.

Cadence and smoothing can still affect the 10 s end; only long flights contribute near 10,000 s. Coverage and local slopes accompany each fit. Changing the fitting interval measures sensitivity to curvature and selection.

Sampling uncertainty needs a separate calculation. Windows from one flight, and flights sharing a site and day, may be dependent. Resampling whole site–day groups with their complete curves applies the bootstrap [21] at that level, assuming exchangeable groups. The baseline curves and slopes below are descriptive; where intervals are reported, their resampling unit and assumptions are stated.

3.1.2 Removing a constant velocity

A constant velocity contributes a quadratic term to an uncentred MSD. If $\mathbf{r}(t) = \mathbf{v}_d t + \mathbf{Y}(t)$ and the residual increment has zero mean, then

$$\mathbb{E}|\mathbf{X}(t,\tau)|^2 = |\mathbf{v}_d|^2\tau^2 + \mathbb{E}|\mathbf{Y}(t+\tau) - \mathbf{Y}(t)|^2. \tag{3.8}$$

Without the zero-mean assumption there is also a cross term. The two contributions add as functions of lag; their fitted exponents do not add.

A useful control is the second difference,

$$A_2\mathbf{r}(t;\tau) = \mathbf{r}(t+2\tau) - 2\mathbf{r}(t+\tau) + \mathbf{r}(t), \tag{3.9}$$

$$V_2(\tau) = \left\langle |A_2\mathbf{r}(t;\tau)|^2 \right\rangle. \tag{3.10}$$

Since $A_2(\mathbf{r}_0 + \mathbf{v}_d t) = 0$, this observable is exactly insensitive to a constant velocity. It avoids estimating drift from the endpoints, which would impose a return constraint on the residual trajectory. Its cost is a window of length 2τ. We retain it for this specific purpose, alongside $V_1(\tau) = \langle R^2 \rangle$.

To relate the second-difference slope to H, suppose the residual is a centred finite-variance process with stationary increments and $\mathbb{E}|\mathbf{Y}(t+\tau) - \mathbf{Y}(t)|^2 = K\tau^{2H}$. Writing adjacent increments as $\mathbf{I}_1$ and $\mathbf{I}_2$ gives $2\mathbb{E}[\mathbf{I}_1 \cdot \mathbf{I}_2] = K[(2\tau)^{2H} - 2\tau^{2H}]$, hence

$$V_2(\tau) = K(4 - 2^{2H})\tau^{2H}. \tag{3.11}$$

For $K > 0$ and $0 < H < 1$, half the slope of V_2 estimates H under these assumptions; fractional Brownian motion is one example [22]. At $H = 1$, the prefactor vanishes and the logarithmic slope is undefined.

On a smooth curved trajectory, $A_2\mathbf{r} = \ddot{\mathbf{r}}\,\tau^2 + O(\tau^3)$, so $V_2 \propto \tau^4$ at sufficiently small lag. Half that slope is not the Hurst exponent of a stationary-increment position process. Agreement between differences of order two and three is a consistency check; it does not prove that all environmental trends have disappeared. In particular, $b_1 - b_2$ is not a measured fraction of transport due to wind or course geometry.

Both disciplines have $b_1 > 1$ (Table 3.2). Restricting the analysis to flights with a segment of at least 20,000 s shifts the slopes: the result depends on the selected population. At 10,000 s, plain increments retain 68,207 paraglider and 3,313 hang-glider flights, with 84,997 and 3,920 non-overlapping windows. Second differences retain only 14,360 and 563 flights because they span 2τ.

Table 3.2. Effective equal-flight slopes on 10 s to 10,000 s. The fixed cohort contains flights with at least one segment of 20,000 s. These are descriptive slopes without calibrated confidence intervals.

Discipline	Population	Flights	b_1	b_2
paraglider	all selected	155085	1.703	2.003
	fixed long cohort	14360	1.790	2.059
hang glider	all selected	6060	1.649	1.985
	fixed long cohort	563	1.744	2.025

The measured b_2 values near or above two, together with the curvature in Fig. 3.2, prevent a direct interpretation as $2H$ for a stationary-increment process. The control removes constant velocity but does not yield an intrinsic Hurst exponent here.

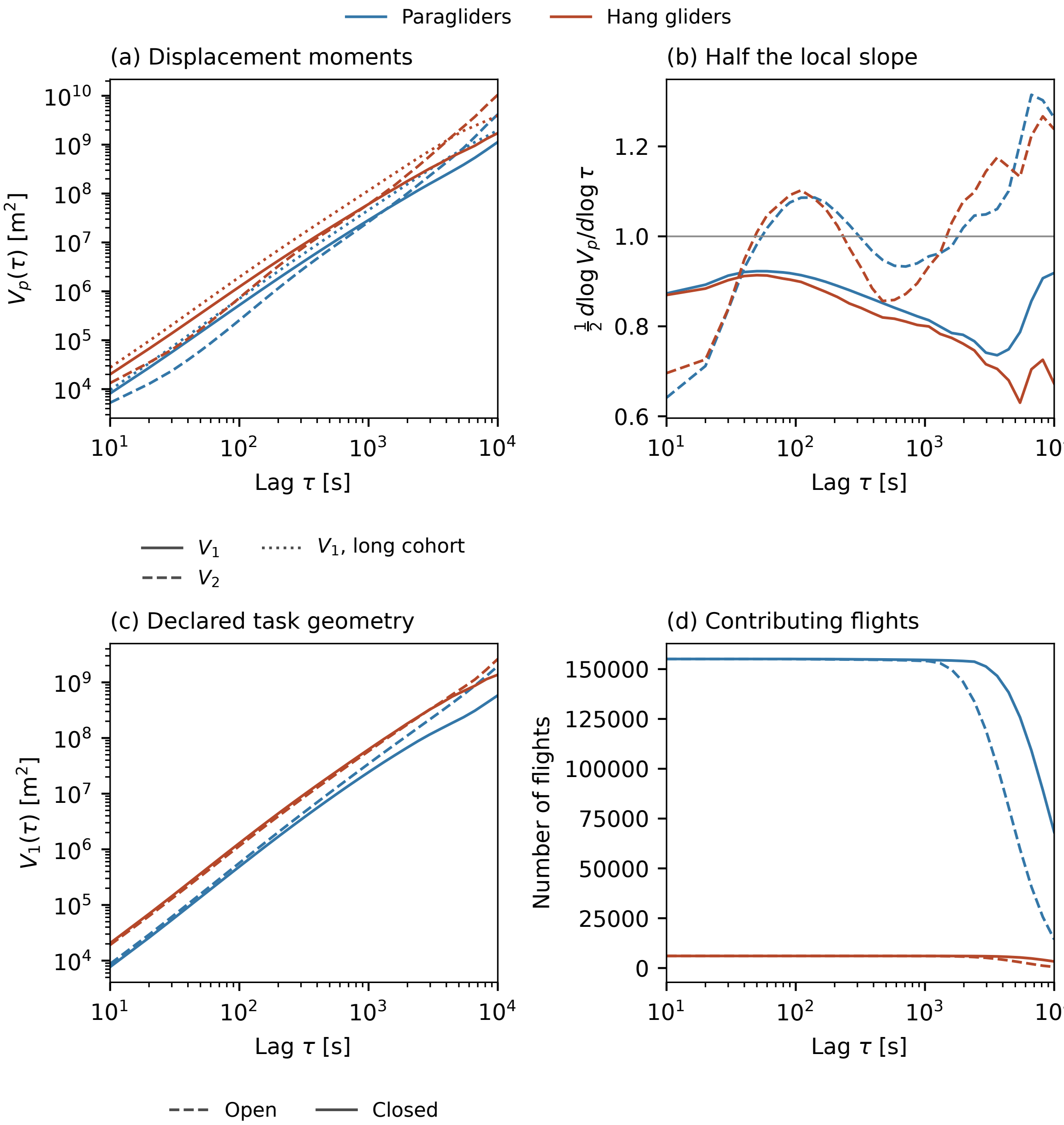


Figure 3.2. Displacement and constant-velocity control over 10 s to 10,000 s in the eligible archive. The figure separates the second moments from their local slopes and the declared open and closed tasks. Coverage is shown explicitly; the support of a second difference is smaller because it spans 2τ. Slopes refer to this interval and this eligible population.

3.1.3 Open and closed routes

A declared closed task rewards return towards the starting region. An open task allows continued progress away from it. This difference can change long-lag displacements even if speed and short-time behaviour are similar. Task classes are compared separately in Figure 3.2; a declaration is not a measurement that a particular retained segment closes exactly.

A circular trajectory makes the distinction precise. For $\mathbf{r}(t) = a(\cos\omega t, \sin\omega t)$ with period $T_c = 2\pi/\omega$,

$$V_1(\tau) = 4a^2 \sin^2\left(\frac{\pi\tau}{T_c}\right), \qquad V_2(\tau) = 16a^2 \sin^4\left(\frac{\pi\tau}{T_c}\right). \tag{3.12}$$

The displacement initially grows, then decreases as the lag approaches a complete return. The second difference also retains this geometry. Removing constant drift cannot remove closure, a turn, or a spatially varying forcing.

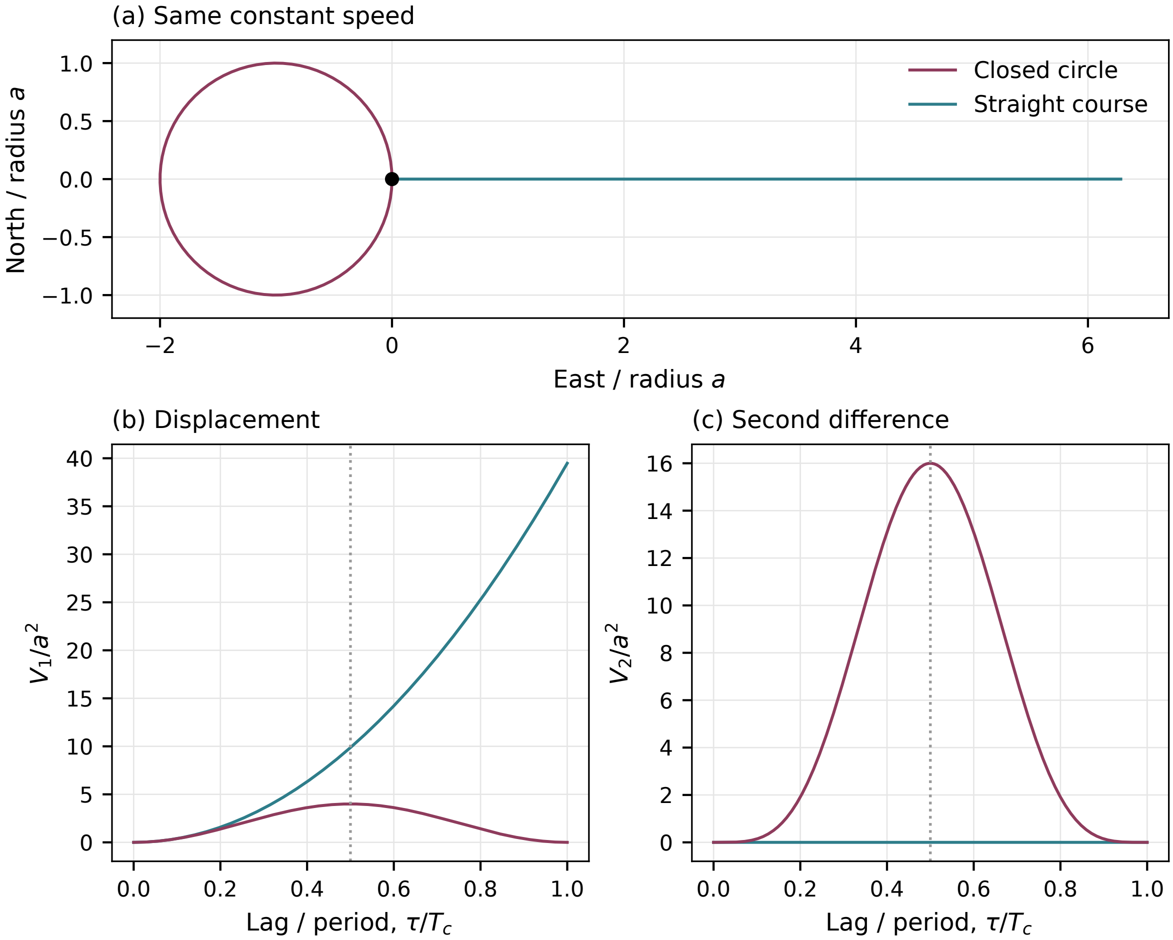


Figure 3.3. Analytical straight and circular paths at the same speed. The circle is continued periodically to evaluate second differences. Both V_1 and V_2 vanish after a complete period: constant-velocity removal preserves the effect of route closure.

The selected paraglider tasks include 67,003 open and 87,807 closed flights; hang gliders contribute 2,183 and 3,877. Their effective b_1 values are respectively 1.770 versus 1.623, and 1.688 versus 1.628. At order two the ordering reverses: 1.937 versus 2.036 and 1.892 versus 2.007. This is consistent with different route geometry affecting the two observables differently; it is not a quantitative decomposition into drift and fluctuations.

A complementary descriptor compares net displacement with the path actually observed on the common grid. Let t_f^- and t_f^+ be the first and last retained grid times, and let L_f^{obs} sum only the within-segment chords:

$$L_f^{\mathrm{obs}} = \sum_{s\in\mathcal{S}_f} \sum_{j=0}^{N_{fs}-2} |\mathbf{r}_{fs,j+1} - \mathbf{r}_{fs,j}|, \qquad \chi_f^{\mathrm{obs}} = \frac{|\mathbf{r}_f(t_f^+) - \mathbf{r}_f(t_f^-)|}{L_f^{\mathrm{obs}}}, \quad L_f^{\mathrm{obs}} > 0. \tag{3.13}$$

For one uninterrupted observed path, the triangle inequality gives $0 \leq \chi_f^{\mathrm{obs}} \leq 1$. Across separated segments, the denominator omits unobserved travel while the numerator spans it, so this bound need not hold. Reacquisition offsets can also affect the endpoint displacement. The full collector contains 181 paraglider and two hang-glider values above one, all among flights with multiple eligible segments. This is an observed-path descriptor, not a reconstruction of the complete distance flown.

In the eligible archive, median χ_f^{obs} is 0.365 for open paraglider tasks and 0.027 for closed ones; hang-glider medians are 0.177 and 0.026. These medians describe the endpoint geometry of retained observations, with the gap limitation above. Restricting to flights with one eligible observed segment gives closely similar open/closed medians: 0.364/0.027 in paragliders and 0.177/0.026 in hang gliders. This sensitivity check preserves the descriptive ordering; one eligible segment need not cover the complete original flight. For an uninterrupted observed path, small χ_f^{obs} means little net progress relative to its length. None of the increment statistics crosses a segment boundary.

Matching task groups by duration and region, and comparing both physical lag and τ/T_f, would help separate route geometry from population composition.

3.2 Quantiles and self-similarity

3.2.1 The scaling prediction

For an H-self-similar process starting at the origin, $(\mathbf{r}(ct_1), \dots, \mathbf{r}(ct_n)) \overset{d}{=} c^H(\mathbf{r}(t_1), \dots, \mathbf{r}(t_n))$ for every $c > 0$ and every finite choice of times. Self-similarity therefore constrains joint distributions, beyond the MSD alone. With stationary increments, it implies the one-lag prediction

$$\mathbf{X}(t,\tau) \overset{d}{=} (\tau/\tau_0)^H \mathbf{X}(t,\tau_0). \tag{3.14}$$

Writing $a_\tau = (\tau/\tau_0)^H$, and assuming the corresponding densities exist, the two-dimensional and radial laws obey different Jacobians:

$$p_{\mathbf{X}}(\mathbf{x};\tau) = a_\tau^{-2} F(\mathbf{x}/a_\tau), \qquad p_R(r;\tau) = a_\tau^{-1} f(r/a_\tau). \tag{3.15}$$

The radial law follows by integrating the joint density over angle, including the polar Jacobian r. It does not require isotropy. In particular, a two-dimensional density at the origin would scale as a_τ^{-2}, not as a_τ^{-1}.

Define the radial quantile at probability p by

$$Q_p(\tau) = \inf\{r : \Pr[R(t,\tau) \leq r] \geq p\}, \qquad 0 < p < 1. \tag{3.16}$$

Equation (3.14) implies

$$Q_p(\tau) = a_\tau Q_p(\tau_0), \qquad H_p = \frac{\mathrm{d}\log Q_p}{\mathrm{d}\log\tau} = H \quad \text{when } 0 < Q_p(\tau_0) < \infty. \tag{3.17}$$

The four probabilities $p = .25, .50, .75, .90$ describe small, typical and larger displacements. Quantiles use ranks, complementing moments without requiring a finite fourth moment. Common quantile slopes are necessary for the proposed one-lag scaling law; they do not establish self-similarity across several times.

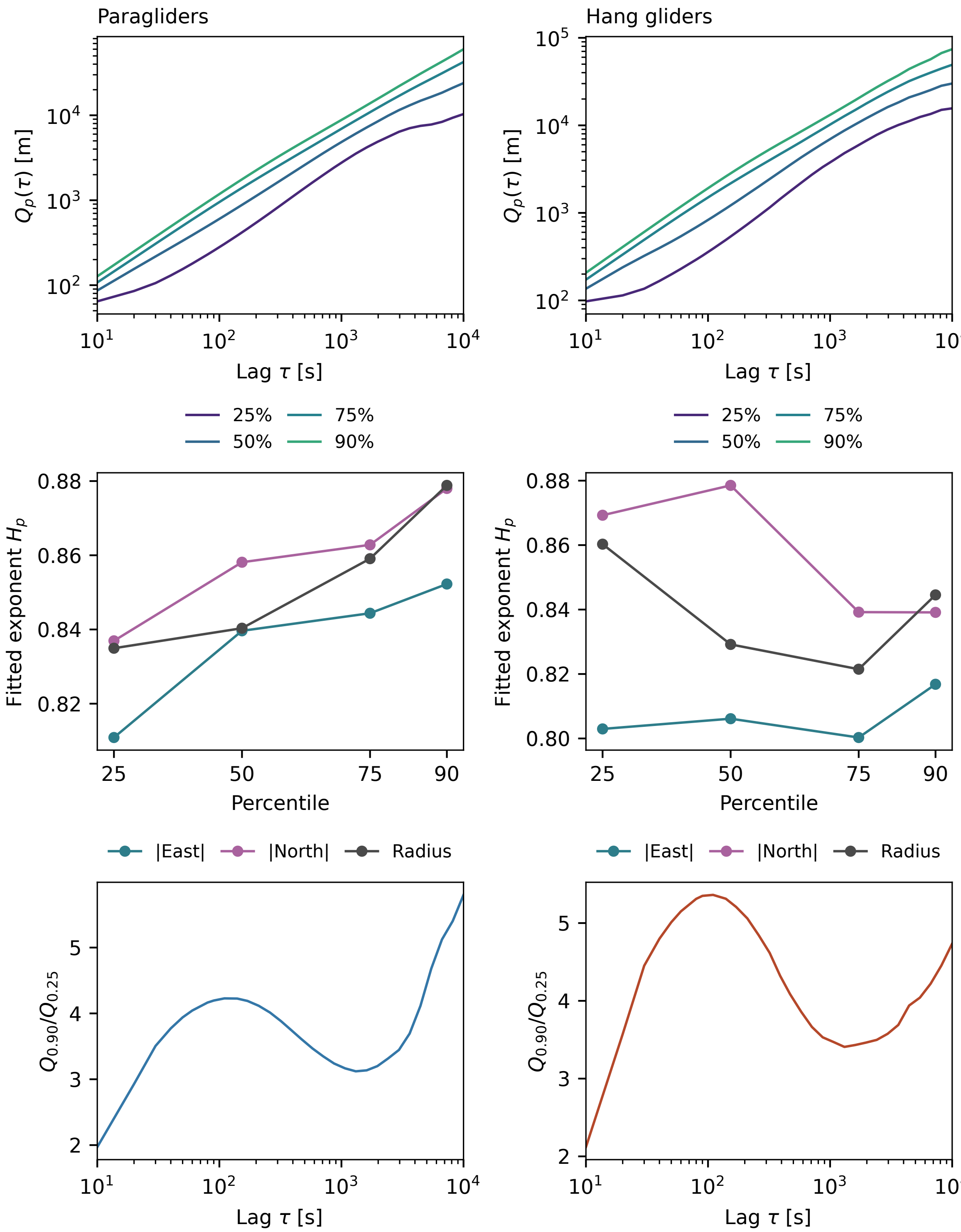


Figure 3.4. Displacement quantiles in the eligible archive over 10 s to 10,000 s, with east and north treated separately from radial displacement. Exponents at individual probability levels and ratios of radial quantiles test whether one scale factor suffices. A common exponent would preserve the quantile ratios; a systematic change indicates shape evolution or a change in the sampled population.

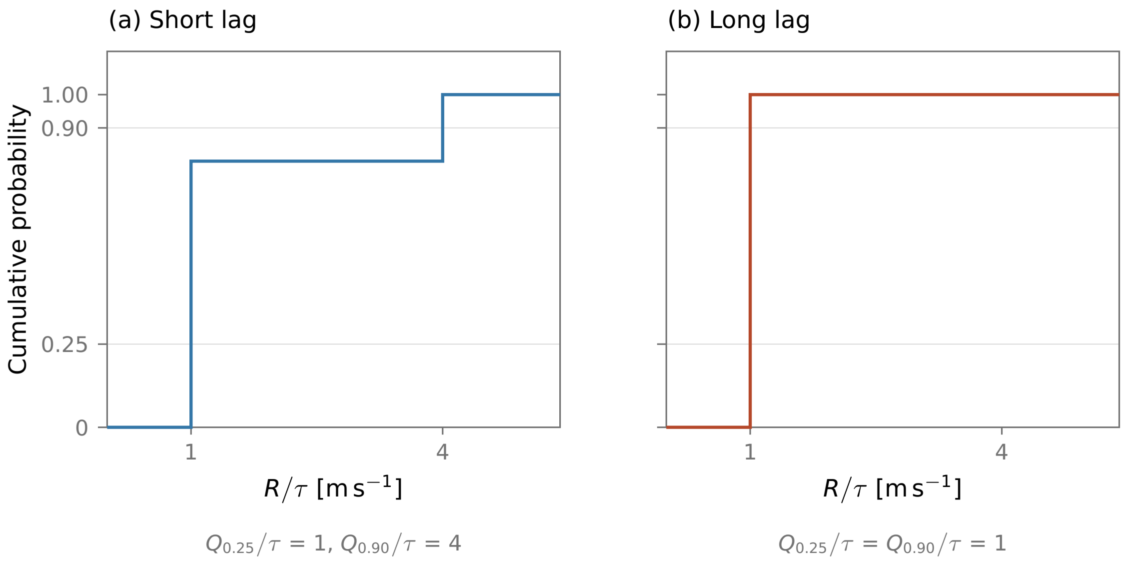


Figure 3.5. A sampling effect can imitate different quantile growth rates. Consider two straight flights at $1\,\mathrm{m\,s^{-1}}$ lasting 20,000 s each and two at $4\,\mathrm{m\,s^{-1}}$ lasting 5,000 s each. At a 10-s lag, the fast flights supply 1/5 of the non-overlapping windows, so the 90th percentile of R/τ is 4. At 10,000 s they supply none, and that percentile is 1. Every flight is exactly ballistic, yet the pooled upper quantile grows more slowly than τ between these lags. A fixed cohort of the two long flights has $Q_p(\tau) = (1\ \mathrm{m\,s^{-1}})\tau$ at every displayed probability level. The example is constructed, not measured from the archive.

3.2.2 A fixed population across lags

At each lag, all percentiles belong to the same displacement distribution. They do not label fixed groups of slow or fast flights. A separate sampling effect arises when that distribution draws on different flights or weights at different lags.

If flight f supplies $n_f(\tau)$ admissible increments, pooling them defines

$$\widehat{F}_\tau^{\mathrm{pool}}(x) = \sum_{f\in\mathcal{F}_\tau} \frac{n_f(\tau)}{\sum_g n_g(\tau)} \widehat{F}_{f,\tau}(x), \qquad \widehat{F}_{f,\tau}(x) = \frac{1}{n_f(\tau)} \sum_k 1\{R_{fk}(\tau) \le x\}. \tag{3.18}$$

As lag grows, short flights disappear and window counts change. Both membership $\mathcal{F}_\tau$ and mixture weights therefore vary. Quantiles depend nonlinearly on these weights: averaging flight-specific quantiles would give a different quantity. Figure 3.5 illustrates the resulting bias.

Figure 3.6 changes one sampling choice at a time:

	Flights	Weight	Origins across lags
1	All available	Equal per window	Changing
2	Fixed cohort	Equal per window	Changing
3	Fixed cohort	Equal per flight	Changing
4	Fixed cohort	Equal per flight	Fixed

The fixed cohort requires at least one retained segment of 20,000 s per flight. This is an additional population control; the changing-pool analysis uses every eligible flight at each supported lag. The 20,000 s threshold is an operational choice, not the minimum duration mathematically required to observe a 10,000 s increment. It supplies at least 1,001 common 10-s origins in one qualifying segment. Other eligible segments of these same flights also contribute when they support the required interval. The cohort therefore fixes flights, not only their longest segments, and its conclusions remain conditional on long flights.

For the last convention, let $\tau_{\max} = 10{,}000\,\mathrm{s}$ and keep every starting time t_{fk} on the 10-s grid whose complete interval $[t_{fk}, t_{fk} + \tau_{\max}]$ lies within one retained segment. Then

$$\widehat{F}_{\tau}^{\mathrm{fixed}}(x) = \frac{1}{|\mathcal{F}_*|} \sum_{f \in \mathcal{F}_*} \frac{1}{n_f^*} \sum_{k=1}^{n_f^*} 1\{R_f(t_{fk}, \tau) \le x\}, \qquad \widehat{Q}_p(\tau) = \inf\{x : \widehat{F}_\tau(x) \ge p\}. \tag{3.19}$$

Flights, origins and weights are now independent of τ. All four conventions use inverse empirical CDFs without interpolation between order statistics. The common origins overlap at most lags; they preserve the sampled population without making its increments independent.

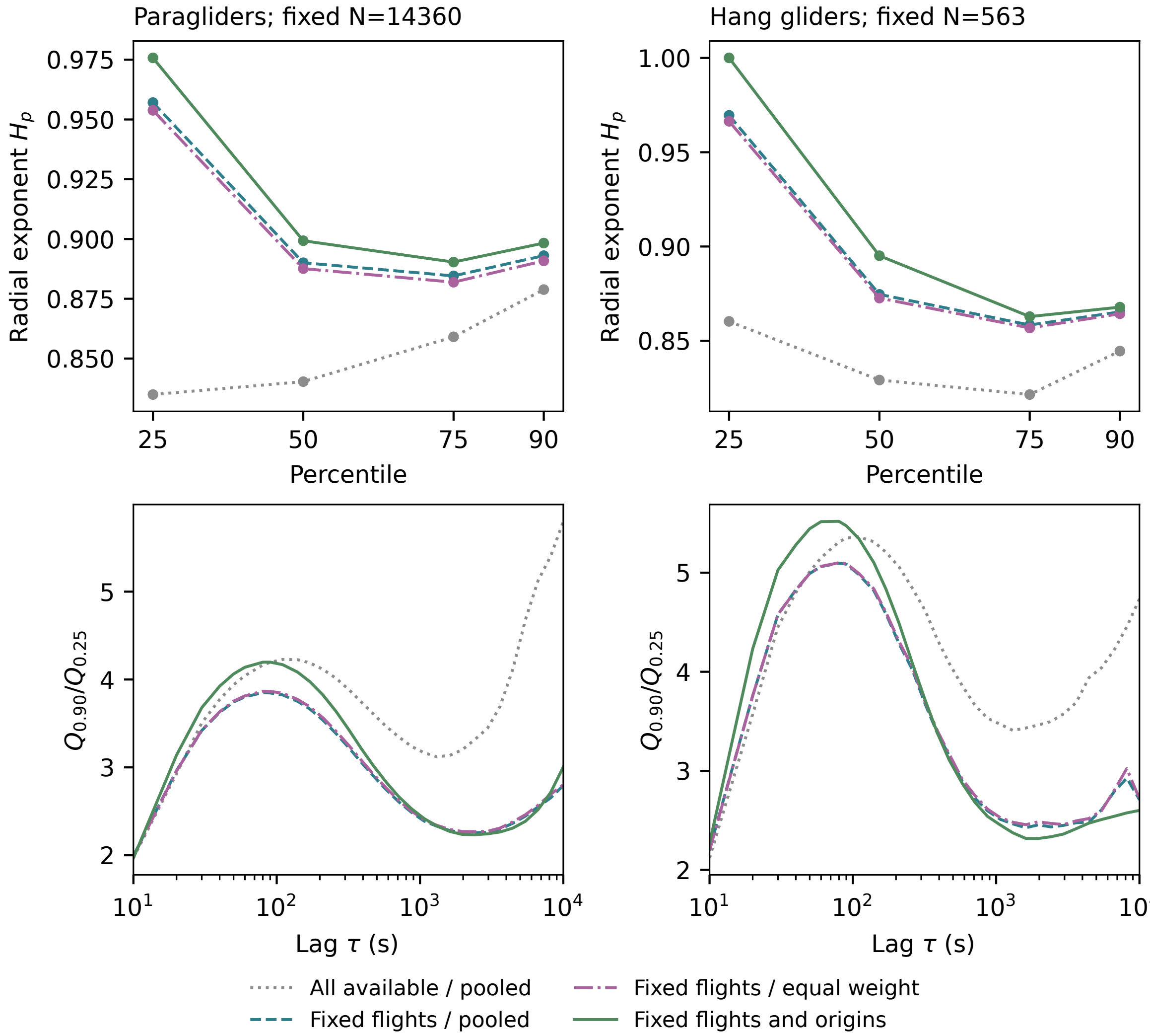


Figure 3.6. Controls for population changes in the quantile analysis. Top: radial quantile slopes under four sampling conventions, fitted on their common positive supported lags. Bottom: the corresponding $Q_{0.90}/Q_{0.25}$ curves. The fixed cohorts contain 14360 paraglider and 563 hang-glider flights; the fully controlled convention retains respectively 21,739,115 and 791,712 origins at every lag. These overlapping origins are not independent replicates. Each displayed estimate requires at least two flights and 30 windows, which are support checks rather than guarantees of statistical precision.

Table 3.3. Radial quantile slopes before and after fixing flights, their weights and time origins. Within each discipline both columns use the same lag set, common also to the intermediate controls and the component quantiles: 10 s to 10,000 s for paragliders and 10 s to 10,000 s for hang gliders. These are descriptive slopes, without calibrated confidence intervals; coverage of the target interval does not establish that a single power describes it.

	Paragliders		Hang gliders	
Percentile	Changing pool	Fixed population	Changing pool	Fixed population
25	0.835	0.976	0.860	1.000
50	0.840	0.899	0.829	0.895
75	0.859	0.890	0.821	0.863
90	0.879	0.898	0.845	0.868

Fixing the paraglider cohort reverses the ordering of the 25th- and 90th-percentile slopes (Table 3.3). Equalising weights has a smaller effect; fixing origins changes the estimates again. Even after all controls, the four slopes are not strictly ordered: $H_{.90} > H_{.75}$ in both disciplines.

The remaining contrast cannot come from changing membership or weights with lag. It can reflect phase changes, route geometry, environment and sampling uncertainty within the selected long flights. The roughly five-and-a-half-hour segment requirement limits this conclusion to that population.

3.2.3 Relative spread

The interpretation of an ordering such as $H_{0.25} > H_{0.90}$ follows directly from

$$\frac{\mathrm{d}}{\mathrm{d}\log\tau}\log\left(\frac{Q_{0.90}(\tau)}{Q_{0.25}(\tau)}\right) = H_{0.90}(\tau) - H_{0.25}(\tau). \tag{3.20}$$

A negative right-hand side means that the ratio of large to small displacements decreases. Both quantiles may increase in absolute distance. The lower one increases faster in *proportion*, so the distribution becomes less spread out by this relative measure. For example, a slope difference of -0.10 maintained over a tenfold increase of lag multiplies the ratio by $10^{-0.10} \simeq 0.79$.

On a common lag grid, the fitted slope of $\log(Q_{.90}/Q_{.25})$ also equals the difference of the fitted quantile slopes, by linearity of least squares. Its sign need not imply monotonic change. In the controlled population, the paraglider ratio rises from 1.97 at 10 s to a sampled maximum of 4.20 at 90 s; the hang-glider ratio rises from 2.27 to 5.52 at 80 s. After these early maxima, the ratios fall to local minima of 2.23 at 2,410 s and 2.32 at 1,970 s, respectively. They then rise to 3.01 and 2.60 at 10,000 s. The negative fitted contrast summarises this curved evolution; it does not describe a monotone narrowing over the full interval.

This ratio measures relative quantile spread, leaving the outer 10% and local density shape unspecified. Observations can also change rank with lag.

Rank-dependent growth in the controlled population challenges a common rescaling exponent. A nearly linear moment spectrum can coexist with it, because moments integrate over the distribution. A model must reproduce the quantile curves and their shape changes as well as their average growth.

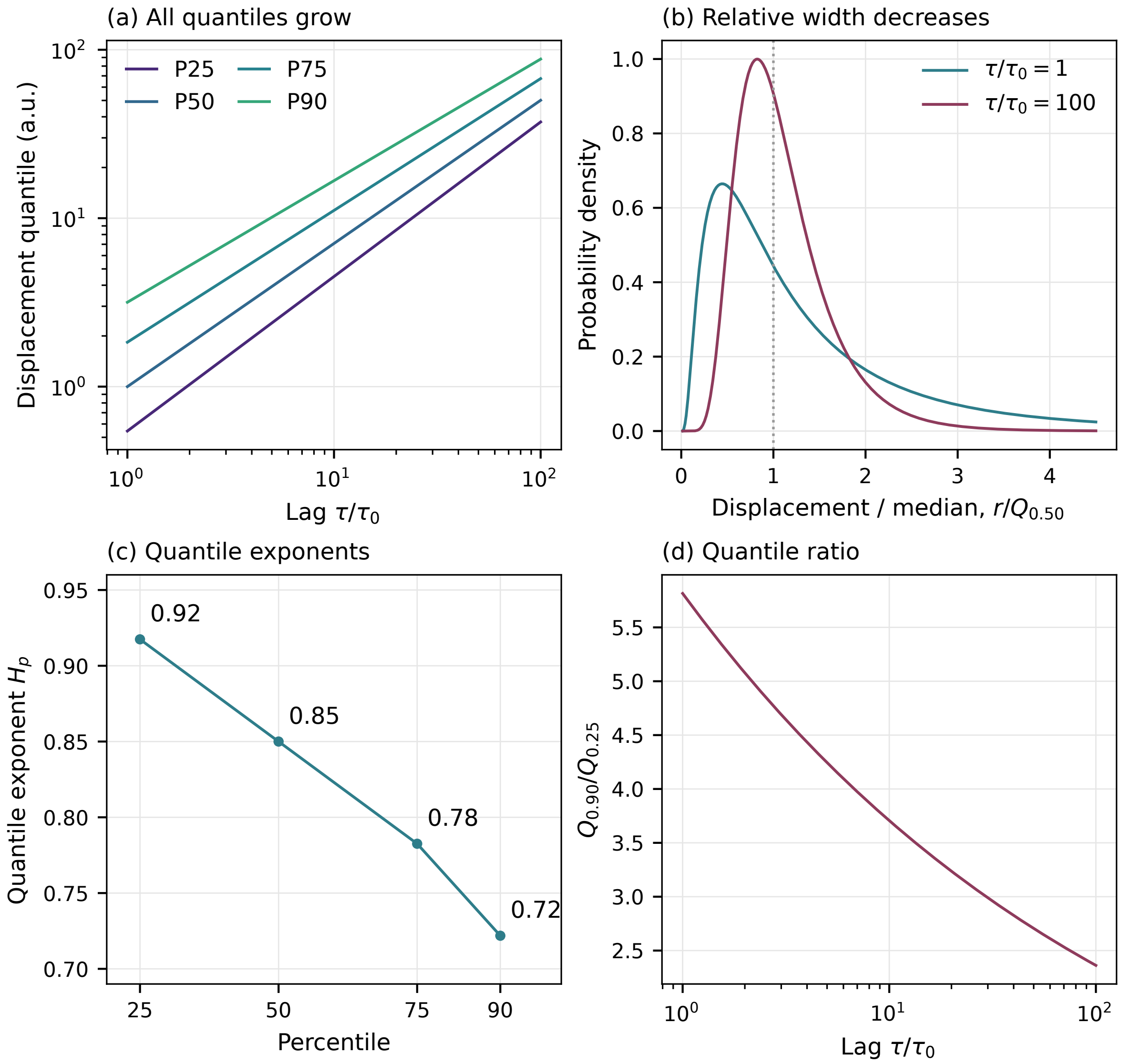


Figure 3.7. Analytical illustration of decreasing quantile exponents, not a fit to flight data. Let $R(\tau) = (\tau/\tau_0)^{0.85} \exp[\sigma(\tau)Z]$, where Z is standard Gaussian and $\sigma(\tau) = 0.9 - 0.1 \log(\tau/\tau_0)$ on the illustrated interval. Then $H_p = 0.85 - 0.1\Phi^{-1}(p)$. All four quantiles grow, but the distribution narrows after division by its median. The declining ratio makes that change explicit.

3.3 Vector scaling and dependence

The radial quantiles describe displacement magnitude. We now ask whether the same rescaling describes the signed east–north vector, including its dependence structure. The population and origin controls of Sec. 3.2.2 remain fixed throughout this comparison. An anisotropic distribution can obey a common scalar rescaling; isotropy is not an assumption of the test.

3.3.1 Testing east, north and radial scaling

The absolute east and north components and the radius change shape over 10 s to 10,000 s, even in the fixed population. To interpret their comparison, first consider two ways marginal scaling can fail to determine vector scaling.

Component exponents can differ. For centred independent Gaussian components with

Table 3.4. Separate effective quantile exponents over 10 s to 10,000 s, using fixed flights, origins and weights. The exponents refer to magnitudes in metres; the corresponding squared-displacement slopes are exactly twice these values.

Discipline	Quantity	$H_{.25}$	$H_{.50}$	$H_{.75}$	$H_{.90}$
Paragliders	$\lvert X_E\rvert$	0.947	0.921	0.882	0.873
	$\lvert X_N\rvert$	0.996	0.948	0.904	0.900
	R	0.976	0.899	0.890	0.898
Hang gliders	$\lvert X_E\rvert$	0.954	0.917	0.862	0.855
	$\lvert X_N\rvert$	1.010	0.965	0.881	0.867
	R	1.000	0.895	0.863	0.868

variances $a\tau^{2H_E}$ and $b\tau^{2H_N}$,

$$\mathbb{E}R^2 = a\tau^{2H_E} + b\tau^{2H_N}, \qquad \frac{1}{2}\frac{\mathrm{d}\log \mathbb{E}R^2}{\mathrm{d}\log\tau} = \frac{H_E a\tau^{2H_E} + H_N b\tau^{2H_N}}{a\tau^{2H_E} + b\tau^{2H_N}}. \tag{3.21}$$

Each Gaussian marginal scales exactly, yet the radial effective exponent varies with lag. No cross-correlation is needed.

Equal marginal exponents also leave component dependence undetermined. For example, the equiprobable pairs

$$\{(1,2),(-1,-2)\} \qquad \text{and} \qquad \{(1,-2),(-1,2)\} \tag{3.22}$$

have identical signed marginals and radius $\sqrt{5}$, but opposite cross-covariance. Thus even matching both marginals and the radial law does not establish a common joint law. The Cramér–Wold characterization uses all signed linear projections [23]. Conversely, scalar scaling of the joint vector always implies radial scaling, including for anisotropic laws. Operator self-similarity, with a matrix exponent, is a broader hypothesis [24].

Paired component and radial quantiles. On the fixed observations of Sec. 3.2.2, compare

$$Y_E = |X_E|, \qquad Y_N = |X_N|, \qquad Y_R = R = \sqrt{X_E^2 + X_N^2}. \tag{3.23}$$

These are magnitudes of displacement over a lag. Absolute values discard direction; R measures the distance between the endpoints, rather than the path flown between them.

Every curve uses fixed flights and 10-s origins, equal total weight per flight, and no drift subtraction. The four probabilities give twelve slopes $H_{j,p}$ per discipline, for $j \in \{E, N, R\}$. The population contains 14360 paraglider and 563 hang-glider flights from the retained snapshot (cleaning version 2.3.0).

We bootstrap whole flights with replacement 400 times. Each resample keeps all origins of a flight together, using its multiplicity at every lag and for all three quantities. Recomputed mixture quantiles preserve the pairing in contrasts such as $H_{N,p} - H_{E,p}$ and $H_{j,.90} - H_{j,.25}$. The pointwise 95% percentile intervals assume independent flights within this fixed population. Shared site–day dependence and simultaneous coverage are not accounted for.

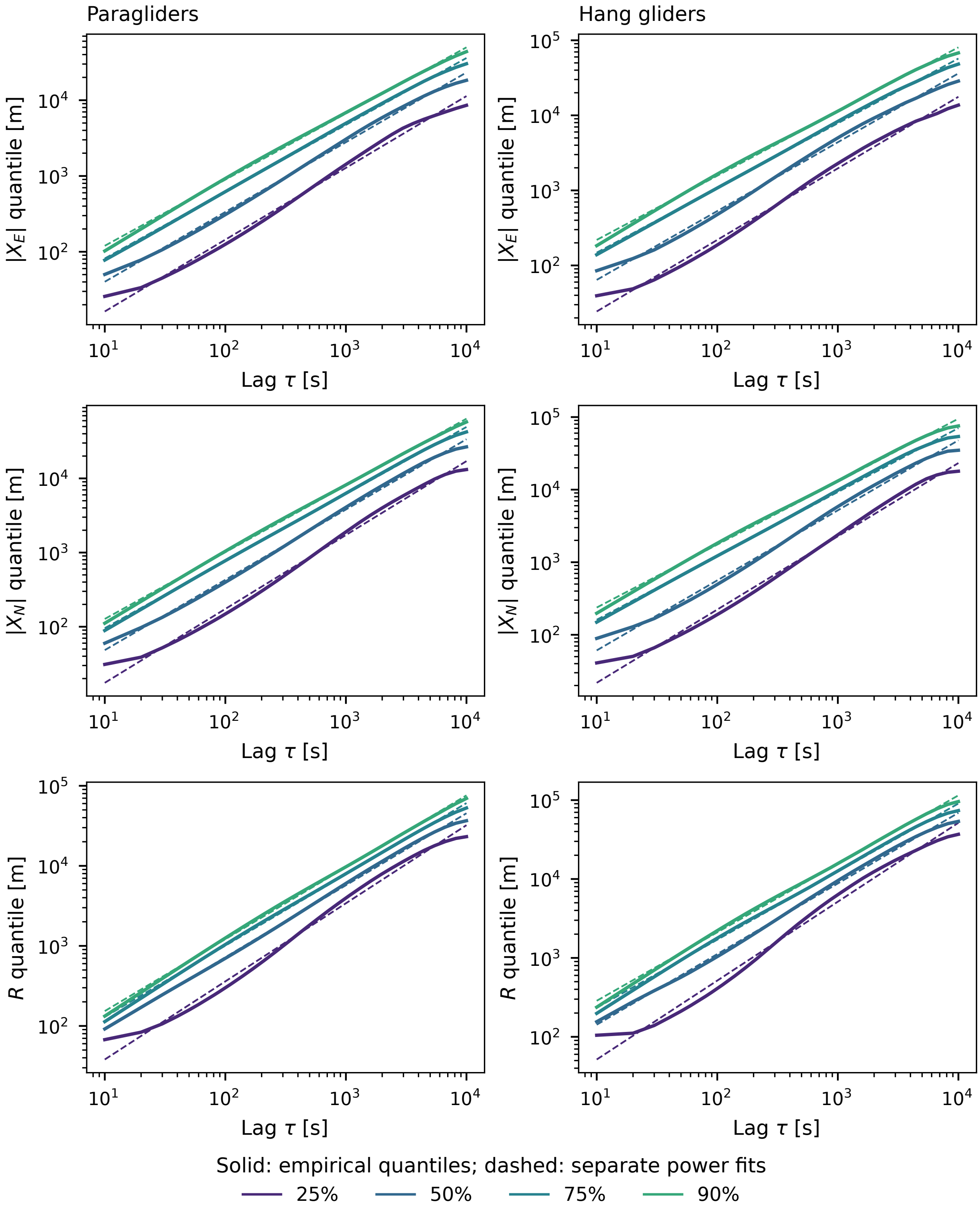


Figure 3.8. Four quantiles for each absolute component and the radius in the fully controlled population. Solid curves are empirical weighted quantiles; dashed lines are their separate log–log fits over 10 s to 10,000 s. Columns distinguish disciplines and rows distinguish quantities. A visibly curved quantile can have a well-defined regression slope without following a single power law.

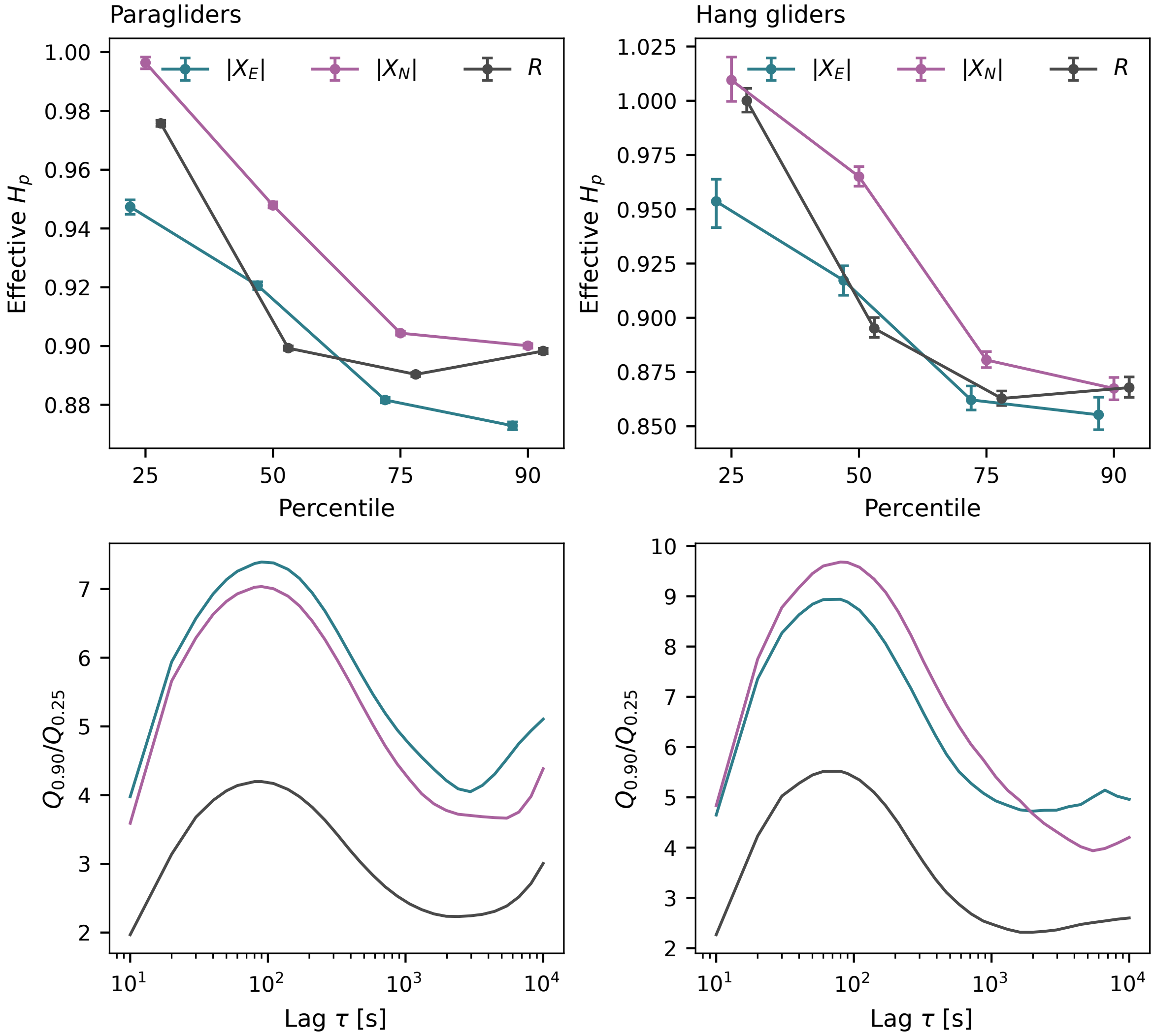


Figure 3.9. Top: the twelve separate magnitude exponents per discipline, with pointwise 95% whole-flight bootstrap intervals. Small horizontal offsets distinguish quantities at the same percentile. Bottom: the corresponding ratios $Q_{.90}/Q_{.25}$ for east, north and radius. Ratio changes test distribution shape without choosing a rescaling exponent. All curves use the fixed population; interval assumptions are stated in the text.

The lower-quartile slope exceeds the 90th-percentile slope for east, north and radius in both disciplines. Two paired contrasts summarize the differences:

Contrast	**Paragliders**	**Hang gliders**
$H_{R,.90} - H_{R,.25}$	-0.077	-0.132
95% interval	$[-0.079, -0.076]$	$[-0.139, -0.124]$
$H_{N,.50} - H_{E,.50}$	0.027	0.048
95% interval	$[0.026, 0.029]$	$[0.040, 0.054]$

Intervals are conditional on the fixed population and the independence assumption above. Figure 3.9 shows the non-monotone spread ratios.

In both disciplines, north–east contrasts are positive at all four ranks, with their paired pointwise intervals excluding zero. The radial rank contrasts are negative, also with intervals excluding zero. These comparisons support rank- and direction-dependent effective growth

Table 3.5. Candidate common magnitude exponent and largest pairwise ECDF separation among the six displayed lags. D_{power} uses the fitted power scale; D_{median} uses each lag's own median. A distance of 0.10 is a ten-percentage-point cumulative probability difference somewhere in the rescaled distribution.

Discipline	Quantity	H_*	D_{power}	D_{median}
Paragliders	$\lvert X_E\rvert$	0.906	0.177	0.147
	$\lvert X_N\rvert$	0.937	0.203	0.178
	R	0.916	0.273	0.194
Hang gliders	$\lvert X_E\rvert$	0.897	0.185	0.179
	$\lvert X_N\rvert$	0.931	0.269	0.213
	R	0.906	0.331	0.229

within the fixed long-flight population under the stated resampling assumptions. They do not establish a single power law at each rank. The radial median slope lies below both component median slopes: different observations can occupy the component and radial ranks.

3.3.2 Distribution collapse

To test a scale family, fit one exponent $H_{j,*}$ per quantity, shared by all four ranks, with separate intercepts $b_{j,p}$, by minimising

$$\sum_{p\in\{.25,.50,.75,.90\}} \sum_{\tau\in\mathcal{L}} \left[\log\frac{Q_{j,p}(\tau)}{1\,\mathrm{m}} - b_{j,p} - H_{j,*}\log\frac{\tau}{\tau_0}\right]^2, \qquad \tau_0 = 1{,}000\,\mathrm{s}. \tag{3.24}$$

Ranks and lags have equal weight. On the common grid, $H_{j,*}$ is the mean of the four separate slopes. The reference time fixes the intercept without changing the exponent.

With $a_j(\tau) = (\tau/\tau_0)^{H_{j,*}}$, a density collapse plots

$$u = y/a_j(\tau), \qquad a_j(\tau)\, p_{Y_j}(a_j(\tau)u;\tau). \tag{3.25}$$

The fitted exponent rescales both axes at every lag, without tuning the histogram overlay. Densities use probability mass divided by bin width over the full positive support; any exact zero is stored separately as an atom. Quantiles and empirical-CDF distances use the observations directly, independently of binning.

The full positive support can approach numerical zero. Summing all bins that intersect $[0, 1\,\mathrm{mm})$, together with the zero atom, bounds the probability in that interval below 3×10^{-5} in every displayed law. Very narrow bins can therefore have a high density while carrying little probability. These values do not establish sub-millimetre physical resolution; the median-normalised CDF panels below make the central probability structure easier to compare than the full-range logarithmic densities.

To distinguish an imperfect power fit from a failure of any single scale factor, we also plot $Y_j/Q_{j,.50}(\tau)$. A shape-preserving scale family must collapse under this normalisation even if its scale is not a power of lag. We measure separation by

$$D = \max_{\tau,\tau'\in\mathcal{L}_{\text{shown}}} \sup_u |\widehat{F}_\tau^{\text{scaled}}(u) - \widehat{F}_{\tau'}^{\text{scaled}}(u)|. \tag{3.26}$$

We evaluate the supremum at the empirical jumps for the six displayed lags. It is a descriptive probability difference. Weighted, paired and overlapping observations do not support an independent-sample Kolmogorov–Smirnov p-value.

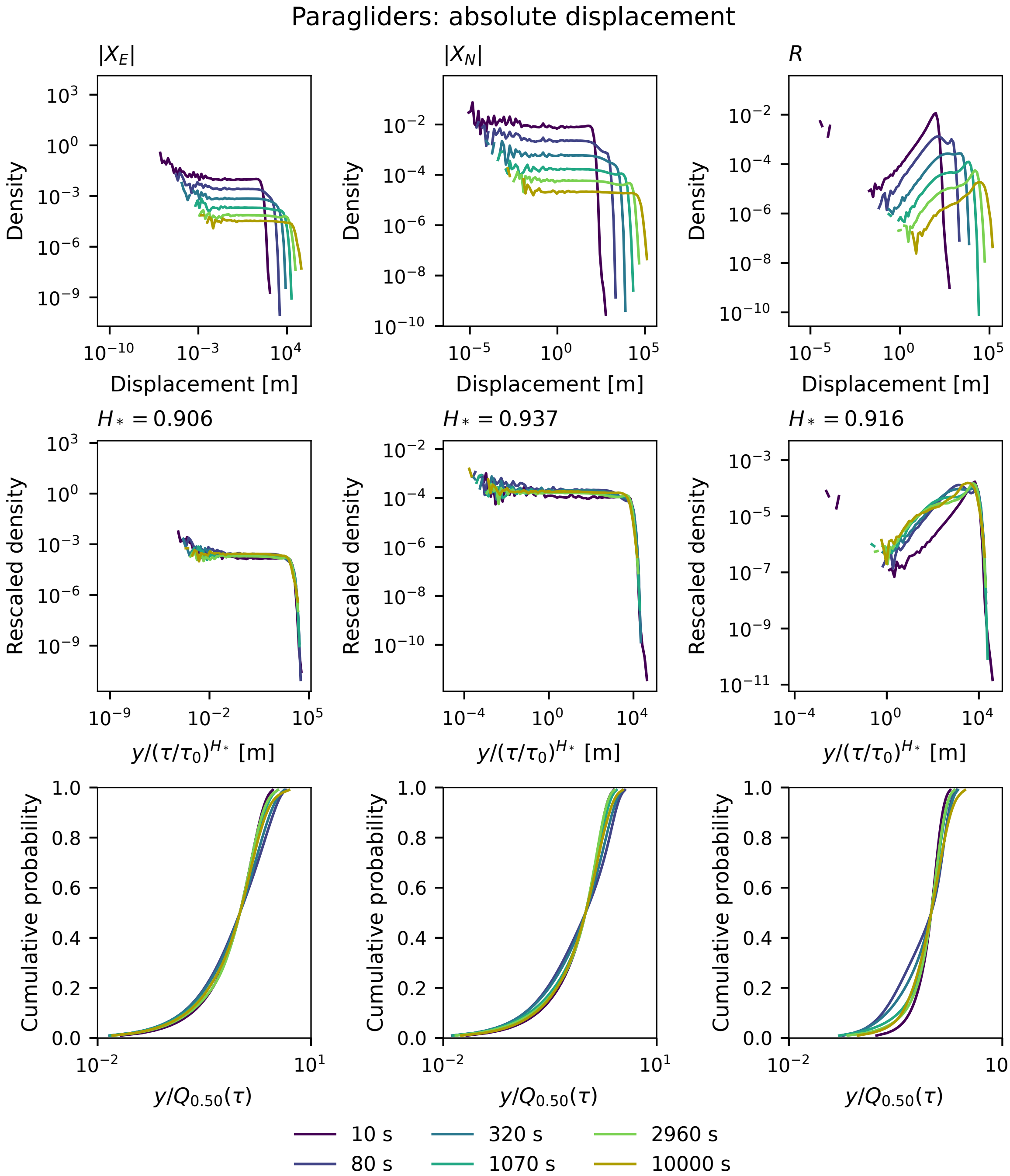


Figure 3.10. Paraglider displacement distributions with fixed flights, weights and origins. Top: raw densities of $|X_E|$, $|X_N|$ and R. Middle: their rescaling by the common exponent fitted separately for each quantity over 10 s to 10,000 s. Bottom: cumulative distributions after division by each lag's own median, drawn from the 1st–99th empirical percentiles. Colours identify six lags selected across the measured interval. Median normalisation removes an arbitrary scale change, including one that is not a power law; remaining separation therefore reveals changes of shape. The quantitative distances use the full ECDFs.

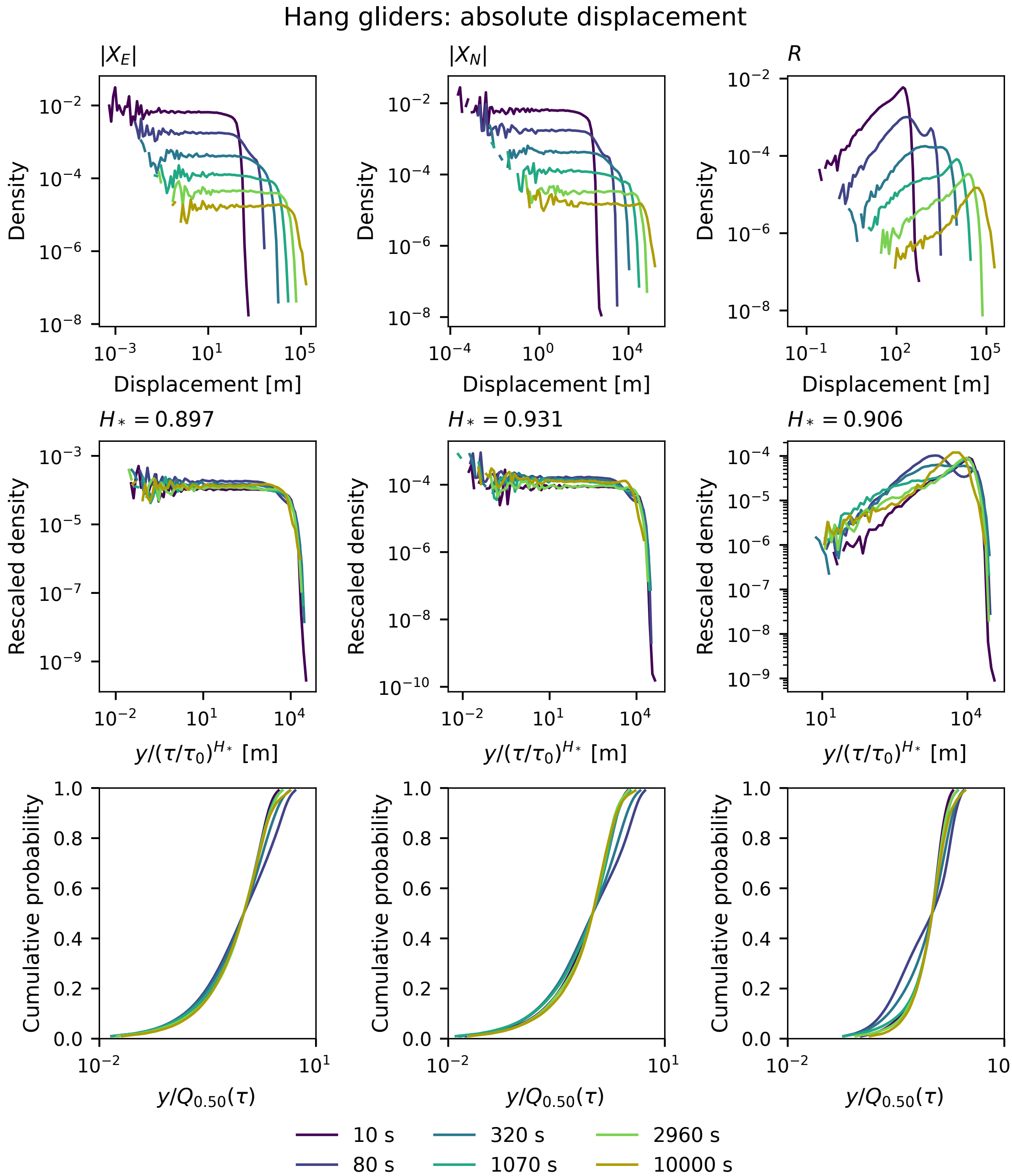


Figure 3.11. The same distribution and collapse diagnostics for hang gliders. The calculation uses the same origins at every lag and the same definition of $H_{j,*}$ as Figure 3.10; it does not borrow the paraglider exponent.

Table 3.6. Common magnitude exponent $H_{j,*}$ fitted on three lag windows with identical observations and weights. The differences expose the cost of summarising curved quantile growth by a single power. All twelve separate slopes and their paired bootstrap contrasts for each window are retained in the numerical report.

Discipline	Quantity	10–10000 s	60–1970 s	210–1970 s
Paragliders	$\lvert X_E\rvert$	0.906	0.952	0.956
	$\lvert X_N\rvert$	0.937	0.977	0.985
	R	0.916	0.953	0.950
Hang gliders	$\lvert X_E\rvert$	0.897	0.951	0.943
	$\lvert X_N\rvert$	0.931	0.981	0.992
	R	0.906	0.957	0.948

The fitted exponents are near 0.9, yet the rescaled distributions remain separated (Table 3.5). The radius has the largest power-rescaling distance in both disciplines, about 0.27 for paragliders and 0.33 for hang gliders. Dividing by each lag's median reduces these radial distances to about 0.19 and 0.23, but leaves separation for all three quantities. Thus allowing an arbitrary scale at each lag still does not remove the observed shape changes. These are empirical discrepancies without a calibrated sampling-error threshold.

Squared displacements. For any of the nonnegative magnitudes Y_j and $S_j = Y_j^2$, the inverse-CDF quantiles satisfy exactly

$$Q_{S_j,p}(\tau) = Q_{Y_j,p}(\tau)^2, \qquad \beta_{j,p} = 2H_{j,p}, \qquad p_{S_j}(s;\tau) = \frac{p_{Y_j}(\sqrt{s};\tau)}{2\sqrt{s}}, \quad s > 0. \tag{3.27}$$

The squared-density collapse therefore uses scale a_j^2 and exponent $2H_{j,*}$; bootstrap slope intervals double too. Squared-law plots appear in Appendix 3.A. Squaring is a monotone bijection for nonnegative values, so corresponding ECDF distances are unchanged. It cannot repair a failed collapse. Radial quantiles must still be computed from paired components: a quantile of $X_E^2 + X_N^2$ is generally not the sum of their quantiles.

Dependence on the fitting interval. We repeat the common-exponent fit on 60 s to 1,970 s and 210 s to 1,970 s, retaining the same flights, origins and probabilities. These windows measure sensitivity; their endpoints are not estimated breakpoints.

The intermediate-window exponents move closer to one (Table 3.6). Growth therefore depends on rank, direction and lag even after fixing the observations. Circling, partial returns, extended glides and weather could contribute, but this analysis does not separate their effects. Since the component laws themselves change shape, the data also do not isolate the special case of scaling marginals with changing dependence alone.

3.3.3 Signed joint distributions

Figures 3.12 and 3.13 retain the signs of (X_E, X_N) on the same flights and origins. Both coordinates are divided by $s_0(\tau/1{,}000\,\mathrm{s})^{H_*}$, where H_* is the mean of all twelve slopes and $s_0 = Q_{R,.50}(\tau_r)/(\tau_r/1{,}000\,\mathrm{s})^{H_*}$, with $\tau_r = 1{,}070\,\mathrm{s}$, the available lag nearest the reference time. Thus the recorded median is adjusted to the common reference by the same power. Geographic axes remain fixed, without lag-specific centring, rotation or whitening.

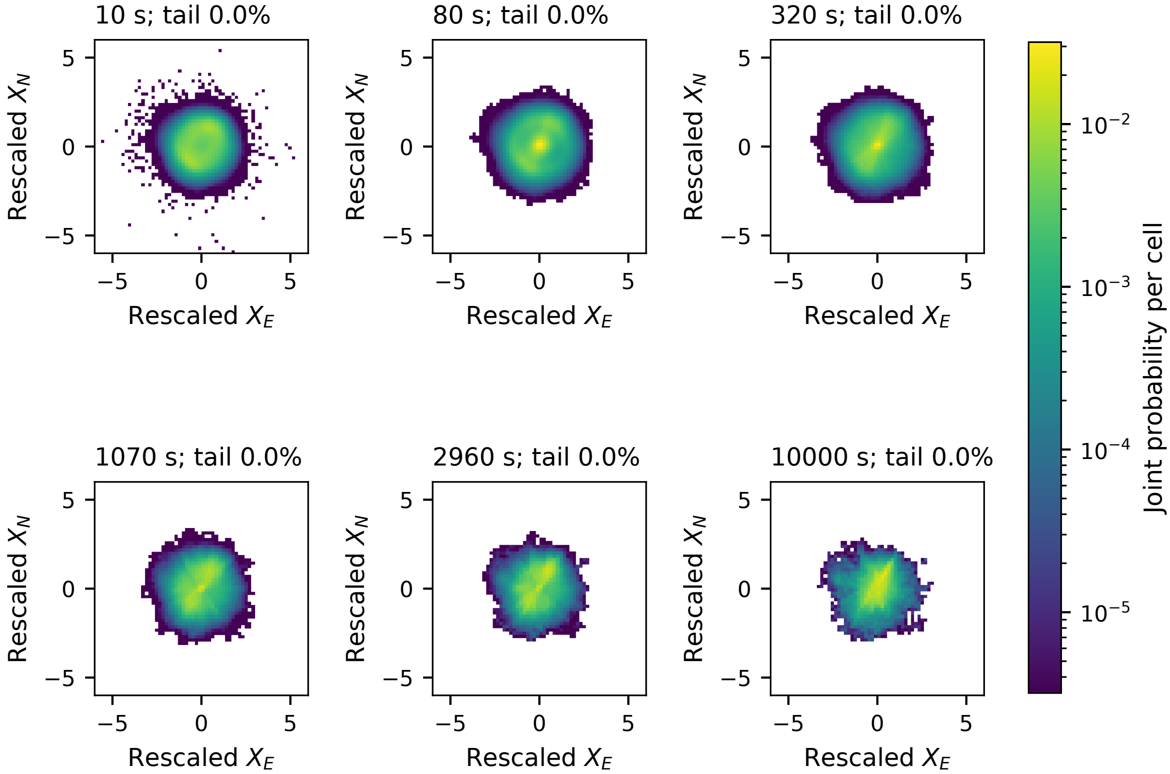


Figure 3.12. Signed joint displacement laws for the fixed paraglider population, after one scalar power rescaling of both axes. Colours show probability per common cell. Each panel reports the mass outside the displayed square; overflow cells are retained in numerical comparisons. These are descriptive distributions, not iid hypothesis tests.

The six lags share 64 bins per axis within $[-6, 6]$, plus overflow cells. With equal total weight per flight, let $p_{\tau,ij}$ be the joint cell mass. The distance between two binned laws is

$$D_{\text{joint}}(\tau, \tau') = \frac{1}{2} \sum_{ij} |p_{\tau,ij} - p_{\tau',ij}|, \tag{3.28}$$

including overflow. This comparison resolves differences at the chosen bin size; it is not a calibrated rejection test. Energy distance is an alternative for full multivariate laws with finite first moments [25], but inference would still require calibration for paired flights and dependent origins. Even exact agreement of every one-lag vector law would leave multi-time self-similarity untested.

The candidate vector exponents are $H_* = 0.920$ for paragliders and 0.911 for hang gliders. After this common rescaling, the central concentration and orientation still evolve with lag. In particular, the long-lag law occupies a smaller spread in the rescaled coordinates; the geographical mean also changes. These changes concern the signed vector distribution, beyond the magnitude comparisons. At the chosen resolution, pairwise distances range from 0.099 to 0.370 for paragliders and from 0.136 to 0.441 for hang gliders. The 10-s versus 10000-s distances are 0.370 and 0.403, respectively. Overflow mass is zero to numerical precision, so the measured separation is not caused by omitting a tail outside the displayed square.

The fixed flights and origins rule out changing membership as the source of these particular differences. They do not distinguish changes of marginal shape, mean motion and component dependence. A larger binned distance also does not rank the disciplines by strength of a physical mechanism: the finite-flight sampling error and sensitivity to bin size have not been calibrated.

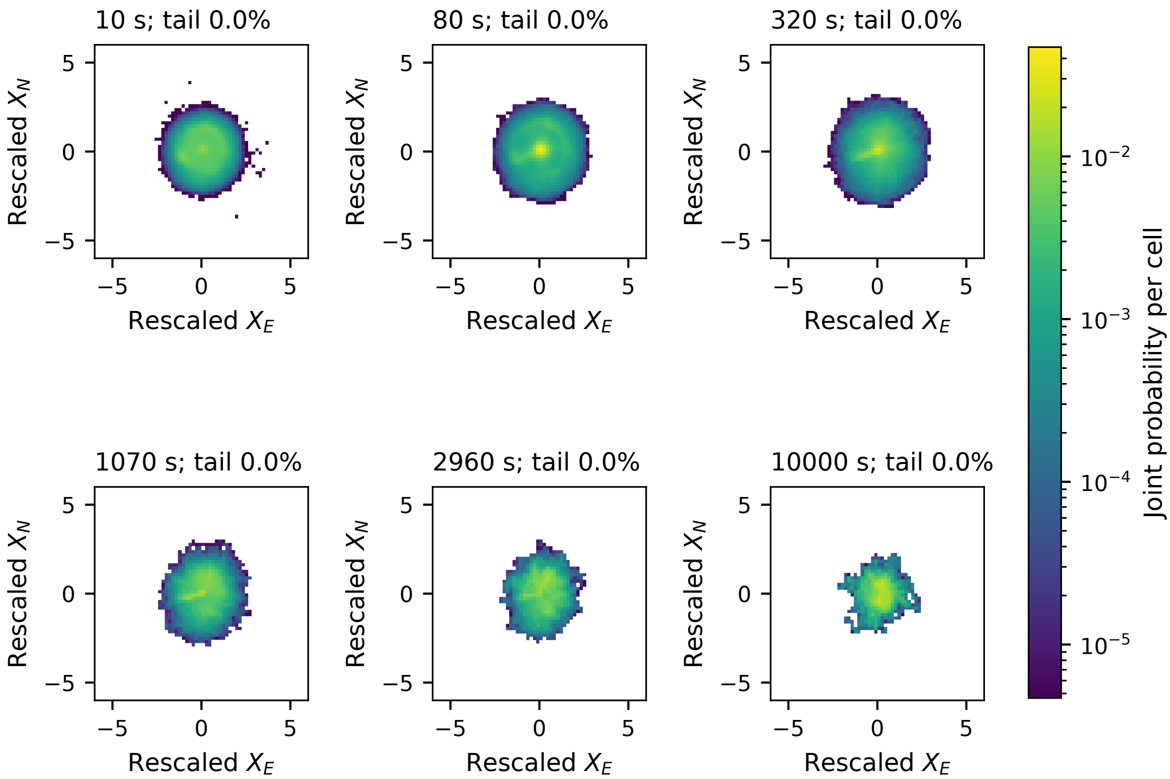


Figure 3.13. The same signed joint-law diagnostic for the fixed hang-glider population. Axes and colour scales remain common across lags within the discipline.

3.4 Directional structure, terrain and equipment

We next ask how directional structure varies across take-off regions and equipment groups, and which environmental interpretations these contrasts can support. The quantities differ from the fixed-population collapse test: launch-time second moments describe mean motion and spread together, whereas regional increment covariances describe centred spread over a lag. Their clocks, weighting and contributing flights must therefore be specified before comparing them.

3.4.1 East–north balance in the retained archive

Isotropy requires the joint displacement law to remain unchanged under rotation. Equality of east and north second moments is one necessary condition. For positions measured from launch, define

$$\rho_{EN}(t) = \frac{\langle E_f(t)^2 \rangle_f}{\langle N_f(t)^2 \rangle_f}, \qquad \langle E_f(t)^2 \rangle_f = \mathrm{Var}_f[E_f(t)] + \langle E_f(t) \rangle_f^2. \tag{3.29}$$

This uncentred ratio combines mean motion and spread. Here t is time since launch, and its contributing flights change with t. Unity checks one moment condition for isotropy.

Over 10 s to 10,000 s, the unresampled ratio ranges from 0.72 to 0.87 for paragliders and from 0.54 to 1.45 for hang gliders. These are curve extrema, not bootstrap-band limits. The directional imbalance supports treating east, north and radial displacement separately. It describes the heterogeneous retained archive; it does not identify whether route choice, terrain or wind produces the imbalance. The centred regional analysis below asks how much directional

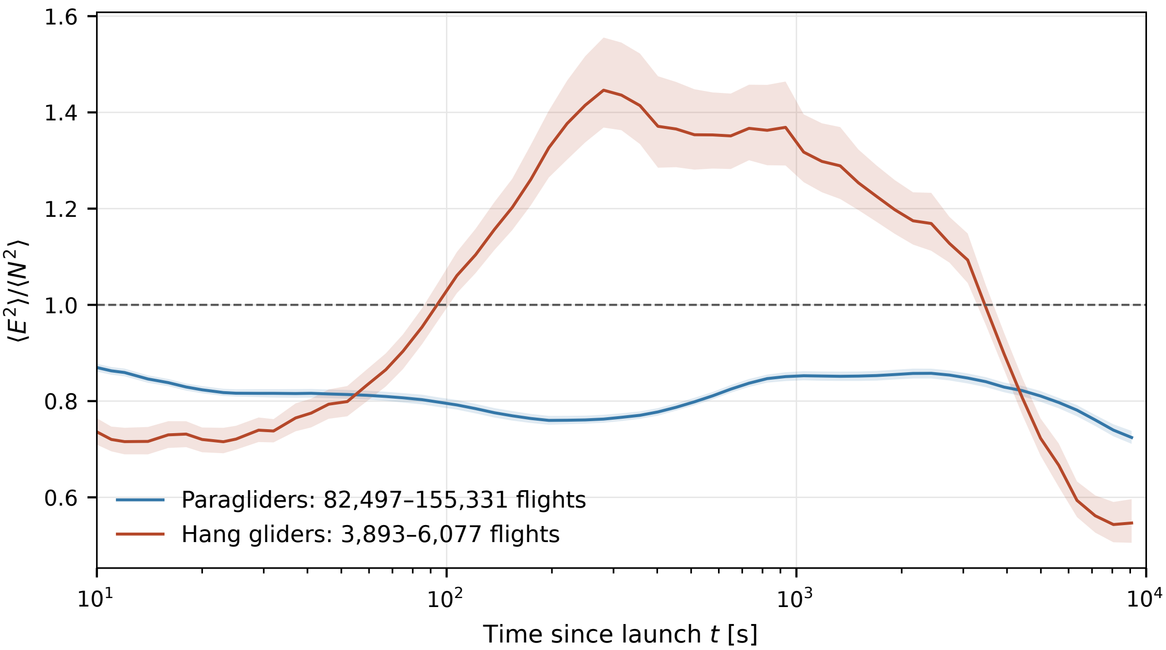


Figure 3.14. Archive east–north second-moment ratio against elapsed time since launch. Curves show the observed ratio; shading gives the pointwise 10–90% range from 200 bootstrap resamples of whole take-off-site-and-day groups. Only times with at least 30 paired flights and 10 contributing clusters are shown; legend counts give the minimum and maximum flight support along each curve. These are display minima, not a precision criterion. Unity is a necessary condition for isotropy, not sufficient evidence of it; these bands are not a simultaneous confidence statement across all times.

structure remains after subtracting each subgroup's mean. Launch-referenced positions can retain reacquisition offsets across missing intervals; within-segment increments avoid bridging those intervals but do not independently validate their endpoints.

3.4.2 Regional principal axes

For a region g, define the centred displacement covariance

$$\boldsymbol{\mu}_g(\tau) = \mathbb{E}[\mathbf{X} \mid g], \qquad \Sigma_g(\tau) = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu}_g)(\mathbf{X} - \boldsymbol{\mu}_g)^\top \mid g] = U_g \operatorname{diag}(\lambda_1, \lambda_2)\, U_g^\top, \tag{3.30}$$

with $\lambda_1 \geq \lambda_2$. The ratio and whitening below require $\lambda_2 > 0$. The leading eigenvector describes the direction of greatest spread, modulo 180°; it is an axis, not a signed wind direction. The ratio λ_1/λ_2 measures anisotropy of the covariance. An axis becomes poorly determined when the eigenvalues are close. Covariances pool all supported baseline increment origins within each take-off box, so flights with more origins receive more weight. They use the eligible regional archive, independently of the fixed long-flight cohort. The four lags are exactly 10, 100, 1000 and 10000 s, evaluated directly on the 10-s coordinate grid. Increment origins are spaced by the lag within each segment; no increment crosses a segment boundary. Each lag uses every flight with supported increments, so the contributing population can shrink as the lag increases. At least eight flights are required; this display condition supplies neither an uncertainty interval nor a precision guarantee.

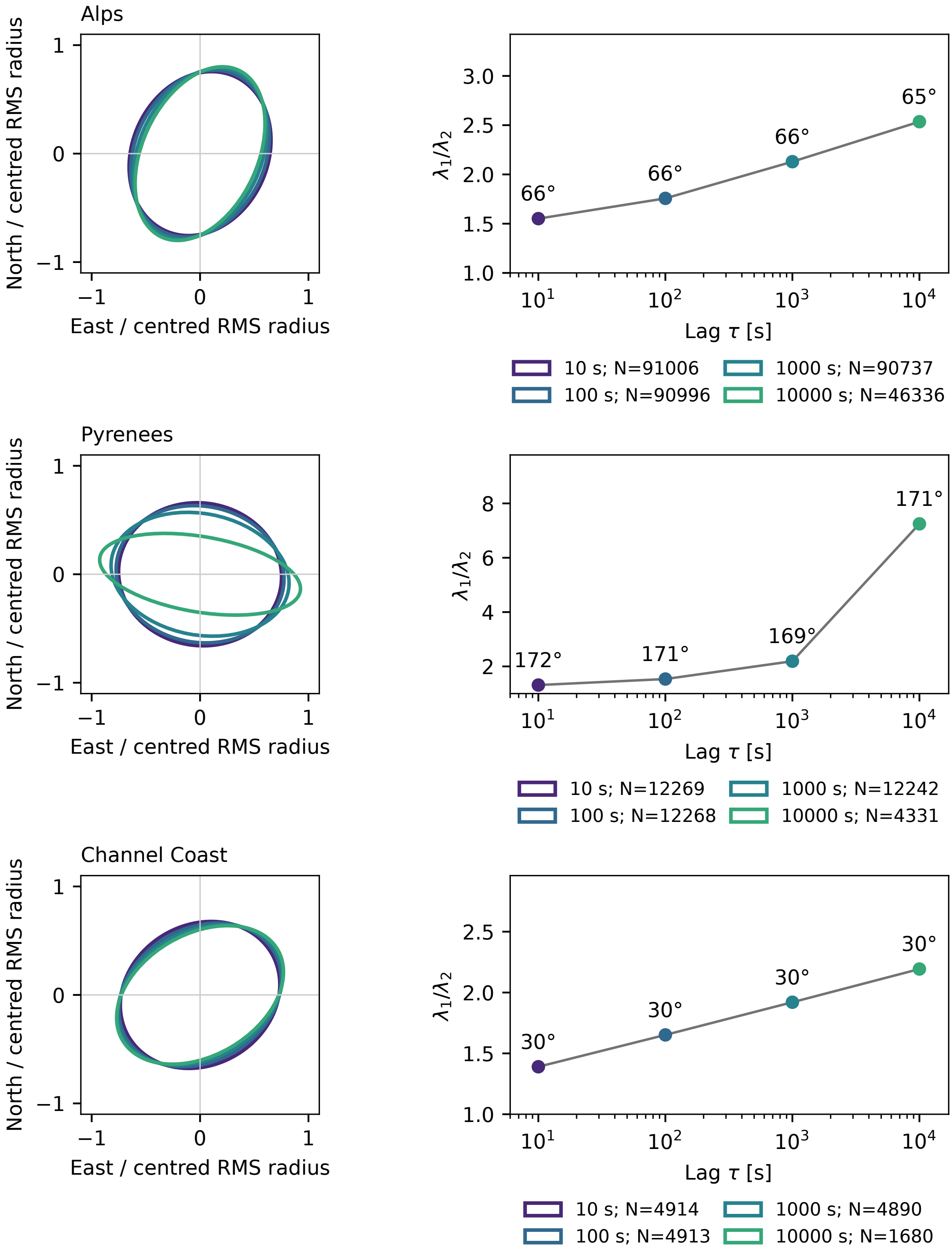


Figure 3.15. Regional principal-axis diagnostics in the paraglider eligible archive. Covariances are centred before diagonalisation. The principal direction is an axis modulo 180° and its interpretation depends on the separation of the two eigenvalues. Region counts and lag dependence delimit the evidence. These measurements describe displacement geometry and do not identify wind. Ellipses and matching coloured markers show all four lags: 10, 100, 1000 and 10000 s. Ellipses are normalised by the square root of the covariance trace, removing amplitude differences. Legends give contributing-flight counts at each lag; angle labels accompany the eigenvalue ratios. Changes across lags can also reflect changes in this population.

At 1,000 s, the regional comparison is

Region	Flights	λ_1/λ_2	Axis from east
Alps	90,737	2.13	65.7°
Pyrenees	12,242	2.19	169.5°
Channel coast	4,890	1.92	29.7°

Angles run counterclockwise from east. The Pyrenean axis lies closer to east–west; its eigenvalue ratio increases towards long lags as support falls, leaving both route geometry and selection as possible contributors. The coastal comparison also has unequal principal variances, with thousands of flights at the reference lag. Its geographical label does not provide an isotropic control.

Rotation into the principal frame gives uncorrelated components at the lag at which the covariance was fitted. It does not establish independence or remove temporal correlations. It cannot improve radial collapse because

$$|U_g^\top \mathbf{X}| = |\mathbf{X}|. \tag{3.31}$$

Whitening with the reference covariance, $\mathbf{Z}(\tau) = \Sigma_g(\tau_0)^{-1/2}[\mathbf{X}(\tau) - \boldsymbol{\mu}_g(\tau)]$, changes lengths. It sets the covariance to the identity at the reference lag τ_0; it need not do so at other lags, and it does not imply a rotationally invariant distribution. Whitening separately at each lag would enforce a part of the desired collapse by construction. A meaningful control instead fits a transformation at a fixed reference scale and tests it on other scales and flights.

3.4.3 Terrain, equipment and wind

Figure 3.16 compares east/north second-moment ratios over 10 s to 10,000 s. Their medians are

Region	Position ratio	Velocity ratio
Alps	0.67	0.81
Pyrenees	1.67	1.19

These uncentred moments retain mean motion, as in Eq. (3.29). The stronger Pyrenean east–west preference is consistent with the massif constraining headings, without identifying geography as its sole cause.

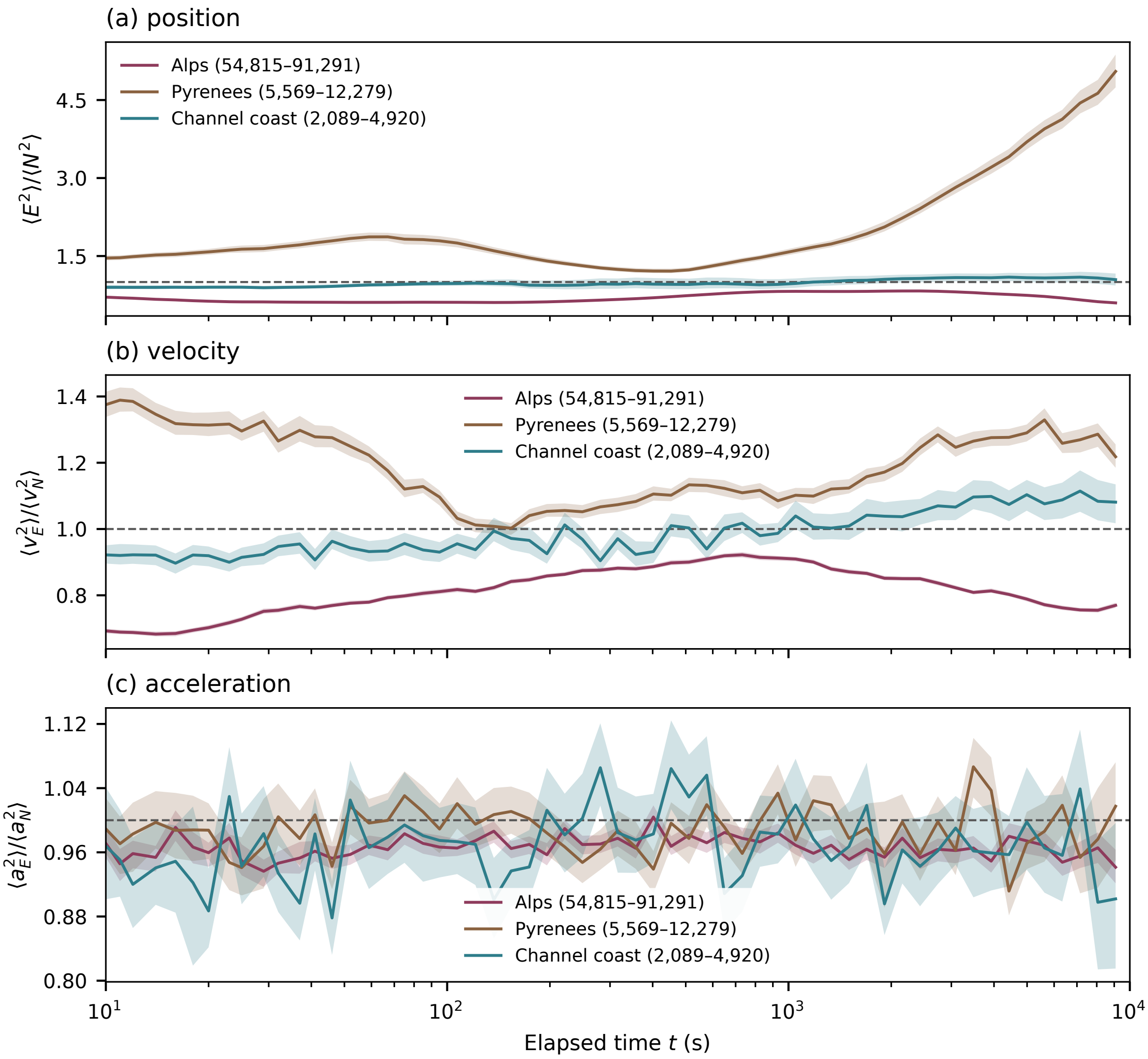


Figure 3.16. Paraglider east/north second-moment ratios by take-off region, against time since launch. Position, velocity and acceleration all retain mean motion. Shading gives pointwise 10–90% site–day bootstrap ranges; times require 30 flights and 10 clusters, with flight-count ranges in the legends. Missing site/date records form singleton groups. Unity marks equal component moments, leaving cross moments and angular structure untested. Panel limits follow each quantity; no scaling exponent is fitted.

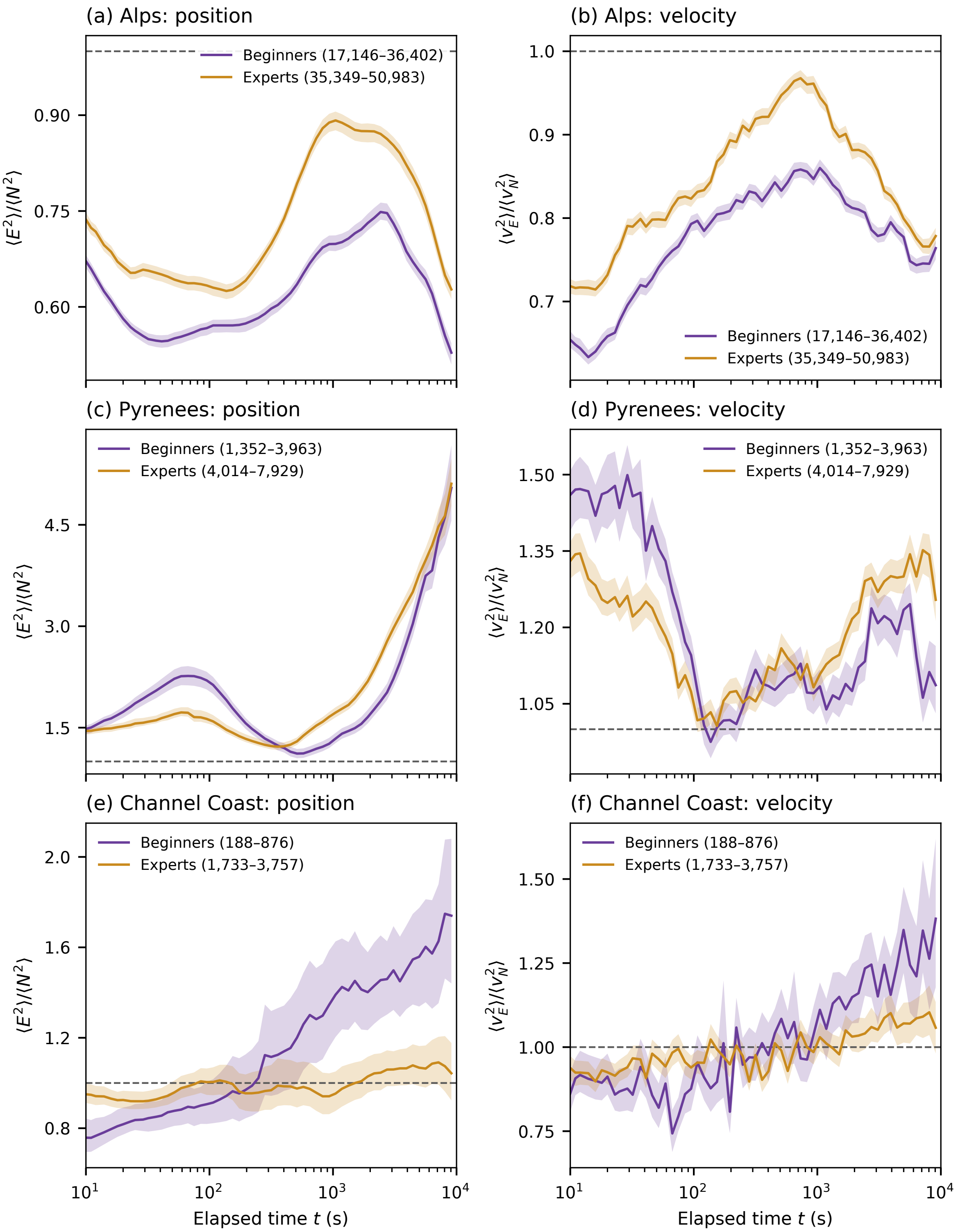


Figure 3.17. Paraglider second-moment ratios within each take-off region: beginners (EN A/B, violet) and experts (EN C/D/CCC, ochre), using equipment as an experience proxy. Tandem, non-certified and unclassified records are excluded. Pointwise bootstrap bands and support minima follow Fig. 3.16.

Equipment within each region. EN A/B and EN C/D/CCC provide the beginners and experts proxies of Sec. 2.4. In the Alps, the C/D/CCC position and velocity ratios remain closer to unity at all displayed supported times. The median position ratio is 0.70, versus 0.61 for A/B. The Pyrenees and coastal contrasts do not have the same ordering across all times and quantities.

An interpretation to test. The smaller east–north imbalance of the C/D/CCC group in the Alps and over much of the coastal comparison motivates a hypothesis: equipment and pilot decisions may moderate the directional constraints associated with wind and terrain. A wing with different flight characteristics may permit different routes, while a pilot's choices may alter how a route follows the local lift and wind. Similar mountain profiles could reflect a shared geographical constraint, with a difference in its magnitude between groups. This is a possible explanation of the observed contrasts, rather than a measured decomposition of their causes.

The Pyrenean curves limit that explanation. Their ordering reverses with elapsed time, and the C/D/CCC group does not remain uniformly closer to one in either position or velocity. Near 9,086 s, the position ratios are about 5.05 for A/B and 5.11 for C/D/CCC. The coastal position contrast is larger at the same time, about 1.74 versus 1.04, but its support is also unequal: 188 versus 1,733 paired flights. The bands are pointwise 10–90% site–day bootstrap ranges, not simultaneous confidence bands or a direct confidence interval for a between-group difference.

What proximity to unity establishes. The ratio compares two ensemble raw second moments. It does not measure the isotropy of each flight or the ability of an individual pilot to make progress against wind. For example, a centred Gaussian with covariance

$$\begin{pmatrix} 1 & 0.9 \\ 0.9 & 1 \end{pmatrix}$$

has equal east and north second moments but a principal-variance ratio of 19. For a general centred law, a covariance proportional to the identity establishes only a second-order condition, rather than rotational invariance of the full distribution. A mixture of strongly east-directed and strongly north-directed routes can also balance ensemble moments while each route remains directional. Deliberately following a downwind cross-country route may be successful flight despite that directionality.

Equipment, terrain and selection. EN A/B and EN C/D/CCC are equipment groups used as experience proxies. Actual pilot skill is unmeasured, and an experienced pilot can fly an A/B wing. Certification classes describe tested flight behaviour and associated pilot requirements; the class is not itself a measured aerodynamic performance ranking [26]. Equipment, pilot selection and intended task therefore remain entangled in these comparisons.

The Channel Coast box is also not a wind-only experiment. It includes coastal cliffs and valleys, including those documented in the Pays de Caux [27]. Wind and terrain may jointly orient lift and routes. Moreover, the boxes classify take-off locations rather than terrain exposure along the entire flight. The additional Poitou-Charente and Champagne-Lorraine comparisons in Fig. 3.18 extend the geographical description, but these broad boxes are not verified terrain polygons or meteorologically matched controls. The Poitou-Charente curves cross with broad overlapping bands; the later Champagne-Lorraine position contrast also leaves the C/D/CCC ratio above one. Thus the coastal pattern does not establish a general flatland mechanism.

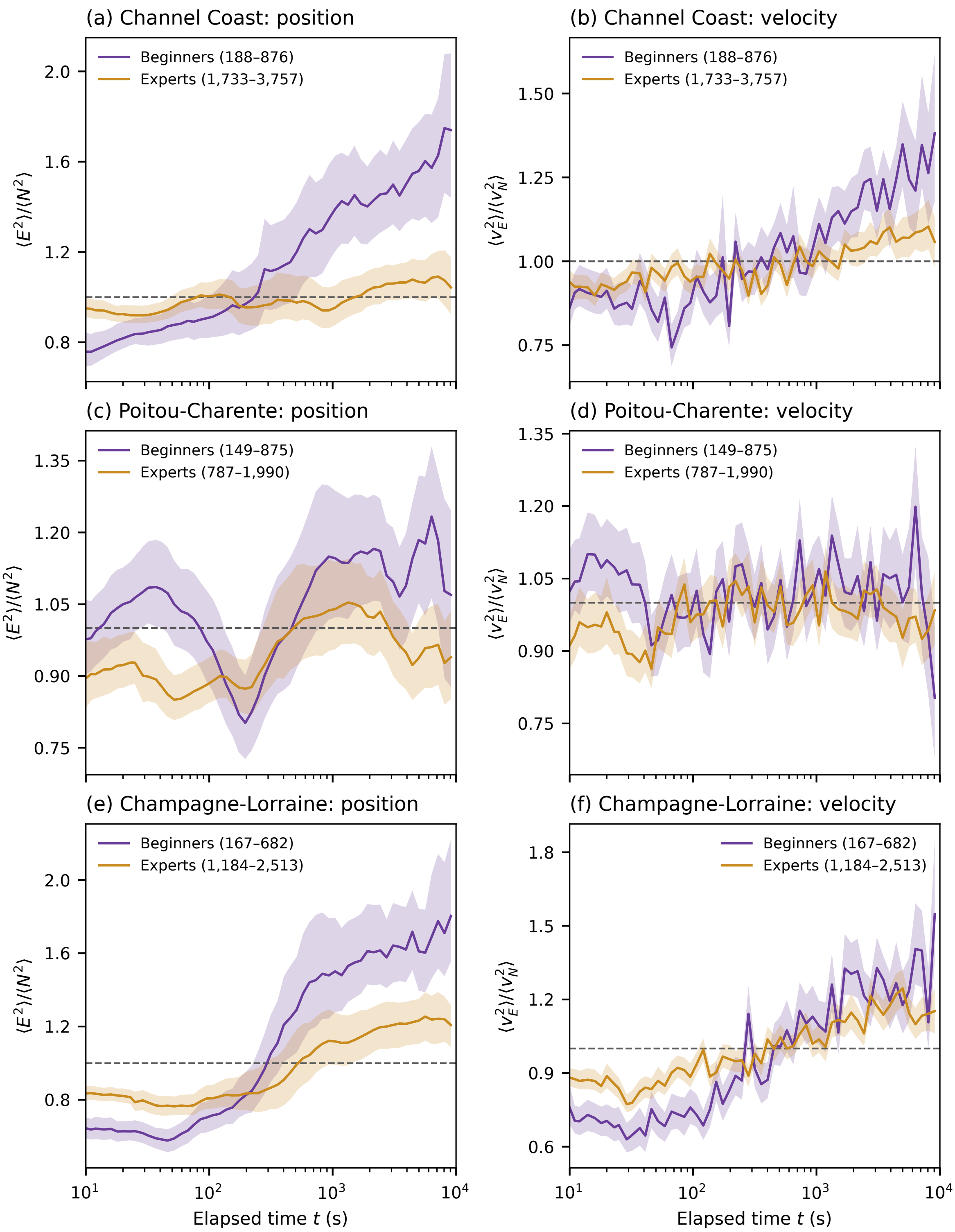


Figure 3.18. Equipment contrasts in three take-off boxes outside the mountain comparison: Channel Coast, Poitou-Charente and Champagne-Lorraine. Position and velocity ratios use the paired support, pointwise 10–90% site–day bootstrap bands and display minima of Fig. 3.17. The coastal row is repeated to compare it directly with the two additional regions. The boxes label take-off locations; they do not measure terrain exposure along each route or match wind conditions between equipment groups.

Discriminating comparisons. A first comparison would restrict both equipment groups to overlapping take-off sites and days, with comparable duration and task. This would test whether the contrast survives closer matching of observed conditions; the current cluster bootstrap adjusts the uncertainty for one grouping structure without performing that matching. Mean ground drift, centred directional variability and the mixture of individual route directions should then be distinguished. Independent wind estimates and local terrain axes would allow an environmental alignment test. Where records permit it, comparisons of the same pilot using different wings could help assess equipment and pilot selection separately. Before describing any of these contrasts as resistance to wind, the target needs to be defined: upwind progress, air-relative performance and the choice of route are different outcomes.

Wind and ground velocity. The kinematic relation

$$\mathbf{v}_{\text{ground}} = \mathbf{v}_{\text{air}} + \mathbf{w}_h \tag{3.32}$$

prevents ground velocity alone from separating still-air motion and wind. Route choice and net progress also enter the mean. Differentiation removes a constant wind contribution but retains changing wind and manoeuvres.

If the wind field is $\mathbf{w}_h(\mathbf{r}, z, t)$, its derivative along the trajectory is

$$\mathbf{a}_{\text{ground}} = \frac{\mathrm{d}\mathbf{v}_{\text{air}}}{\mathrm{d}t} + \partial_t \mathbf{w}_h + (\mathbf{v}_{\text{ground}} \cdot \nabla_h)\mathbf{w}_h + \dot{z}\,\partial_z \mathbf{w}_h. \tag{3.33}$$

Thus even a time-independent wind field contributes to acceleration when the glider crosses spatial gradients.

Separating these effects requires terrain alignment and independent wind estimates within matched region and site–day groups. Thermal-centre drift may help after segmentation, subject to assumptions about thermal motion. Randomly rotating flights offers an alignment control: it balances pooled directions while preserving every radial displacement, without reconstructing still-air motion.

3.4.4 Regional changes of velocity and direction

Second differences permit a regional comparison that is insensitive to an added constant velocity. Their physical scale can be expressed without fitting an exponent. For the two adjacent interval-mean ground velocities,

$$\overline{\mathbf{v}}_1 = \frac{\mathbf{r}(t+\tau) - \mathbf{r}(t)}{\tau}, \qquad \overline{\mathbf{v}}_2 = \frac{\mathbf{r}(t+2\tau) - \mathbf{r}(t+\tau)}{\tau}, \qquad D_2(\tau) = \frac{\sqrt{V_2(\tau)}}{\tau} = \sqrt{\mathbb{E}|\overline{\mathbf{v}}_2 - \overline{\mathbf{v}}_1|^2}. \tag{3.34}$$

Thus D_2, in $\mathrm{m\,s^{-1}}$, measures changes of the mean velocity vector between adjacent intervals. Both changes of direction and changes of its magnitude contribute. This identity requires neither stationary increments nor self-similarity. It does not measure wind speed.

The focused measurement uses all 108,189 paraglider and 4,747 hang-glider flights in the three regional take-off boxes. It reads the verified common-grid coordinates used by the regional PCA, without repeating cleaning or limiting the number of flights. Eighteen physical lags span 10 s to 10,000 s; the four display lags agree with the updated PCA at 10, 100, 1000 and 10,000 s. Segment boundaries remain mandatory.

Three support conventions answer different questions. The available-origin curves use every supported origin for each order separately, spaced by τ within each segment. The matched-order comparison uses the same origins and flights for orders one, two and three, and therefore requires 3τ of continuous support. These stencils overlap even though consecutive origins are separated by τ. A further common-origin control uses every 10-s origin supporting a continuous 3000-s window, and retains that exact set for every order and every lag through 1,000 s. The 1000-s limit is a declared support convention; it avoids selecting only flights with 30,000-s continuous records merely to compare the smaller scales.

Within each flight, eligible origins are pooled before flights receive equal weight. The report also retains the pooled-origin result used as a weighting sensitivity. This distinguishes the estimator here from the pooled-origin PCA in Sec. 3.4. Uncertainty uses 500 bootstrap draws of complete site–day groups, jointly across lags and difference orders. Missing group keys occur in three Alpine paraglider flights and are treated as singleton groups. Pointwise 95% intervals require at least 20 contributing groups and 90% finite replicates; this display threshold is not a guarantee of coverage. Regional RMS-change ratios additionally receive approximate simultaneous 95% bands over their supported lag grid, obtained from the bootstrap maximum standardised log-ratio deviation. The interpretation assumes independently resampleable groups; dependence across sites, dates or repeated pilots can remain.

For directionality, retain the vector $\mathbf{d}_2 = A_2\mathbf{r}$ before taking its squared norm. Its mean and centred covariance satisfy

$$\boldsymbol{\mu}_2 = \mathbb{E}\mathbf{d}_2, \qquad C_2 = \mathrm{Cov}(\mathbf{d}_2), \qquad V_2 = \operatorname{tr} C_2 + |\boldsymbol{\mu}_2|^2. \tag{3.35}$$

The covariance includes variation between flight means as well as within flights. The ratio of its eigenvalues measures directional imbalance of the dispersion; the scalar V_2 alone cannot do so. As before, the leading axis has no signed direction and becomes poorly determined near equal eigenvalues.

Table 3.7. Paraglider diagnostics at 1,000 s, with equal flight weights and origins shared by all three difference orders. The eigenvalue ratios refer to centred covariances at the indicated order.

Region	Flights	D_2 [m/s]	Ratio, order 1	Ratio, order 2	V_3/V_2
Alps	89,181	5.19	2.28	1.83	2.60
Pyrenees	12,020	4.78	2.17	1.42	2.79
Channel Coast	4,679	3.77	1.92	1.37	2.42

At this lag, second differencing reduces the covariance anisotropy relative to ordinary increments in all three regions, but leaves a directional contrast (Table 3.7). The order-two axes are about 65.45°, 171.09° and 30.61° counterclockwise from east. At 10 s the second-difference covariance is much closer to equal principal variances, especially on the coast; an angle there should not be interpreted as a resolved environmental orientation.

Holding origins fixed through 1000 s gives D_2 values of $5.31\,\mathrm{m\,s^{-1}}$ in the Alps, $4.89\,\mathrm{m\,s^{-1}}$ in the Pyrenees and $3.72\,\mathrm{m\,s^{-1}}$ on the coast at 1000 s. The Alpine-to-coastal ratio is 1.426, with simultaneous band 1.403–1.449; the Alpine-to-Pyrenean ratio is 1.086, with band 1.074–1.097. The ordering changes with lag: at 10 s the coastal and Pyrenean values exceed the Alpine value. Similar-looking mountain curves therefore do not establish regional equality, and a single regional amplitude ratio does not describe the whole interval.

A descriptive standardisation further restricts the common-origin population to shared strata of declared open/closed task, retained duration (up to 2 h, 2–4 h, over 4 h), season period (through 2009, 2010–2019, 2020 onward) and equipment (EN A/B or EN C/D/CCC). Each retained stratum needs at least ten flights per region; its target mass is proportional to its smallest regional count. Unknown context labels are excluded from this control. The resulting 23 strata retain 65,537, 9,933 and 4,366 flights. Their standardised 1000-s values are respectively $5.13\,\mathrm{m\,s^{-1}}$, $4.62\,\mathrm{m\,s^{-1}}$ and $3.72\,\mathrm{m\,s^{-1}}$. The coastal contrast remains in these point

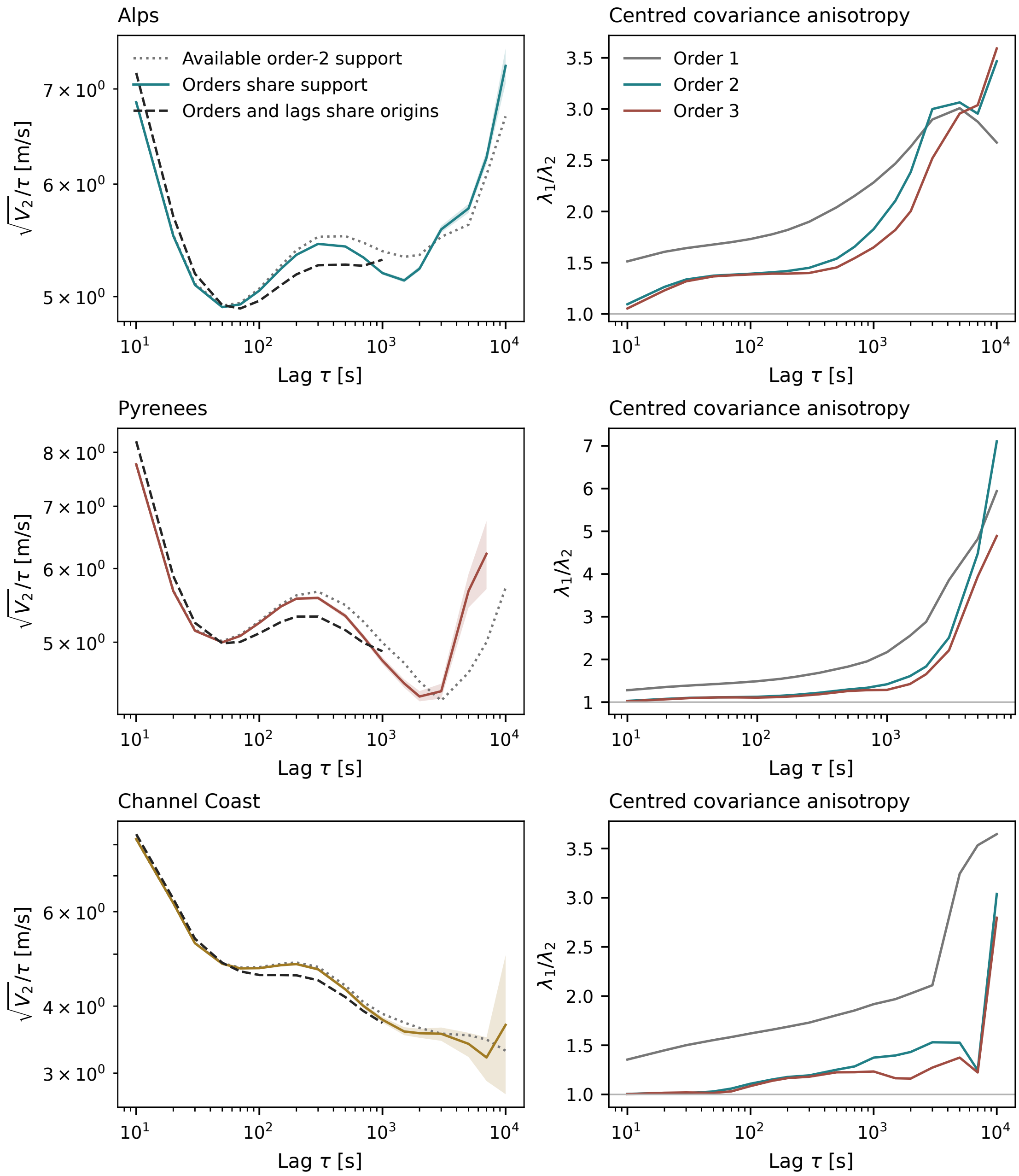


Figure 3.19. Regional changes of ground velocity in paragliders. Left: D_2 on available order-two support, on support shared by orders one to three, and on origins common to all orders and lags through 1000 s. Shading is the pointwise site–day interval for the matched-order curve. Right: covariance eigenvalue ratios on the same matched origins. Curves are omitted where fewer than eight flights contribute; all numerical cells remain in the report. Support at the long end is shown in Fig. 3.24.

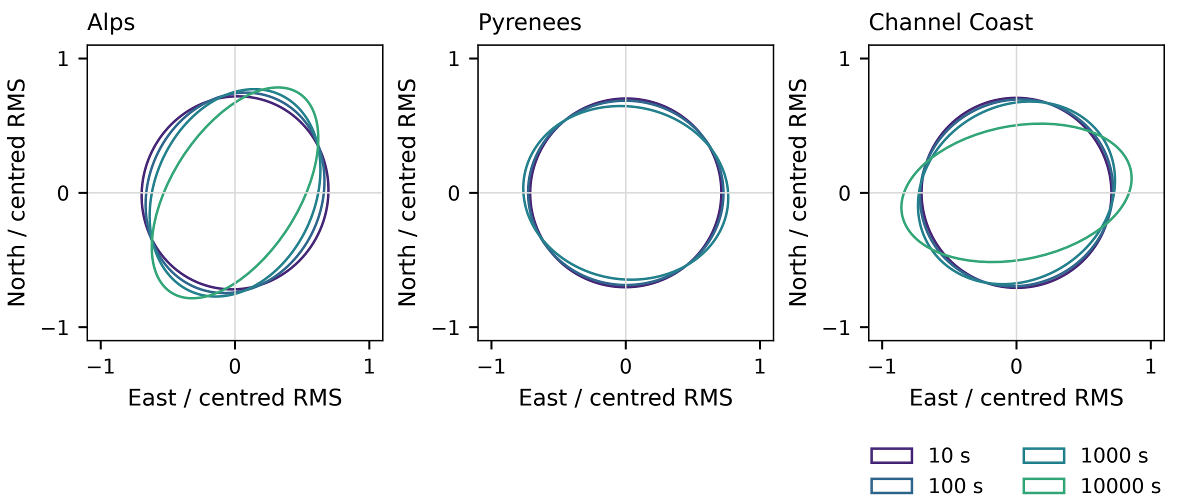


Figure 3.20. Centred second-difference covariance ellipses, normalised by their own trace, on matched-order origins. Colours use the same four physical lags as the regional PCA. Normalisation removes amplitude and the ellipses omit the mean. The Pyrenean 10,000-s ellipse is not shown because only three flights contribute.

estimates; calibrated intervals for the standardised contrasts are not supplied. No three-region common equipment/context strata survive for hang gliders.

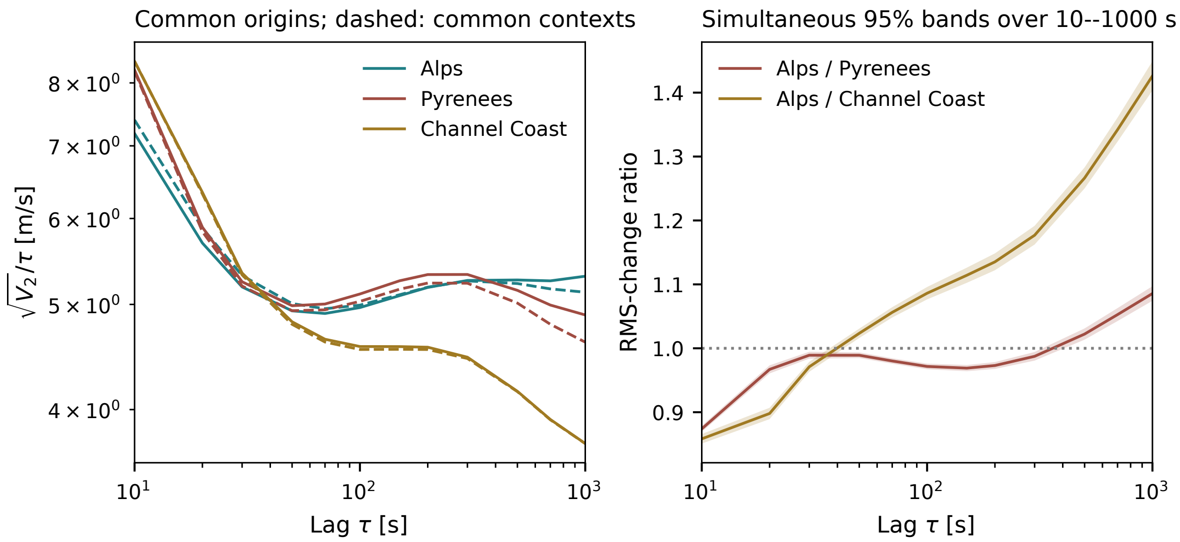


Figure 3.21. Paraglider population controls. Left: common-origin curves, with the common-context standardisation dashed. The dashed curves are descriptive point estimates. Right: regional ratios of the unstandardised common-origin curves; shading is simultaneous over the 18-lag grid's supported subset through 1000 s. These controls do not match meteorology, local terrain exposure or pilot identity.

These findings constrain a wind explanation without identifying wind. If populations differed only by an added constant velocity and otherwise shared the same residual dynamics, their second differences would agree. Their measured differences require more than that particular explanation. Spatial or temporal wind variation, route geometry, manoeuvres and population selection remain possible contributors. The coast has smaller changes of interval-mean velocity at 1000 s, rather than a larger signal that could simply be labelled stronger wind. An environmental attribution requires independent wind information and terrain alignment, ideally within sites under contrasting conditions.

3.4.5 Third differences and the distribution of changes

The third difference extends the cancellation by one polynomial degree:

$$A_3\mathbf{r} = \mathbf{r}(t+3\tau) - 3\mathbf{r}(t+2\tau) + 3\mathbf{r}(t+\tau) - \mathbf{r}(t), \qquad V_3 = \mathbb{E}|A_3\mathbf{r}|^2. \tag{3.36}$$

For $\mathbf{r}(t) = \mathbf{r}_0 + \mathbf{v}_0 t + \frac{1}{2}\mathbf{a}t^2$ with constant vector acceleration, $V_2 = |\mathbf{a}|^2\tau^4$ while $V_3 = 0$. A smooth curved trajectory generally retains both signals: locally $A_3\mathbf{r}$ is proportional to $\mathbf{r}^{(3)}\tau^3$. Constant-speed circular motion is therefore not eliminated by taking order three.

Under the same centred stationary-increment second-moment law used in Eq. (3.11), algebra gives

$$V_3 = K\bigl(15 - 6\,2^{2H} + 3^{2H}\bigr)\tau^{2H}, \qquad \frac{V_3}{V_2} = \frac{15 - 6\,2^{2H} + 3^{2H}}{4 - 2^{2H}}, \quad 0 < H < 1. \tag{3.37}$$

The power law must hold through 3τ. Thus the order ratio, in addition to the two slopes, is a consistency condition on that model. It equals three for Brownian motion plus an arbitrary constant velocity. Agreement between orders would not constitute independent data or certify removal of every environmental trend.

The observed order ratios change with lag (Fig. 3.22). At 100 s they are about 2.29 in both mountain regions and 2.38 on the coast; at 1000 s they are 2.60, 2.79 and 2.42. These curves supply a sensitivity and model-consistency diagnostic, without establishing a scale-independent Hurst parameter. Order three also increases the effect of independent position errors: variance σ^2 per coordinate adds $6\sigma^2$ to the order-two scalar-component moment and $20\sigma^2$ to order three. Correlated errors and smoothing alter those expressions. The 10-s end therefore remains sensitive to acquisition and preprocessing resolution; a calibrated sensor-noise subtraction is not performed here.

The distribution of $|A_2\mathbf{r}|/\tau$ supplies information lost by its second moment. We retain equal-flight histogram probabilities and the squared-change contribution of every bin, including zeros and both tails. Reported quantiles are bin bounds, not interpolated exact sample quantiles. At 1000 s, the largest tenth of changes accounts for approximately 42.7–46.5% of the squared-change signal in the Alps, 39.8–43.7% in the Pyrenees and 38.4–42.8% on the coast. Each range reflects the unresolved boundary bin, not sampling uncertainty. This measures concentration of the observable; it does not identify a flight phase, turbulence or a tail-generating mechanism.

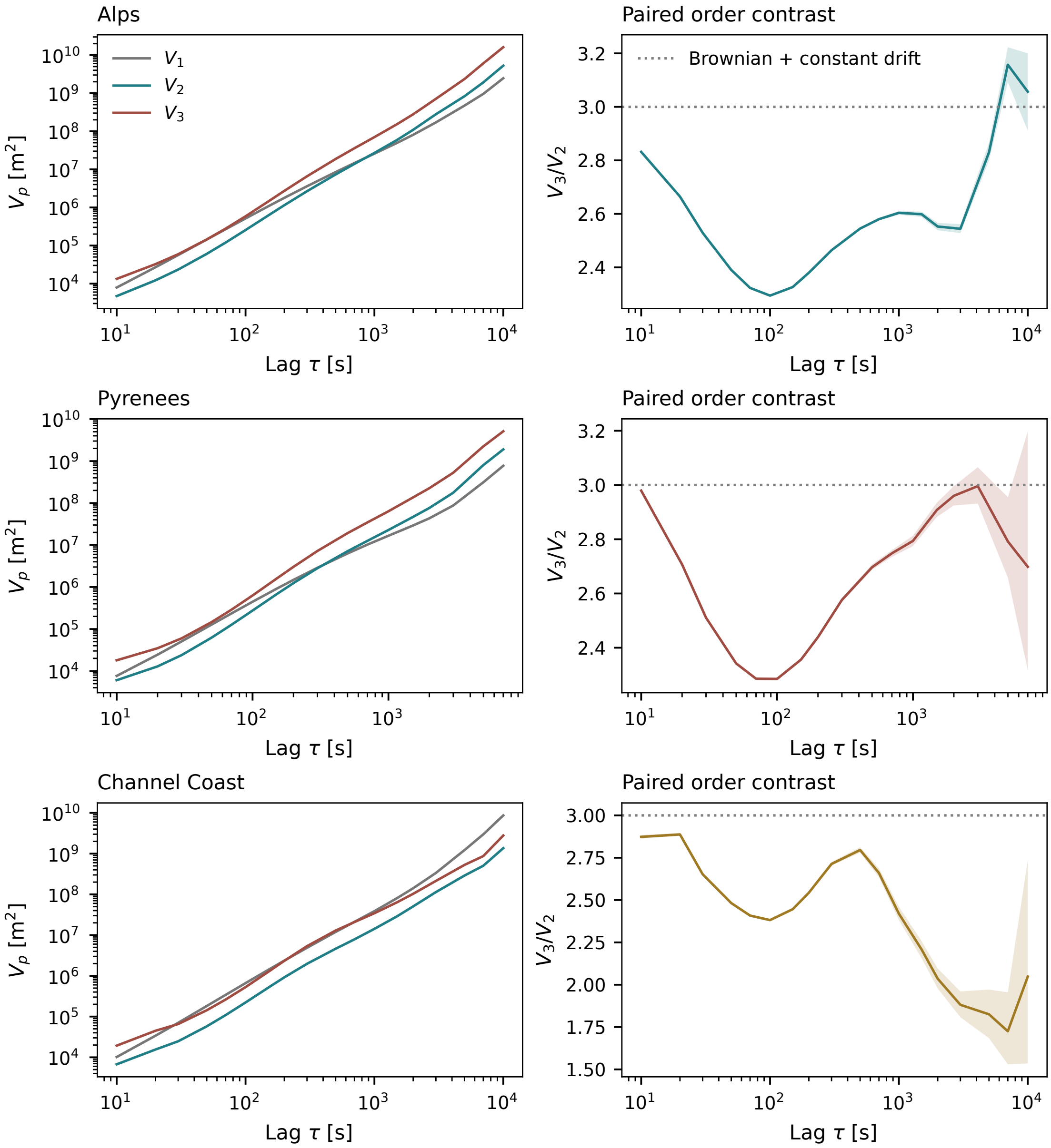


Figure 3.22. Paraglider variations and their paired order ratio on identical origins. Different prefactors make the amplitudes of V_2 and V_3 directly unequal even under one scaling model. Right: V_3/V_2 with pointwise site–day intervals; the dotted line is the Brownian-plus-constant-velocity reference, not a fit. Fewer than eight contributing flights suppress a displayed curve cell.

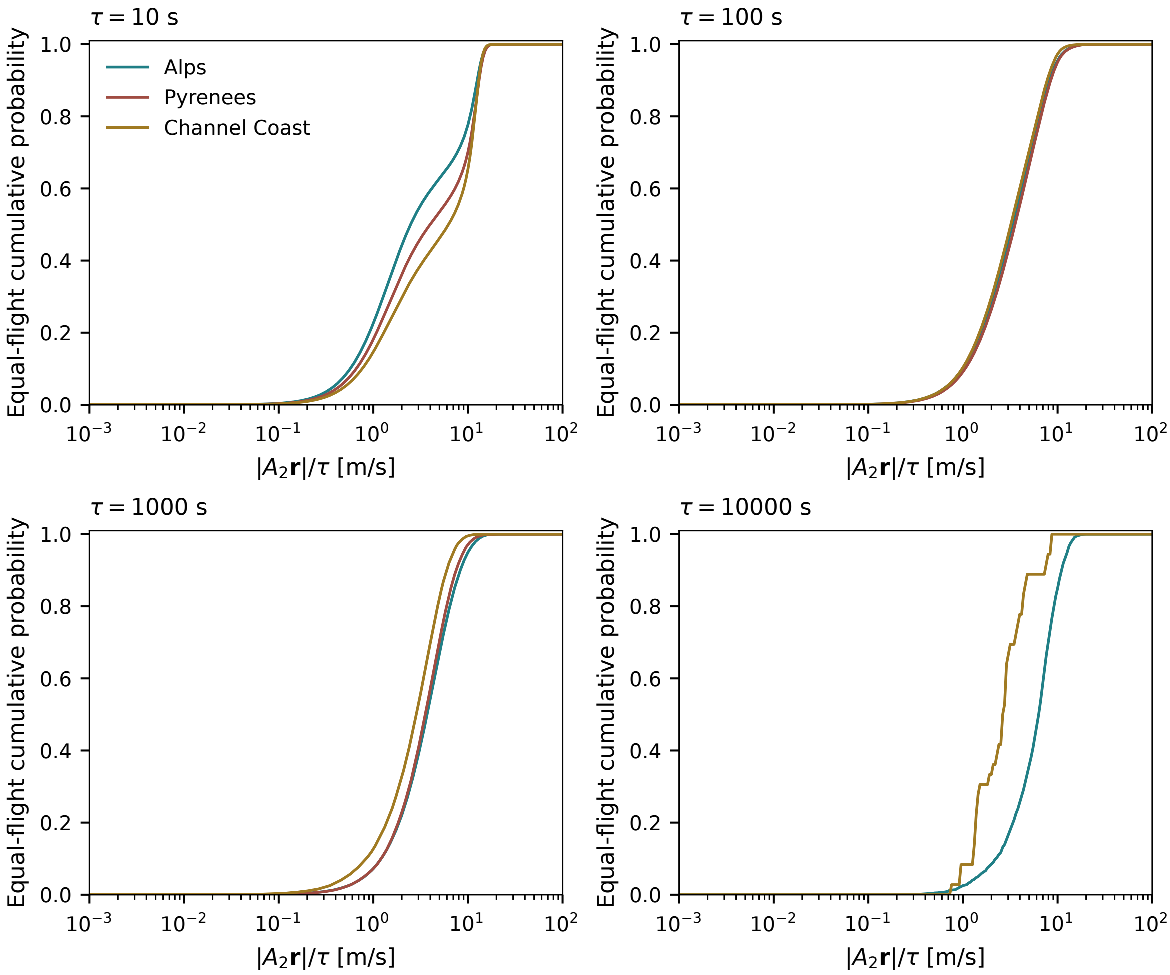


Figure 3.23. Equal-flight distributions of changes of interval-mean velocity at the four PCA display lags, using matched-order support. Curves are accumulated from fixed logarithmic bins; the complete report retains zero and overflow mass. No distribution is drawn for fewer than eight flights. The changing long-lag population must accompany comparisons between panels.

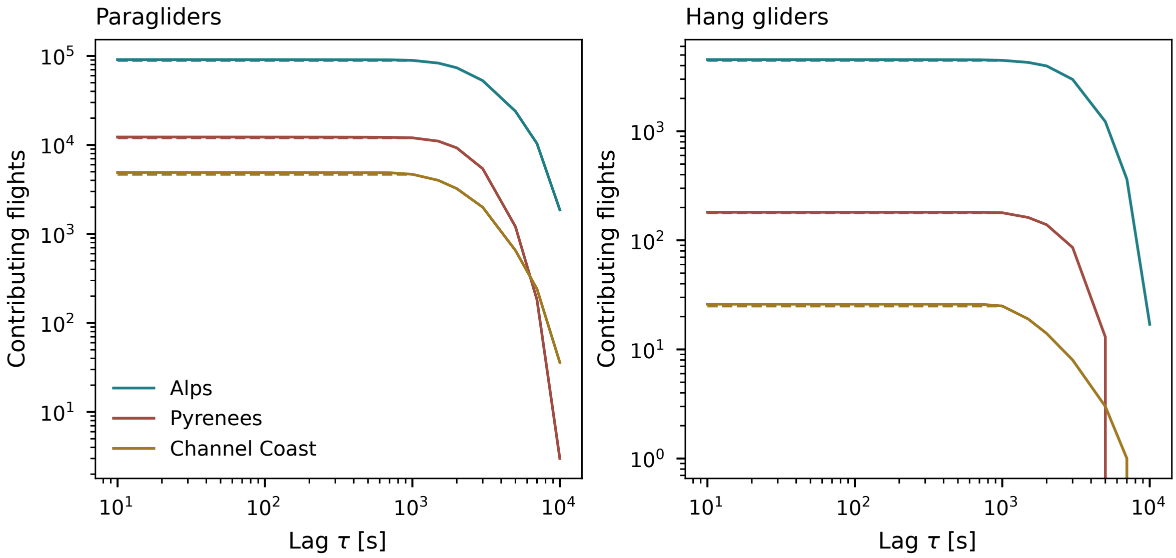


Figure 3.24. Contributing flights for matched orders (solid) and common origins through 1000 s (dashed). A matched observation requires 3τ of continuous support. At 10,000 s the paraglider counts are 1,865, 3 and 36; hang-glider counts are 17, 0 and 0. All regional eligible flights enter where supported.

3.5 Flight duration and equipment composition

Longer flights may select different equipment, pilots and conditions. A change in their MSD can therefore reflect a change of population rather than an effect of observation length. We examine both duration thresholds and their equipment composition, using the identified segment curves from the full archive: 155,085 paraglider and 6,060 hang-glider flights have segments of at least eight fixes with native cadence at most 10 s. Here T_f is the sum of the retained segment spans of flight f; gaps and discarded time are not part of this observed duration.

First combine the available segment curves within each flight, weighting them by the number of admissible time origins, then give every contributing flight equal weight:

$$m_f(\tau) = \frac{\sum_s n_{fs}(\tau) m_{fs}(\tau)}{\sum_s n_{fs}(\tau)}, \qquad M_{T_0}(\tau) = \frac{1}{N_{T_0}(\tau)} \sum_{f: T_f \geq T_0,\, n_f(\tau)>0} m_f(\tau). \tag{3.38}$$

For an eligible segment fs of N_{fs} fixes at interval Δt_{fs}, $n_{fs} = N_{fs} - \mathrm{round}(\tau/\Delta t_{fs})$; it is zero outside that segment's stored lag support, which ends at $\lfloor N_{fs}/2 \rfloor \Delta t_{fs}$. No displacement crosses a segment boundary. This equal-flight average differs from the equal-segment archive average introduced earlier. Requested lags are rounded to each segment's native grid.

Figure 3.26 repeats the duration comparison within the beginners (EN A/B) and experts (EN C/D/CCC) groups. This separates within-class contrasts from changes in their relative abundance.

Without duration conditioning, the fitted slopes over 10 s to 10,000 s are 1.728 for beginners and 1.788 for experts. Table 3.8 compares all eight group–duration cells on 10 s to 1,695 s, ending before half the one-hour threshold. Fit residuals measure departures from the line, independently of sampling uncertainty.

Experts have the larger slope in every duration cell. From all durations to $T \geq 4$ h, the slope rises from about 1.755 to 1.806 for beginners and from 1.803 to 1.836 for experts.

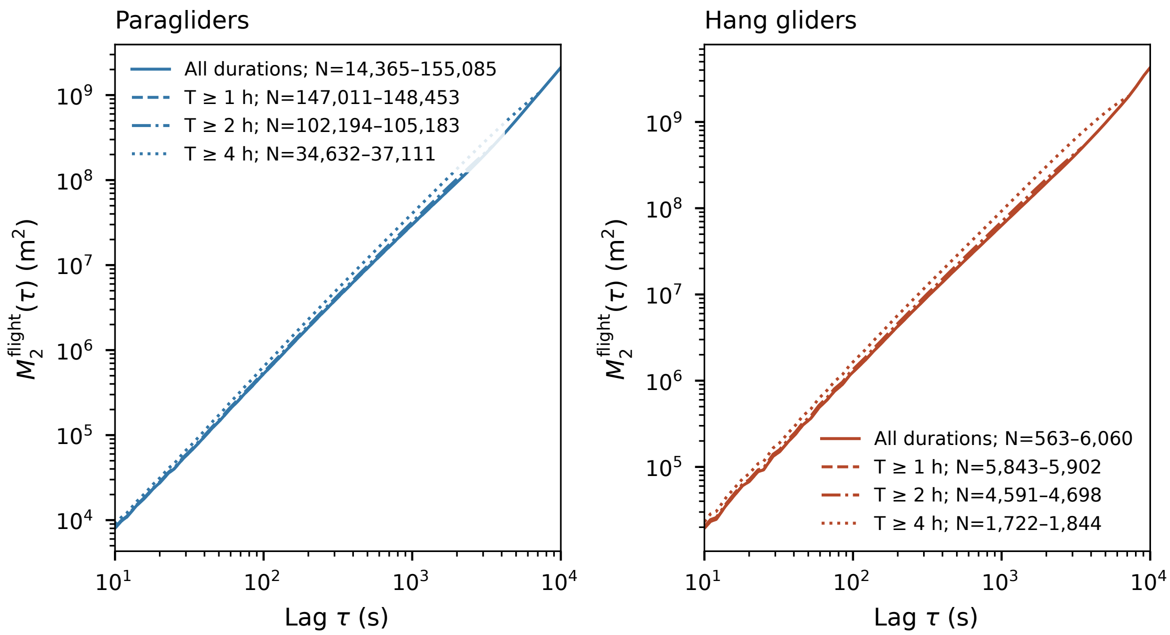


Figure 3.25. Time-averaged squared displacement by retained flight duration. Cohorts are nested; legend counts give the range of contributing flights. Each lag requires at least 30 flights. Threshold curves end at $T_0/2$, while the all-duration curve extends to 10,000 s. Shorter flights progressively leave the reference near $T_0/2$, bringing the populations closer by selection.

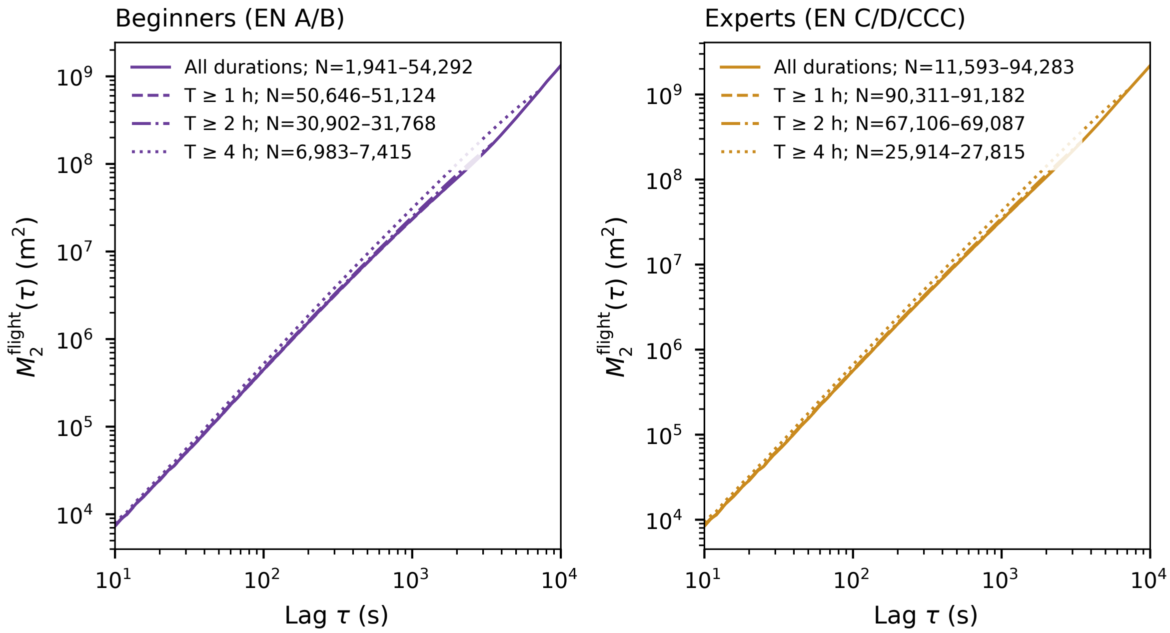


Figure 3.26. The duration comparison repeated for beginners (EN A/B) and experts (EN C/D/CCC), using the same estimator, lag limits and minimum support as Fig. 3.25. Counts refer to distinct flights, and overlapping groups do not constitute independent samples. Tandem, non-certified and unknown equipment is omitted from this contrast.

Table 3.8. Duration and equipment contrasts on a common supported lag interval, 10 s to 1,695 s. Flight counts include every eligible flight in the cell; the contributing count decreases with lag. The residual is the root-mean-square deviation in $\log_{10} M_2$ from the fitted line.

Experience proxy	Duration	Flights	b	RMS residual (dex)
Beginners	All	54,292	1.755	0.022
Beginners	$T \geq 1$ h	51,124	1.754	0.022
Beginners	$T \geq 2$ h	31,768	1.767	0.020
Beginners	$T \geq 4$ h	7,415	1.806	0.017
Experts	All	94,283	1.803	0.017
Experts	$T \geq 1$ h	91,182	1.802	0.017
Experts	$T \geq 2$ h	69,087	1.811	0.016
Experts	$T \geq 4$ h	27,815	1.836	0.015

Duration contrasts thus persist within both classes. The nested cohorts remain dependent and the comparisons descriptive.

To separate within-class behaviour from class proportions, write the exact mixture identity over the two experience proxies,

$$M_{T_0}^{\text{known}}(\tau) = \sum_c p_c(\tau \mid T_0) M_{T_0,c}(\tau), \qquad p_c(\tau \mid T_0) = \frac{N_{T_0,c}(\tau)}{\sum_j N_{T_0,j}(\tau)}. \tag{3.39}$$

The weights depend on lag as well as duration. A standardized comparison instead holds them fixed at $\pi_c = N_{0,c}/\sum_j N_{0,j}$, the class proportions among all eligible known-equipment flights:

$$M_{T_0}^{\text{fixed}}(\tau) = \sum_c \pi_c M_{T_0,c}(\tau). \tag{3.40}$$

Figure 3.27 compares the four-hour/all-duration ratio before and after this adjustment. Both ratios exclude the same unknown or other equipment; otherwise changing the population would

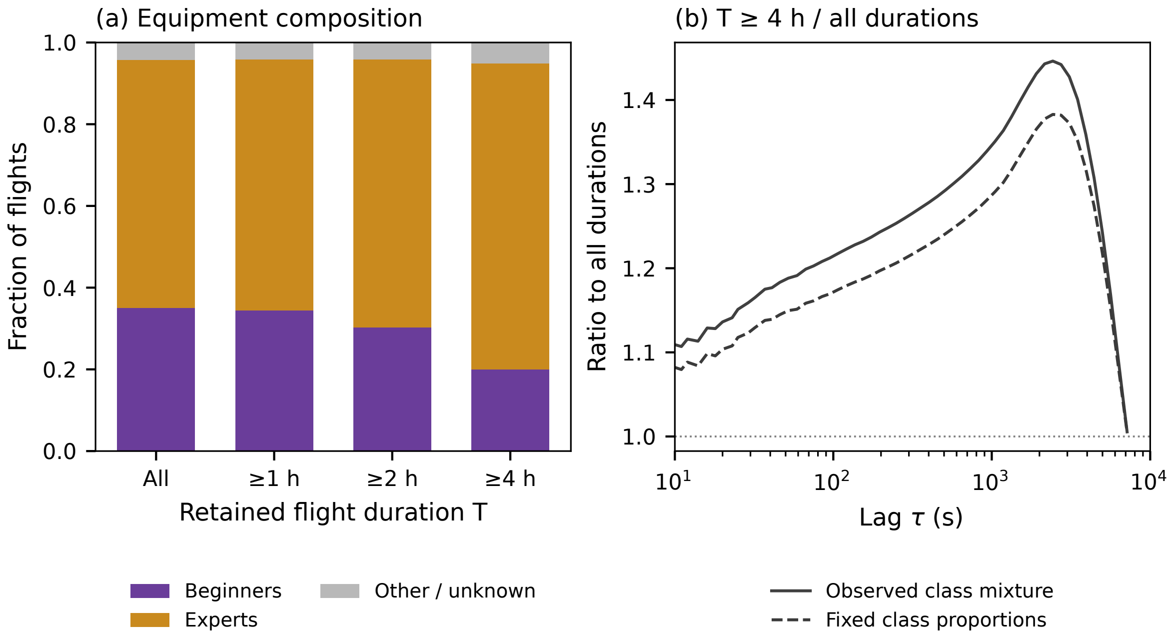


Figure 3.27. Equipment composition and its contribution to the duration comparison. Left: proportions of flights in the nested duration cohorts, including other or unknown equipment (6,510 flights in the full eligible sample). Right: the four-hour/all-duration MSD ratio among known classes, using the observed lag-dependent mixture or fixed class weights. Unity denotes equal MSD amplitudes at that lag, not equal trajectories or established statistical independence.

be confounded with changing the weights. The adjustment is displayed only where every class has at least 30 contributing flights.

Among the two known classes, the expert share rises from about 63.5% in all durations to 78.9% for $T \geq 4$ h. The latter cohort still includes 7,415 beginner-proxy flights. Fixing the class mixture reduces the MSD amplitude contrast:

	Four-hour/all-duration MSD ratio	
Lag [s]	**Observed mixture**	**Fixed class weights**
576	1.30	1.25
1695	1.42	1.35

Composition explains part of the contrast under this adjustment; a substantial within-class difference remains. The return towards unity near 7,200 s coincides with shorter flights leaving the reference population and cannot establish duration independence.

Conditions, routes and pilot selection can still vary within each equipment class, so these associations do not isolate pilot skill or a causal effect of duration.

3.6 Comparison with stochastic models

3.6.1 The moment spectrum and the Lévy-walk benchmark

For $q > 0$, define a moment and its growth exponent by

$$M_q(\tau) = \mathbb{E}[R(t,\tau)^q], \qquad M_q(\tau) \propto \tau^{\zeta(q)}, \qquad \nu(q) = \zeta(q)/q. \tag{3.41}$$

A common scaling law with finite moments gives $\zeta(q) = qH$. Increasing q gives large displacements more weight, so the spectrum tests a different aspect of the distribution from any one quantile. Plain increments are used here: second differences of a straight leg vanish and would change the physical observable being compared.

For an unbiased renewal Lévy walk with independent isotropically chosen headings, constant speed and leg-duration density $\psi(t) \sim t^{-1-\beta}$, the superdiffusive regime $1 < \beta < 2$ predicts

$$\zeta_{\mathrm{LW}}(q) = \begin{cases} q/\beta, & 0 < q < \beta, \\ q + 1 - \beta, & q > \beta, \end{cases} \qquad \zeta_{\mathrm{LW}}(2) = 3 - \beta. \tag{3.42}$$

At $q = \beta$ a logarithmic correction accompanies the common power. The bulk and the rare long legs contribute different branches; their meeting produces strong anomalous scaling [4,28]. This benchmark is an asymptotic model prediction with stated renewal and speed assumptions.

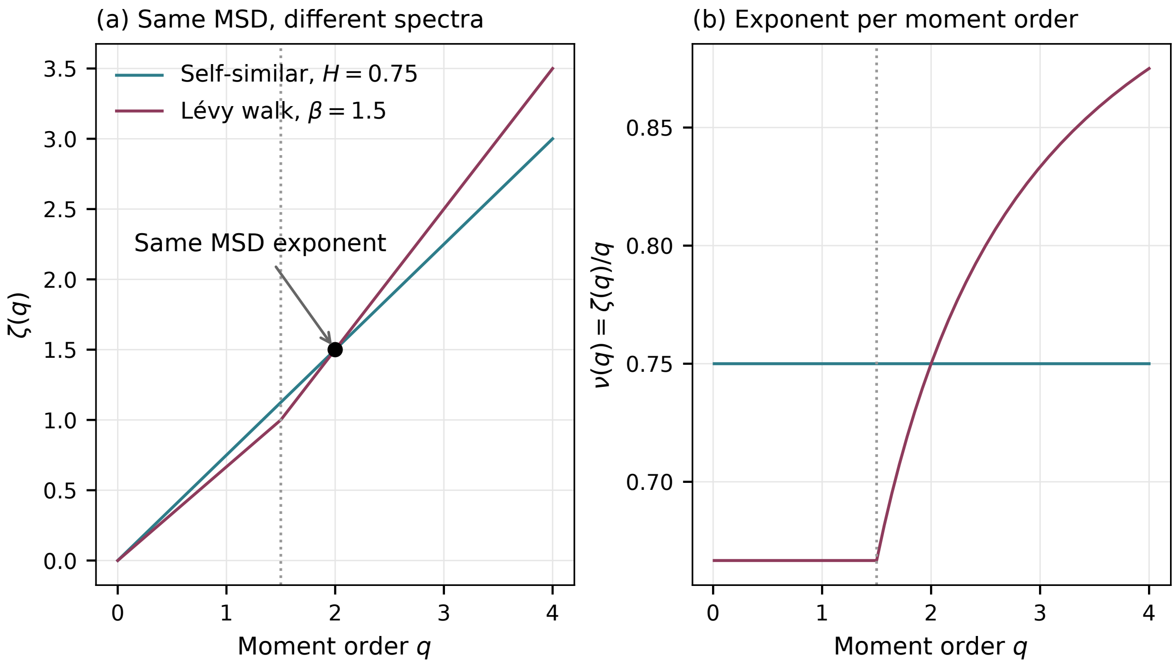


Figure 3.28. Theoretical comparison. A self-similar process with $H = 0.75$ and a standard Lévy walk with $\beta = 1.5$ have the same MSD exponent, $\zeta(2) = 1.5$, but different moment spectra. The dashed vertical line marks $q = \beta$. These are asymptotic predictions, not simulated or empirical estimates.

For $q = 0.25$ to 4, the measured $\nu(q)$ ranges from 0.843 to 0.883 in paragliders and from 0.832 to 0.854 in hang gliders (Fig. 3.29). The pooled values $\zeta(2) = 1.737$ and 1.682 differ from the equal-flight b_1 because the weights differ. The plotted $\beta = 1.5$ curve is a theoretical reference, not a fitted null.

The near-linear spectrum lacks the predicted branch change, challenging this asymptotic benchmark where the observations can resolve it. Ballistic Lévy walks, truncated legs, correlated steps and finite-time crossovers need separate predictions. A quantitative test requires simulations with the observed durations, sampling and preprocessing, compared jointly with moments and controlled quantiles. High-order moments also need flight-level sensitivity checks; the 90th percentile alone need not track the ballistic front.

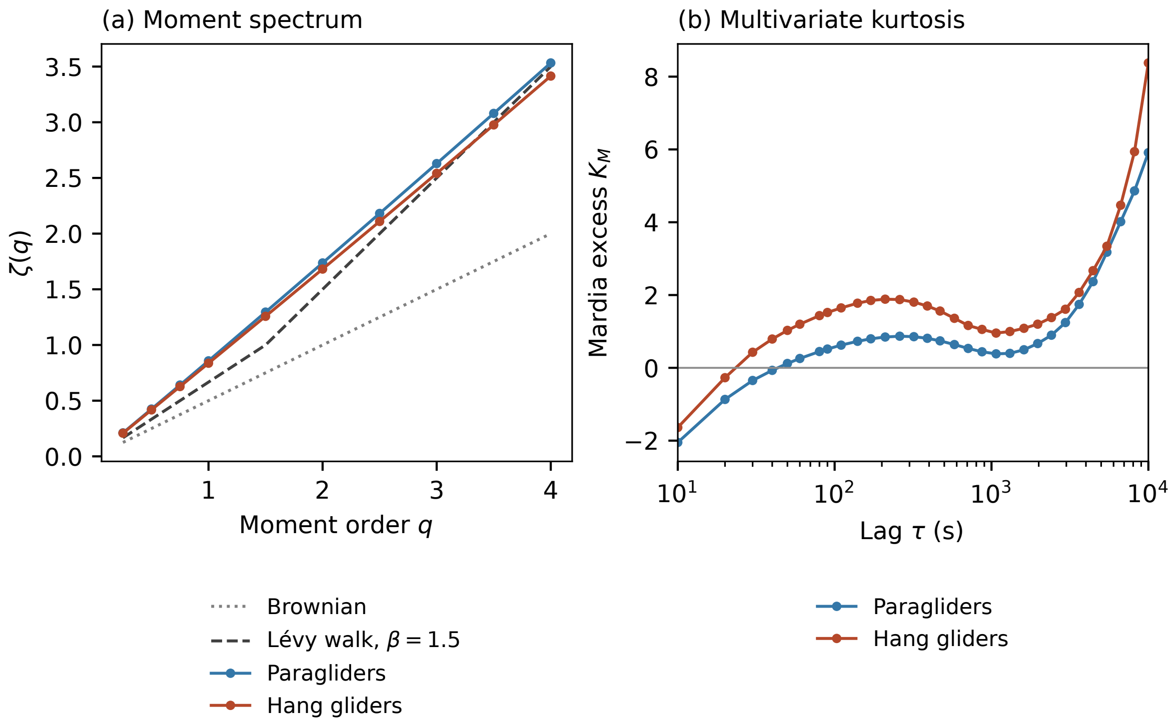


Figure 3.29. Displacement moment spectra and centred multivariate kurtosis in the diagnostic eligible archive, with theoretical moment spectra shown as reference curves. These observables constrain the growth and shape of the increment distribution, respectively, but neither alone identifies the generating process.

3.6.2 Centred kurtosis and heterogeneity

At each lag with a finite, positive-definite covariance, centre the increment and standardise by that full covariance:

$$Q = (\mathbf{X} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{X} - \boldsymbol{\mu}), \qquad K_M(\tau) = \mathbb{E}[Q^2] - d(d+2), \quad d = 2. \tag{3.43}$$

Mardia's kurtosis excess [29] vanishes for a nonsingular Gaussian vector, including an anisotropic one, since $Q \sim \chi^2_d$. Positive excess means a larger fourth moment of standardised radius. It neither requires a power-law tail nor makes zero excess sufficient for Gaussianity.

For the n pooled increments at a lag, the calculation uses

$$\begin{aligned} \widehat{\boldsymbol{\mu}} &= \frac{1}{n}\sum_{i=1}^{n} \mathbf{X}_i, \qquad \widehat{K}_M = \frac{1}{n}\sum_{i=1}^{n} \widehat{Q}_i^2 - 8, \\ \widehat{\Sigma} &= \frac{1}{n}\sum_{i=1}^{n} (\mathbf{X}_i - \widehat{\boldsymbol{\mu}})(\mathbf{X}_i - \widehat{\boldsymbol{\mu}})^\top. \end{aligned} \tag{3.44}$$

Here $\widehat{Q}_i$ uses these empirical moments. With covariance denominator n, $n^{-1}\sum_i \widehat{Q}_i = 2$, so Jensen's inequality gives $\widehat{K}_M \geq -4$. Negative excess is possible, and even a Gaussian sample need not give exactly zero.

The measured K_M is not constant. Across the displayed interval its range is -2.05 to 5.91 for paragliders and -1.64 to 8.38 for hang gliders. The medians over sampled lags are 0.67 and 1.52, respectively; these lag medians are descriptive summaries, not time averages. The change of sign near the shortest lags and the rise where support is smallest discourage replacing the curve with one claim about a tail.

Excess can also arise from heterogeneity even when the conditional motion is Gaussian. For example, let $\mathbf{X} = \sqrt{A}\,\mathbf{G}$, where $\mathbf{G}$ is a centred Gaussian vector with fixed covariance, $A > 0$ is independent, and both required moments exist. Then

$$K_M = d(d+2)\left[\frac{\mathbb{E}A^2}{(\mathbb{E}A)^2} - 1\right]. \tag{3.45}$$

This identity applies to the stated scale mixture; flights with different means and covariance orientations require a fuller calculation. The pooled excess therefore compares the archive with one homogeneous Gaussian law. Conditional Gaussian models need separate centring and covariance estimates within environment or phase groups, with simulations accounting for finite samples and dependence. A peak in pooled kurtosis alone cannot identify a thermal timescale.

3.6.3 Velocity memory across resolutions

A fix-to-fix velocity mixes recorder cadence, smoothing and physical turning. To examine longer scales, define velocity averaged over a duration h,

$$\mathbf{v}_h(t) = \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h}, \qquad C_h(\tau) = \frac{\langle \delta\mathbf{v}_h(t)\cdot\delta\mathbf{v}_h(t+\tau)\rangle}{\langle|\delta\mathbf{v}_h|^2\rangle}. \tag{3.46}$$

We compare $h = 10$, 60 and 300 s. In each retained segment fs, form n_{fs} adjacent velocity blocks and subtract their segment mean. The segment estimate and archive average are

$$\begin{aligned}
\widehat{C}_{fs,h}(kh) &= \frac{(n_{fs}-k)^{-1}\sum_{j=0}^{n_{fs}-k-1}\delta\mathbf{v}_{fs,j}\cdot\delta\mathbf{v}_{fs,j+k}}{n_{fs}^{-1}\sum_{j=0}^{n_{fs}-1}|\delta\mathbf{v}_{fs,j}|^2},\\
\widehat{C}_h(kh) &= \frac{1}{|\mathcal{F}_h(k)|}\sum_{f\in\mathcal{F}_h(k)}\frac{1}{|\mathcal{S}_{f,h}(k)|}\sum_{s\in\mathcal{S}_{f,h}(k)}\widehat{C}_{fs,h}(kh).
\end{aligned} \tag{3.47}$$

Here $\mathcal{S}_{f,h}(k)$ contains the supported segments of flight f:

$$n_{fs} \geq 8, \qquad \sum_j |\delta\mathbf{v}_{fs,j}|^2 > 0, \qquad 1 \leq k \leq \lfloor n_{fs}/4\rfloor.$$

Flights with a nonempty set form $\mathcal{F}_h(k)$. Segments receive equal weight within each flight, then flights receive equal weight. This differs from normalising one pooled covariance.

The separation $\tau = kh \geq h$ avoids overlap between the two velocity blocks. Finite-segment centring can nevertheless depress long-lag correlations.

Figure 3.30 places the positive correlations on logarithmic axes, where a power-law interval would satisfy

$$C_h(\tau) \simeq A_h\tau^{-\gamma_h} \quad\Longrightarrow\quad \log C_h(\tau) \simeq \log A_h - \gamma_h\log\tau. \tag{3.48}$$

The linear-ordinate panels retain zero crossings and negative correlation.

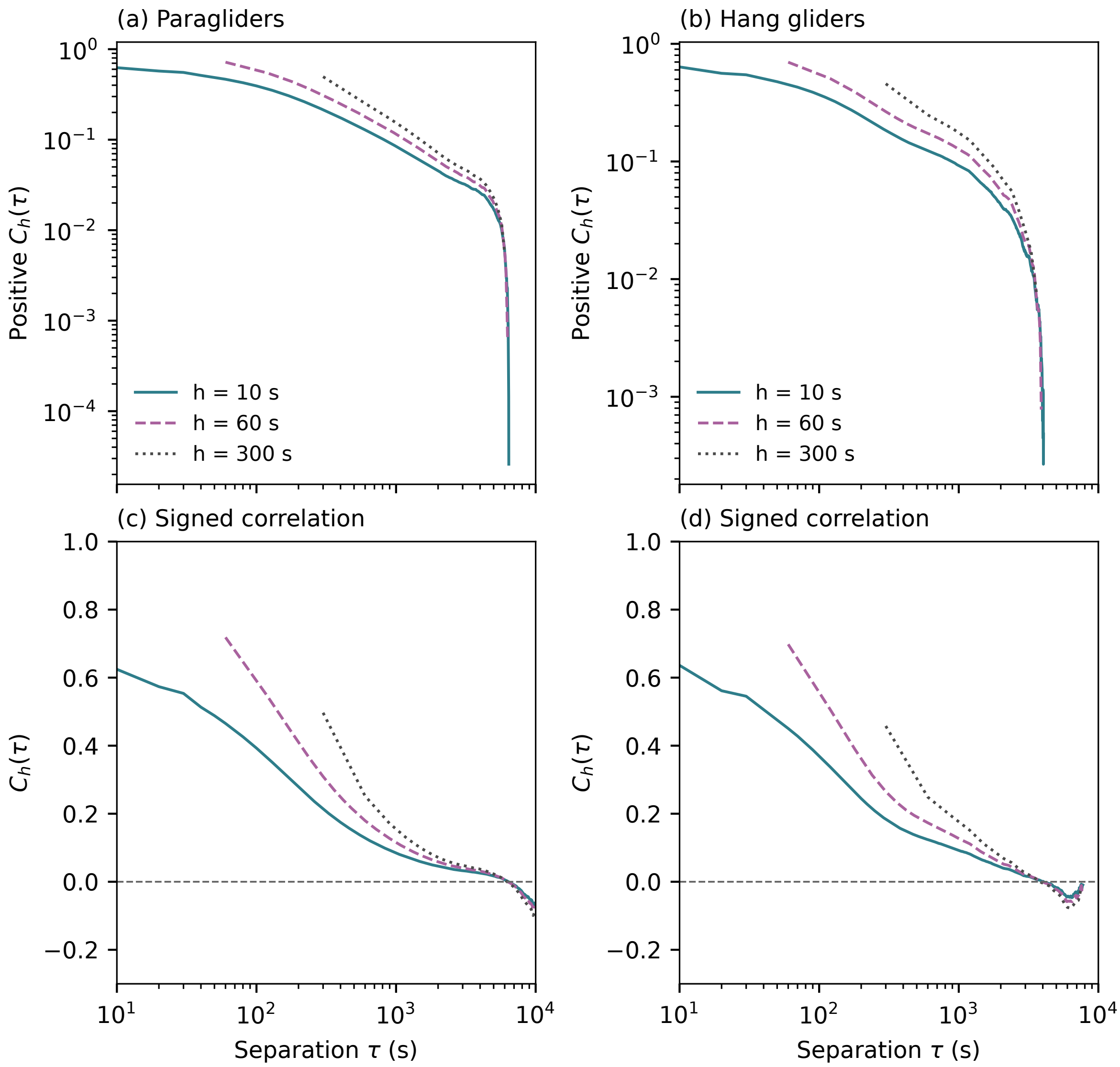


Figure 3.30. Velocity memory at averaging widths h = 10, 60 and 300 s. The upper panels use log–log axes for positive correlations; nonpositive values interrupt the curves. The lower panels show the same estimates with a linear vertical axis, preserving zero crossings and anticorrelation. At least 20 contributing flights are required, with separation no shorter than h and no longer than one quarter of each contributing velocity record. A visually straight part of the upper curves is a possible scaling interval, not evidence of an asymptotic law.

At a ten-minute separation, all three resolutions retain positive sample means:

Width h [s]	**Paraglider** $\widehat{C}_h(600\,\mathrm{s})$	**Hang-glider** $\widehat{C}_h(600\,\mathrm{s})$
10	0.128	0.124
60	0.180	0.173
300	0.252	0.249

No calibrated intervals are reported here. Coarse averaging suppresses fast fluctuations and can increase normalized correlation without lengthening the underlying memory. The log–log curves bend, and estimates in both disciplines turn negative at long separation; one decay exponent does not describe the full interval.

Memory and displacement growth. For a stationary centred velocity with variance $\sigma_v^2 = \langle|\delta\mathbf{v}|^2\rangle$, integrating velocity over the displacement interval and using stationarity gives Taylor's relation [30],

$$M_{2,c}(\tau) = 2\sigma_v^2 \int_0^\tau (\tau - s)C(s)\,\mathrm{d}s. \tag{3.49}$$

The factor σ_v^2 restores the velocity units because C is normalised. Taylor's relation concerns centred displacement in the same stationary population.

For example, an exponentially correlated velocity, $C(s) = e^{-s/t_p}$, gives

$$M_{2,c}(\tau) = 2\sigma_v^2 t_p \left[\tau - t_p(1 - e^{-\tau/t_p})\right]. \tag{3.50}$$

Its limits are

$$M_{2,c}(\tau) \simeq \begin{cases} \sigma_v^2\tau^2, & \tau \ll t_p \quad \text{(ballistic)}, \\ 2\sigma_v^2 t_p\tau, & \tau \gg t_p \quad \text{(diffusive)}. \end{cases}$$

An intermediate superdiffusive slope can therefore arise from finite persistence. Different t_p across flights broaden the crossover. A fitted MSD slope above one does not exclude this model.

An asymptotic tail $C(s) \sim s^{-\gamma}$ with $0 < \gamma < 1$ instead gives $M_{2,c} \propto \tau^{2-\gamma}$. Finite records cannot establish whether the correlation integral diverges. Correlation between successive glides also requires the episode boundaries introduced in Chapter 4.

3.6.4 Model predictions and remaining tests

Table 3.9 collects the useful exclusions and the hypotheses that remain open. The scope is the pooled data over the observed interval. A candidate can fail there and still describe a restricted phase, environment or limiting time regime.

A glider moves while climbing. Treating climb as a waiting phase in a renewal model therefore needs tests of episode durations, displacement coupling and independence in the segmented analysis.

Table 3.9. Model predictions and the scope of the present comparison. "Challenged" means that a displayed diagnostic disagrees descriptively with the simple prediction; it does not stand for a calibrated statistical rejection.

Candidate	**Prediction to confront**	**Present scope**
Homogeneous Brownian motion	$\zeta(q) = q/2$, Gaussian increments, independent disjoint increments	Challenged as a description across the measured interval by displacement growth, shape and memory. A long-time diffusive limit remains untested.
Constant drift plus Brownian motion	$M_2 = \lvert\mathbf{v}_d\rvert^2\tau^2 + 4D\tau$ in 2D; drift-free $V_2 \propto \tau$	A control for apparent superdiffusion. Log–log curvature in V_2 or non-Gaussian residuals require more than a fitted mean velocity; environmental mixtures remain possible.
Homogeneous fractional Brownian motion	Gaussian joint law, common $H_p = H$, $\zeta(q) = qH$	Challenged as a single pooled law when quantile slopes differ and centred kurtosis is nonzero. Conditional Gaussian models are not thereby excluded.
Exponential velocity persistence	Eq. (3.50): ballistic–diffusive crossover	Remains a candidate to test jointly against MSD and coarse velocity correlations; a broad distribution of persistence times is a separate hypothesis.
Standard renewal Lévy walk, $1 < \beta < 2$	Two-branch spectrum in Eq. (3.42)	The missing expected branch change challenges this benchmark. Finite-record simulations are needed before claiming exclusion; generalisations are not covered.
Unbounded Lévy flight	Instantaneous jumps; divergent second moment for stable index below two	Unsuitable as a literal finite-speed flight mechanism. Truncated or effective coarse models require their own finite-scale predictions.

3.7 Constraints for the phase analysis

A model must account jointly for moment growth, changing quantile spread, regional anisotropy, velocity memory, and the contrasts by route, duration and equipment. An effective exponent alone cannot describe this set of observations.

Chapter 4 introduces phase labels. Connecting those phases to global transport requires durations, displacements and dependencies within and between episodes. If c partitions all admissible windows into phase histories, including windows crossing a phase boundary, then

$$M_q(\tau) = \sum_c w_c(\tau) M_q(\tau \mid c), \qquad w_c = \frac{n_c}{\sum_{c'} n_{c'}}, \qquad \sum_c w_c = 1, \tag{3.51}$$

for window weighting. With flight weighting, apply the identity within each flight first. Windows wholly inside individual phases omit transitions and cannot generally reconstruct the global moment. Since both weights and conditional amplitudes depend on lag, global exponents are not weighted phase exponents.

Any exhaustive partition satisfies this identity, including one with incorrect labels. Reconstructing the global moments therefore checks the calculation; independent labels are still needed to assess the phase assignment.

Chapter 4

Flight phases

The global observables of Chapter 3 constrain the motion of a complete flight, but because they mix directed travel, searching and altitude recovery, they do not identify which behaviour contributes to a particular scaling law. Segmentation provides the missing conditional description: given a phase, how does the glider move, how long does that phase last, and what happens next?

The aim here is to assign these phases reproducibly and establish how the labels can be checked, bearing in mind that kinematics provide indirect evidence of behaviour: a coloured climb on a trajectory remains an inference. We first build the probabilistic model and define its measured features, then examine its numerical behaviour and the independent annotations needed to assess its accuracy.

4.1 What the three phases mean

We retain the three-state description used by Vilpellet et al. [2], with labels $\mathcal{S} = \{\mathrm{T}, \mathrm{S}, \mathrm{C}\}$:

Transition (T).
Sustained directed travel between lift regions, usually with limited turning and altitude loss.

Search (S).
Exploratory turns, reversals or recentering without a sustained established climb. Search can be brief or absent between a transition and a climb.

Climb (C).
Sustained exploitation of lift, usually combining positive mean vertical speed and circling.

These definitions concern observable flight geometry rather than a pilot's unrecorded intention, and allow departures from a fixed cycle: a glider may encounter lift directly, leave an unsuccessful search, or interrupt a climb. Ridge soaring and straight flight in rising air may also be ambiguous within this three-state vocabulary. Such cases should remain visible during validation instead of being forced to confirm the proposed interpretation.

4.2 From a Markov chain to hidden-state inference

The phase of a flight is not observed directly: a recorder measures motion, and different behaviours can produce overlapping ranges of speed and turning. A hidden Markov model represents this distinction through two coupled sequences. The latent sequence Q_k describes the state at decision time k, while the observed vector $\mathbf{Y}_k$ carries the measurements associated with

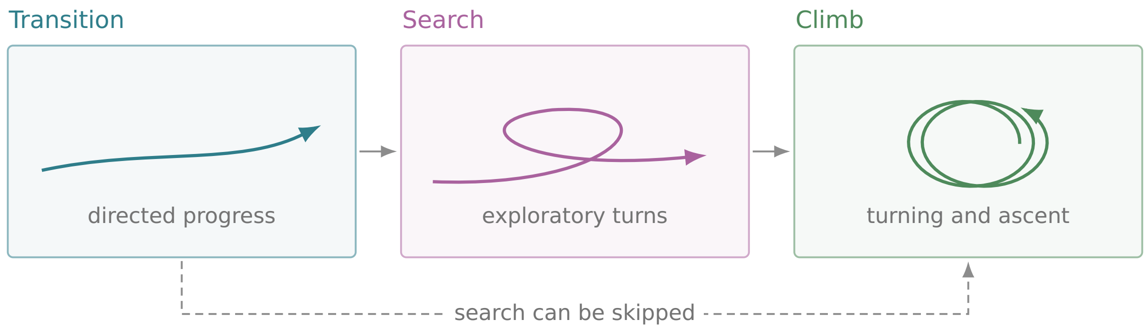


Figure 4.1. Schematic plan views of the three phase interpretations. The drawings are illustrations, not measured trajectories or decision rules. Climbing circles can drift, and searching can cover substantial ground; neither phase is assumed stationary.

that state. The construction and inference below follow the standard HMM formulation [31, Secs. II–III]; its use for behavioural interpretation requires additional care, as discussed by McClintock et al. [32].

4.2.1 Constructing the probability law

Begin with a chain on a finite state space $\mathcal{S}$. Its initial distribution and transition probabilities satisfy

$$\begin{aligned} \Pr(Q_1 = i) &= \pi_i, & \sum_i \pi_i &= 1, \\ \Pr(Q_{k+1} = j \mid Q_{1:k-1}, Q_k = i) &= a_{ij}, & \sum_j a_{ij} &= 1. \end{aligned} \tag{4.1}$$

Here $a_{ij} = \Pr(Q_{k+1} = j \mid Q_k = i)$ and all probabilities are nonnegative. The Markov assumption concerns the hidden state: its present value contains the information about its history used to predict the next state. A constant matrix A adds time homogeneity; it is a modelling choice, not a consequence of the observations being sampled regularly.

Now attach an emission density $b_i(\mathbf{y})$ to each state, with $\int b_i(\mathbf{y})\,\mathrm{d}\mathbf{y} = 1$. Conditional on the complete state sequence, the emissions are assumed independent and the density at time k depends only on Q_k. The chain rule then gives

$$p(q_{1:K}, \mathbf{y}_{1:K} \mid \theta) = \pi_{q_1} b_{q_1}(\mathbf{y}_1) \prod_{k=1}^{K-1} a_{q_k q_{k+1}} b_{q_{k+1}}(\mathbf{y}_{k+1}), \tag{4.2}$$

where θ collects π, A and the emission parameters. This is also a generative recipe: draw the first state, generate its observation, then alternate a state transition with a new observation. Although emissions are independent conditional on the states, the observed sequence is generally correlated and need not itself be first-order Markov.

For continuous features, a common emission model is a multivariate Gaussian,

$$b_i(\mathbf{y}) = \frac{\exp[-\frac{1}{2}(\mathbf{y} - \boldsymbol{\mu}_i)^{\mathsf{T}} \Sigma_i^{-1} (\mathbf{y} - \boldsymbol{\mu}_i)]}{(2\pi)^{d/2} |\Sigma_i|^{1/2}}, \qquad \Sigma_i \succ 0. \tag{4.3}$$

Its full covariance describes dependence among simultaneous features within a state. Gaussianity is an additional assumption about the emission shape; it is not part of the definition of an HMM. Nor does a state number determine a behavioural name: permuting all state labels consistently leaves the observation law unchanged.

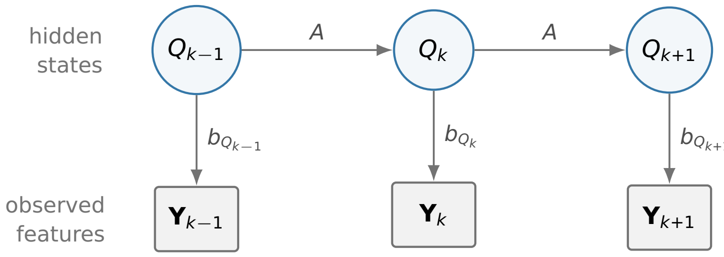


Figure 4.2. Conditional structure of an HMM. Horizontal arrows carry state persistence and transitions; vertical arrows describe how a state generates a measurement. Absence of an arrow between observations represents conditional independence given the hidden sequence, not independence of the observed features over time.

4.2.2 Likelihood and uncertainty about the states

Three tasks must be distinguished: evaluating how well given parameters explain an observed sequence, inferring its hidden states, and estimating the parameters themselves. For the first, the states must be summed out,

$$\mathcal{L}(\theta) = p(\mathbf{y}_{1:K} \mid \theta) = \sum_{q_{1:K} \in \mathcal{S}^K} p(q_{1:K}, \mathbf{y}_{1:K} \mid \theta). \tag{4.4}$$

With continuous observations this is a density evaluated at the recorded vectors, rather than the probability of those exact vectors. Direct evaluation would enumerate $|\mathcal{S}|^K$ paths, but the Markov factorization allows their shared prefixes and suffixes to be accumulated recursively [31, Sec. III-A].

For one valid block of K observations, define forward and backward quantities

$$\alpha_1(i) = \pi_i b_i(\mathbf{y}_1), \qquad \alpha_{k+1}(j) = b_j(\mathbf{y}_{k+1}) \sum_i \alpha_k(i) a_{ij}, \qquad \mathcal{L} = \sum_i \alpha_K(i), \tag{4.5}$$

$$\beta_K(i) = 1, \qquad \beta_k(i) = \sum_j a_{ij} b_j(\mathbf{y}_{k+1}) \beta_{k+1}(j). \tag{4.6}$$

Their cost is $O(K|\mathcal{S}|^2)$, in contrast to enumerating $|\mathcal{S}|^K$ paths. The posteriors used in Baum–Welch are

$$\gamma_k(i) = \frac{\alpha_k(i)\beta_k(i)}{\mathcal{L}}, \qquad \xi_k(i,j) = \frac{\alpha_k(i) a_{ij} b_j(\mathbf{y}_{k+1}) \beta_{k+1}(j)}{\mathcal{L}}. \tag{4.7}$$

Here $\alpha_k(i)$ is the joint density of the observations up to k and state i, whereas $\beta_k(i)$ is the conditional density of the remaining observations given that state. Their product combines evidence on both sides of the decision. Thus $\gamma_k(i)$ is a smoothed state probability, while $\xi_k(i,j)$ gives the expected weight of a transition $i \to j$. Summing γ_k over states, or ξ_k over both indices, gives one; no most-probable path needs to be selected to calculate them.

4.2.3 Decoding a whole sequence

Choosing $\arg\max_i \gamma_k(i)$ separately at every time maximizes the expected number of correct individual states, but those choices need not form the most probable joint sequence and can even violate a forbidden transition. Viterbi instead retains the best path ending in each state [31, 33]. In logarithmic form,

$$d_1(j) = \log \pi_j + \log b_j(\mathbf{y}_1), \tag{4.8}$$

$$d_{k+1}(j) = \log b_j(\mathbf{y}_{k+1}) + \max_i [d_k(i) + \log a_{ij}], \tag{4.9}$$

$$\psi_{k+1}(j) = \arg\max_i [d_k(i) + \log a_{ij}]. \tag{4.10}$$

After choosing $\widehat{q}_K = \arg\max_j d_K(j)$, backtracking through $\widehat{q}_k = \psi_{k+1}(\widehat{q}_{k+1})$ recovers the optimal sequence. Zero probabilities have log score $-\infty$; ties require a consistent convention. The forward recursion adds competing paths, while Viterbi takes their maximum, which explains why a decoded path and posterior uncertainty are different outputs. Both are conditional on the adopted model, so a high posterior does not independently establish that a behavioural interpretation is correct.

4.2.4 Learning from unlabelled observations

If states were observed, transitions could be counted and each Gaussian fitted to its assigned measurements. With hidden states, Baum–Welch replaces those unknown assignments by their posterior weights and iterates the calculation [34]. This is an expectation–maximization procedure: at iteration r, compute

$$\mathcal{Q}(\theta \mid \theta^{(r)}) = \mathbb{E}_{Q \mid \mathbf{y}, \theta^{(r)}}[\log p(Q, \mathbf{y} \mid \theta)], \qquad \theta^{(r+1)} \in \arg\max_{\theta} \mathcal{Q}(\theta \mid \theta^{(r)}). \tag{4.11}$$

The E step uses forward–backward probabilities; the M step maximizes the resulting expected complete-data log likelihood. Its monotonicity follows from the EM lower-bound argument under an exact M step [35], but does not imply a global optimum. For B observation blocks treated as independent in the likelihood and indexed by b, the unregularized Gaussian M step gives the following updates, provided each denominator is positive:

$$\pi_i^{\text{new}} = \frac{1}{B} \sum_{b=1}^{B} \gamma_{b,1}(i), \qquad a_{ij}^{\text{new}} = \frac{\sum_b \sum_{k=1}^{K_b-1} \xi_{b,k}(i,j)}{\sum_b \sum_{k=1}^{K_b-1} \gamma_{b,k}(i)}, \tag{4.12}$$

$$\boldsymbol{\mu}_i^{\text{new}} = \frac{\sum_{b,k} \gamma_{b,k}(i) \mathbf{y}_{b,k}}{\sum_{b,k} \gamma_{b,k}(i)}, \qquad \Sigma_i^{\text{new}} = \frac{\sum_{b,k} \gamma_{b,k}(i) (\mathbf{y}_{b,k} - \boldsymbol{\mu}_i^{\text{new}}) (\mathbf{y}_{b,k} - \boldsymbol{\mu}_i^{\text{new}})^{\mathsf{T}}}{\sum_{b,k} \gamma_{b,k}(i)}. \tag{4.13}$$

Initial-state counts use only the first observation of each block, and transition counts stop at its penultimate observation, so no transition is invented across a gap. These formulas explain the statistical fit; numerical scaling, regularization, initialization and stopping rules must also be specified for the actual estimator. The next sections make those choices for flight observations, rather than treating the general HMM theory as evidence that this particular three-state interpretation is correct.

4.3 From the cleaned flight to observations

The input is the final trajectory of Chapter 2, with local position (E, N, z) and the smoothed velocity and acceleration components. Each preprocessing segment defines a separate observation sequence, within which feature windows and hidden-state transitions cannot cross an acquisition gap or an excluded quality interval.

A common decision interval $\Delta = 10\,\text{s}$ gives the transition matrix the same physical meaning for every flight, restricting the analysis to native cadences no slower than this interval. At each decision time t_k, we first compute each scalar feature at native fixes and linearly interpolate those values. Denoting this interpolant by $\mathcal{I}[f]$, we average it over a centred window of width $W = 30\,\text{s}$:

$$\overline{f}(t_k) = \frac{1}{W} \int_{t_k - W/2}^{t_k + W/2} \mathcal{I}[f](u)\, \mathrm{d}u, \qquad \mathbf{X}_k = (\overline{v}_z, \overline{v}_h, \overline{|\omega|}, C_\omega)(t_k). \tag{4.14}$$

The window suppresses fix-scale fluctuations while preserving manoeuvres over tens of seconds; its width is a chosen measurement scale rather than a measured universal thermalling period. Although boundaries lie on a 10-s grid, the overlapping 30-s windows prevent a claim of independent 10-s temporal resolution.

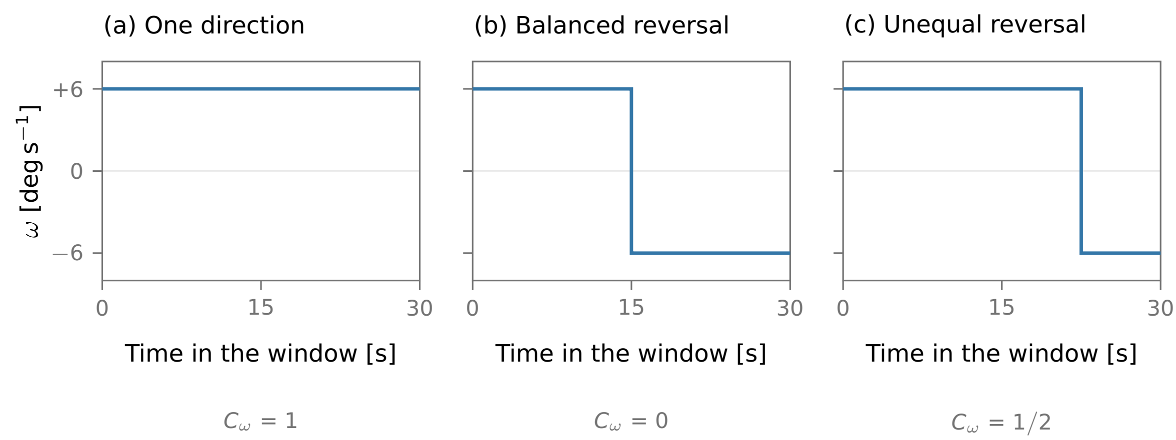


Figure 4.3. Turning amount and directional coherence measure different properties. All three idealized windows have $\overline{|\omega|} = 6$ degrees per second, but their coherence differs because opposite turns cancel in the signed integral. The jumps illustrate limiting waveforms; the data calculation integrates piecewise-linear interpolants of native-fix values as specified in Eqs. (4.14) and (4.16). No phase label follows from coherence alone.

The horizontal speed and signed turn rate are

$$v_h = \sqrt{v_E^2 + v_N^2}, \qquad \omega = \frac{v_E a_N - v_N a_E}{v_E^2 + v_N^2} \quad (v_h \geq 1\,\mathrm{m\,s^{-1}}). \tag{4.15}$$

Below the stated speed floor the implementation sets $\omega = 0$, because numerical heading is unreliable when the horizontal velocity nearly vanishes. The fourth feature measures whether turning retains its sign:

$$C_\omega(t_k) = \frac{\left|\int_{t_k-W/2}^{t_k+W/2} \mathcal{I}[\omega](u)\,\mathrm{d}u\right|}{\int_{t_k-W/2}^{t_k+W/2} \mathcal{I}[|\omega|](u)\,\mathrm{d}u}, \qquad 0 \leq C_\omega \leq 1, \tag{4.16}$$

The denominator is an accumulated angle in radians. The code sets $C_\omega = 0$ if it is at most machine epsilon (2^{-52} in float64), and clips floating-point round-off to $[0, 1]$. Turning in one direction throughout the window gives $C_\omega = 1$ even when its rate varies: coherence is not a measure of circularity or of lift by itself. At a reversal between two fixes, $\mathcal{I}[|\omega|]$ differs from $|\mathcal{I}[\omega]|$: the former is the quantity the code integrates. The pointwise inequality $|\mathcal{I}[\omega]| \leq \mathcal{I}[|\omega|]$ proves the stated bound on C_ω.

For example, equal-duration turns at $+6$ and -6 degrees per second cancel in the numerator and give $C_\omega = 0$, despite a large mean absolute turn rate. Keeping the positive sign throughout gives $C_\omega = 1$; spending three quarters of a window at $+6$ and one quarter at -6 gives $C_\omega = 1/2$ in the ideal piecewise-constant limit. Thus two different questions are measured: how much the heading turns, and how much opposite turns cancel. Figure 4.3 makes this distinction explicit.

A complete safe window is required. Windows touching reconstructed altitude, a derivative influenced by that reconstruction, or an edge derivative from preprocessing are unclassified, as are unsupported feature edges. Influence includes the entire Savitzky–Golay vertical fitting window, including its off-centre boundary fits. The support includes the native fixes bracketing interpolated window endpoints. Consequently even an unsafe fix just outside the nominal integration interval can invalidate it. Grey in the figures means unavailable support; there is no posterior confidence threshold that turns uncertain classifications grey.

Table 4.1. Quantities shown in the phase time-series figures. Feature windows are centred and 30 s wide; the plotted quantities retain physical units, before standardization.

Axis	Mathematical quantity	Unit / range
Altitude	$z(t_k)$ from the cleaned trajectory	m
Vertical speed	$\overline{v}_z(t_k)$, Eq. (4.14)	$\mathrm{m\,s^{-1}}$, signed
Horizontal speed	$\overline{\sqrt{v_E^2 + v_N^2}}(t_k)$	$\mathrm{m\,s^{-1}}$, ≥ 0
Absolute turn rate	$(180/\pi)\overline{\lvert\omega\rvert}(t_k)$	$\mathrm{deg\,s^{-1}}$, ≥ 0
Coherence	$C_\omega(t_k)$, Eq. (4.16)	dimensionless, $[0,1]$
Posterior	$\gamma_k(i)$ for $i \in \{\mathrm{T,S,C}\}$	$[0,1]$, sum $=1$
Phase strip	$\widehat{q}_k$ from the joint Viterbi path	T, S, C or unavailable

4.4 A probabilistic segmentation, and its assumptions

4.4.1 The fitted hidden Markov model

A separate continuous Gaussian HMM is fitted for each discipline, since speed and turning scale differ between paragliders and hang gliders. Write $Q_k \in \mathcal{S}$ for the hidden phase and $\mathbf{Y}_k$ for the features standardized by training-set means and standard deviations. More explicitly, for the n observations actually selected for fitting and feature j,

$$\bar{x}_j = \frac{1}{n}\sum_{k=1}^{n} x_{kj}, \qquad s_j^2 = \frac{1}{n}\sum_{k=1}^{n}(x_{kj} - \bar{x}_j)^2, \qquad Y_{kj} = \frac{X_{kj} - \bar{x}_j}{s_j}. \tag{4.17}$$

A numerically constant feature is rejected; the population divisor n, rather than $n-1$, matches the stored standardizer. The same $\bar{x}_j, s_j$ are used on held-out observations. The model parameters are an initial distribution π, a transition matrix A, and a full Gaussian covariance and mean for each component:

$$a_{ij} = \Pr(Q_{k+1} = j \mid Q_k = i), \quad \sum_j a_{ij} = 1, \qquad b_i(\mathbf{y}) = \mathcal{N}_d(\mathbf{y}; \boldsymbol{\mu}_i, \Sigma_i). \tag{4.18}$$

Full covariance allows simultaneous vertical speed, horizontal speed and turning to be correlated within a phase. It does not model residual correlation *between* successive observations. The factorization in Eq. (4.2) assumes first-order phase memory, time-homogeneous parameters and conditionally independent emissions. The last is only approximate for overlapping windows; posterior confidence may therefore be sharper than independent observations would justify. Gaussian emissions also place some probability on physically impossible feature values, such as negative speed. They are a classification approximation, not a generative model of flight physics.

Baum–Welch applies expectation–maximization to the unlabelled training likelihood [34,35]. Ten fixed-seed initializations are compared by training likelihood. Whole valid blocks enter the fit; the final observation of one block never contributes a transition to the next. The fitted scaler is frozen for validation and test flights.

A numerical component has no behavioural name. At present the component with greatest mean vertical speed is provisionally called climb. Among the other two components, transition maximizes $\mu_{v_h} - \mu_{|\omega|} - \frac{1}{2}\mu_{C_\omega}$ in standardized feature coordinates; the remaining component is called search. Human train labels will resolve this permutation. In particular, "search" currently also collects observations unexplained by the other two components: its behavioural interpretation requires a direct check.

4.4.2 Numerical estimation in the fitted model

Equations (4.12)–(4.13) describe unregularized maximum likelihood. The stored models were fitted with `hmmlearn` 0.3.3 [36]. Its default initial/transition Dirichlet parameters are one, and its mean weight is zero, so those updates reduce to the expressions above. For full covariance the default scalar `covars_prior` is 0.01, broadcast to every matrix entry. With $N_i = \sum_{b,k} \gamma_{b,k}(i)$, the implemented covariance update is therefore

$$\Sigma_i^{\text{new}} = \frac{\sum_{b,k} \gamma_{b,k}(i)(\mathbf{y}_{b,k} - \boldsymbol{\mu}_i^{\text{new}})(\mathbf{y}_{b,k} - \boldsymbol{\mu}_i^{\text{new}})^\mathsf{T} + 0.01\,\mathbf{1}\mathbf{1}^\mathsf{T}}{N_i}. \tag{4.19}$$

The added matrix is rank one: it does not impose a positive lower bound on every covariance eigenvalue. In particular, `min_covar`$= 10^{-3}$ is added to the initial covariance diagonal; it is not enforced as an eigenvalue floor after every M step. This distinction matters for the very narrow coherence component.

Exact unregularized EM cannot decrease its likelihood, but convergence of the parameters to a stationary point requires further regularity conditions; a mixture can approach a singular covariance. The software's stopping flag is also true when the iteration cap is reached or the last likelihood change is negative. That flag in an old artifact is not evidence of convergence. New fits record convergence only for a finite, non-negative last gain below the tolerance, reject non-finite final scores, and still retain all restart scores for inspection. For restart r, write $\ell_r = \sum_b \log \mathcal{L}_b(\widehat{\theta}_r)$ on the same training observations. Its reported relative score is $(\ell_r - \max_u \ell_u)/n \le 0$, in natural-log units per fitted observation, whereas the final EM gain is $\ell^{(m)} - \ell^{(m-1)}$ before normalization. These are numerical fit diagnostics, not accuracy scores. Ten fixed-seed restarts are ranked by finite training likelihood; a restart can be selected without satisfying the stated convergence criterion. Training uses scaled forward–backward arithmetic; current Viterbi and posterior decoding use logarithms to avoid underflow.

4.4.3 Why the current decoder differs from the original fit

The original fit used all four features. Its climb component concentrates coherence extremely close to one: the physical standard deviations are approximately 0.0000889 for paragliders and 0.0000604 for hang gliders (Table 4.3). A circling climb with a small turn reversal can therefore receive negligible climb likelihood. More persistent transitions alone cannot repair this emission problem.

The current decoder integrates coherence out of the original four-dimensional Gaussian. If $\mathbf{U} = (Y_1, Y_2, Y_3)$ and $V = Y_4$, then

$$b_i^{(3)}(\mathbf{u}) = \int_{\mathbb{R}} b_i^{(4)}(\mathbf{u}, v)\,\mathrm{d}v = \mathcal{N}_3(\mathbf{u}; \boldsymbol{\mu}_{i,U}, \Sigma_{i,UU}). \tag{4.20}$$

Thus the retained mean entries and principal covariance submatrix define the new likelihood. This is marginalization, not conditioning on perfect coherence, and not a new three-dimensional fit. Coherence remains recorded and plotted for inspection. A separately refitted three-dimensional model remains a validation comparison: the original parameters were selected using the problematic fourth feature.

The coherence mean and width in Table 4.3 undo the training standardization: $\mu_{C,i}^{\text{phys}} = \bar{x}_4 + s_4 \mu_{i,4}$ and $\sigma_{C,i}^{\text{phys}} = s_4 \sqrt{\Sigma_{i,44}}$. They describe a fitted Gaussian component, not the sample mean and standard deviation of decisions labelled climb.

A second, independent choice softens the phase sequence. With $s_i = a_{ii} < 1$ and $q_{ij} = a_{ij}/(1 - s_i)$ for $j \neq i$, the decoding matrix is

$$\widetilde{a}_{ii} = \widetilde{s}_i, \qquad \widetilde{a}_{ij} = (1 - \widetilde{s}_i)\big[(1 - w)q_{ij} + w q_{ij}^\star\big], \quad j \neq i. \tag{4.21}$$

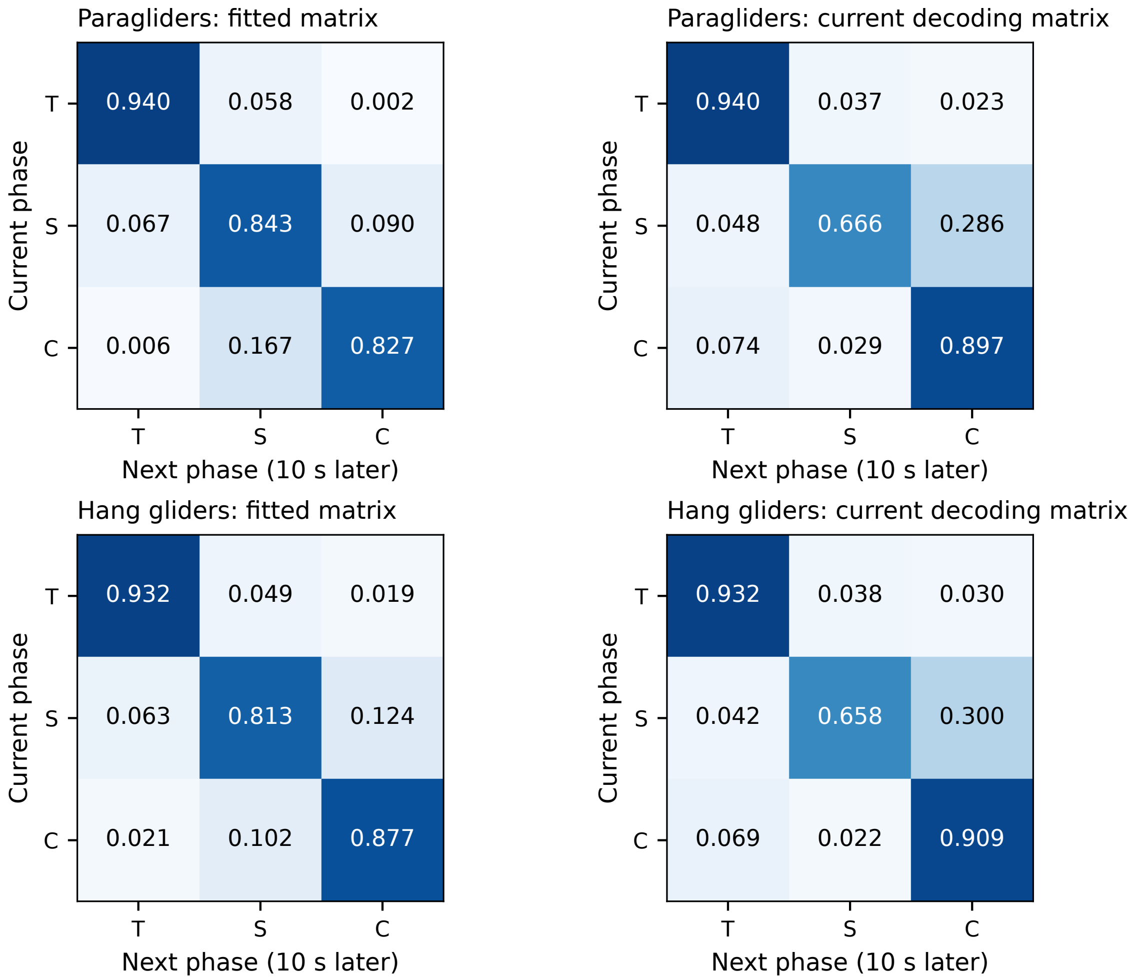


Figure 4.4. Original fitted and current decoding transition matrices, ordered as transition (T), search (S), climb (C). Every entry is a probability per 10-s decision. The right column includes the declared sequence preference and cannot be read as an empirical frequency of pilot behaviour.

Here $w = 0.75$. The target exits from transition split equally between search and climb, allowing direct entry into lift; from search and climb, the next phase in $\mathrm{T} \to \mathrm{S} \to \mathrm{C} \to \mathrm{T}$ receives target probability 0.95. The persistence rule is

$$\widetilde{s}_i = \begin{cases} s_i + w(e^{-\Delta/\theta_S} - s_i), & i = \mathrm{S}, \\ s_i + w\max(0, e^{-\Delta/\theta} - s_i), & i = \mathrm{T}, \mathrm{C}, \end{cases} \qquad \theta_S = 20\,\mathrm{s}, \quad \theta = 120\,\mathrm{s}. \tag{4.22}$$

For an absorbing fitted row ($s_i = 1$), the code starts from equal exit weights and caps the new persistence at $1 - \epsilon_{\mathrm{mach}}$ to retain possible exits. These scales and weights are provisional decoder settings. They are neither fitted behavioural durations nor minimum run lengths. Figure 4.4 shows both matrices so that learned structure and imposed preference can be distinguished. Once a semantic prior is active, changing component names requires decoding again, rather than merely recolouring an existing path.

For both the original and modified model, Viterbi chooses the entire most probable path rather than the most likely phase at each isolated time [33]:

$$\widehat{q}_{1:K} = \arg\max_{q_{1:K}} p(q_{1:K} \mid \mathbf{y}_{1:K}), \qquad \gamma_k(i) = \Pr(Q_k = i \mid \mathbf{y}_{1:K}). \tag{4.23}$$

The hard path and posterior γ use the same effective transition matrix and marginal emission.

Table 4.2. Archive-scale HMM fit and decoding census. A sequence is a contiguous classifiable feature block; restart numbers are one-based.

Discipline	Fit flights	Fit sequences	Fit points	Decoded points	Restart
Paragliders	972	11,598	952,274	171,663,465	6/10
Hang gliders	874	8,775	1,000,000	7,301,634	8/10

Table 4.3. Support and numerical diagnostics of the segmentation. Excluded time uses all retained cleaned-segment duration as its denominator. Coherence widths belong to the original four-dimensional climb component.

Quantity	Paragliders	Hang gliders
Emitted archive decisions	171,663,465	7,301,634
Excluded cleaned duration	8.24%	5.88%
Climb C_ω mean (original fit)	0.9999986	0.9999996
Climb C_ω standard deviation	0.0000889	0.0000604
Final EM log-likelihood change	-0.0755	-2.15
Annotated intervals supplied	0	0

"Smoothed posterior" refers to inference using observations before and after k, not to an extra filter applied to probabilities.

4.5 What has been checked

4.5.1 Archive support and reproducible examples

The original models were fitted and applied to both archives. Table 4.2 records the actual fitting and decoding counts. Fitting uses a deterministic subset of whole training blocks up to the memory cap; longer selected blocks contribute more observations, so this is not equal weighting of flights. The saved archive uses the current decoder. The four-dimensional path is reconstructed from the same fitted parameters as a comparison.The figures in this chapter use the current configuration and saved features, with complete valid blocks decoded before an eight-minute display excerpt is taken. A shortened figure therefore does not shorten the sequence used for inference.

The selected fits do not both satisfy a finite, non-negative last gain below tolerance (Table 4.3). The software stopping flags therefore cannot establish convergence of these fits; covariance and refitting sensitivity remain to be checked.

Coverage has a separate denominator from classification accuracy:

$$f_{\text{unclassified}} = 1 - \frac{\text{duration covered by classified decision cells}}{\text{total retained cleaned-segment duration}}. \tag{4.24}$$

Decision cells $[t_k - \Delta/2, t_k + \Delta/2)$ are clipped at segment boundaries; acquisition gaps contribute to neither duration. For exclusion reason r, the plotted percentage is $100 \sum_k |I_k| 1\{r_k = r\}/D_{\text{clean}}$, with I_k the clipped decision cell and D_{clean} the denominator in Eq. (4.24). Feature-edge status takes precedence over a quality mask if both apply. Segments with native cadence above 10 s contribute their full duration to the cadence category; remaining tails outside any cell form the final category. The categories are disjoint. The denominator does include retained segments too sparsely sampled for classification. The excluded fractions are 8.24% for paragliders and 5.88% for hang gliders, chiefly from unsafe feature support (Fig. 4.5). These masks are shared by the two decoders compared on that snapshot. The feature masks include the propagated reconstruction support of the vertical Savitzky–Golay derivatives.Their

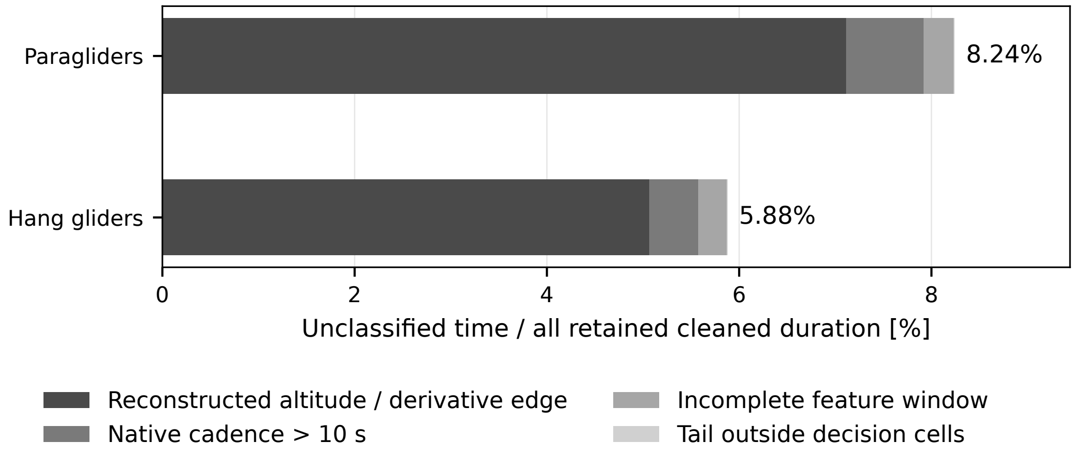


Figure 4.5. Why retained cleaned time lacks a phase label. Fractions use the same duration denominator in Eq. (4.24); they are not fractions of flights or fixes. Windows touching reconstructed altitude or edge derivatives dominate the excluded time.

coverage says how much time is available for subsequent phase statistics, not whether any phase label is correct or whether exclusion is behaviourally unbiased.

The two examples are selected by a fixed priority from validation flights, independently of their apparent phase composition. Recomputing labels with the stored model reproduces the archived paths. Their current labels and posterior curves then use Eqs. (4.20)–(4.22). The audit records the selected identities, model hashes, configuration, and changed-label counts. Thus each plotted label can be traced to its observations and decoder settings.

The paraglider excerpt shows sustained turning and altitude recovery after an initial transition, with a climb label that persists through brief changes in vertical speed and coherence. The hang-glider excerpt is less clear-cut: part of a sustained ascent is assigned to search, and the posterior divides between search and climb. Positive vertical speed is therefore not equivalent to the fitted climb state. This could reflect a difference in turning behaviour, but the figure alone cannot resolve the behavioural name of that component. It gives a concrete interval for human validation rather than an example selected to display apparently perfect classification.

The development example in Fig. 4.8 separates the two decoder changes. On its full preprocessing segment, the four-dimensional reference assigns 6 decisions to climb, the sequence-only variant assigns 6, and the current three-dimensional marginal assigns 32. The comparison shows how the coherence coordinate and sequence preference affect the same observations. The example helped motivate the change, so the comparison is a regression diagnostic, not independent evidence of improved accuracy. Neither a longer green interval nor a smaller number of switches is, by itself, a success criterion.

4.5.2 The manual check that is still needed

The prepared blinded pack contains 40 candidate flights: per discipline, four training candidates, eight validation candidates and eight test candidates. Human reference intervals are needed before reporting accuracy, confusion matrices or a definitive physical meaning for the search state. No human reference intervals were supplied to this audit, so it reports no independent classification scores.The interactive tool is available: select an altitude interval and mark transition, search or climb; edits are saved automatically on the 10-s decision grid. The plan view and kinematics are visible, while model predictions are hidden. The pack must match

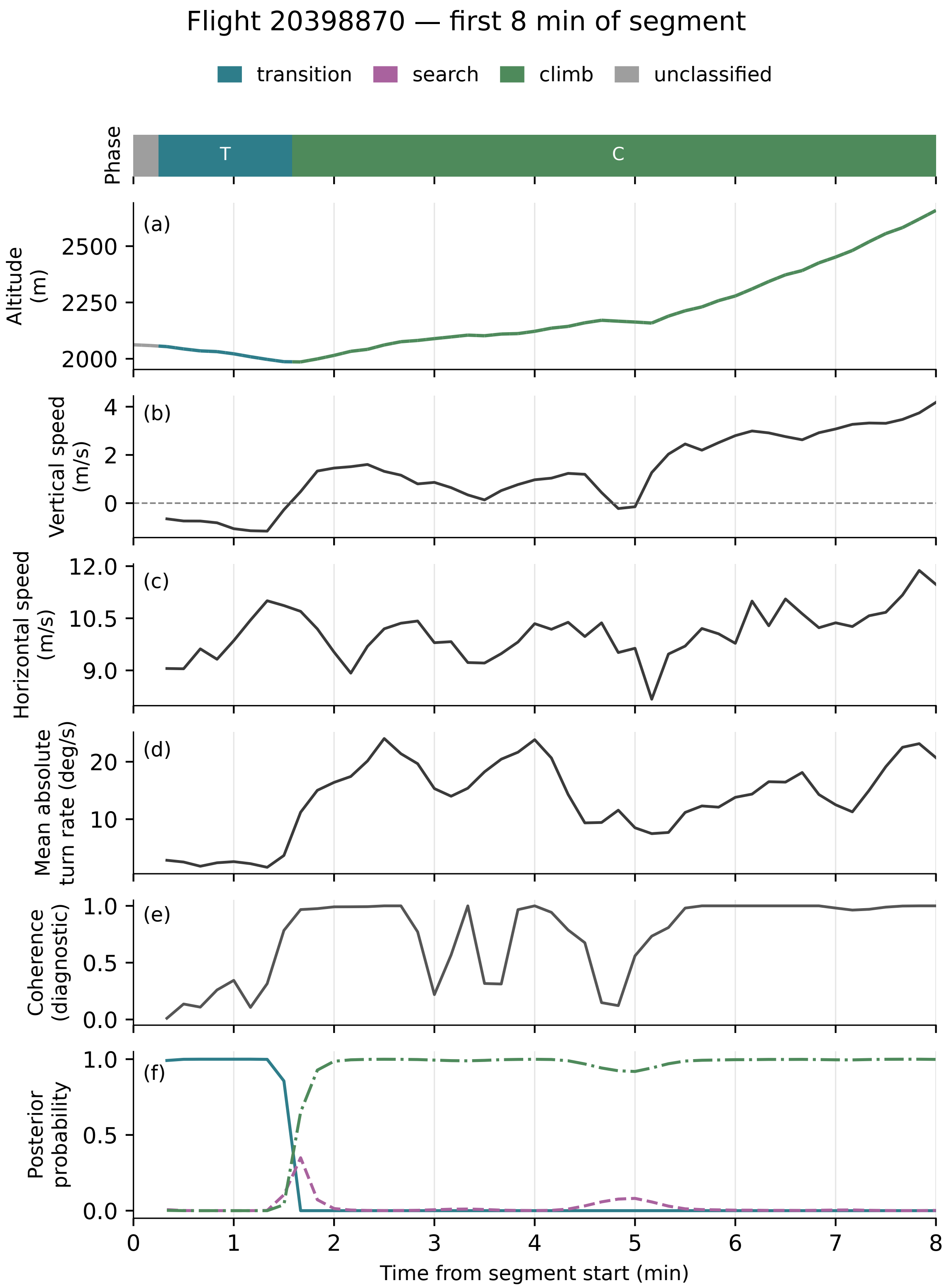


Figure 4.6. Eight minutes of a validation paraglider segment, decoded using the current three-feature marginal and sequence policy. The phase strip, altitude, separate feature axes and posterior probabilities share a time axis marked at one-minute intervals; decisions remain 10 s apart. Table 4.1 defines every displayed quantity. Coherence is shown as a diagnostic and does not enter the current emission likelihood. Grey indicates unavailable support. Phase names remain provisional.

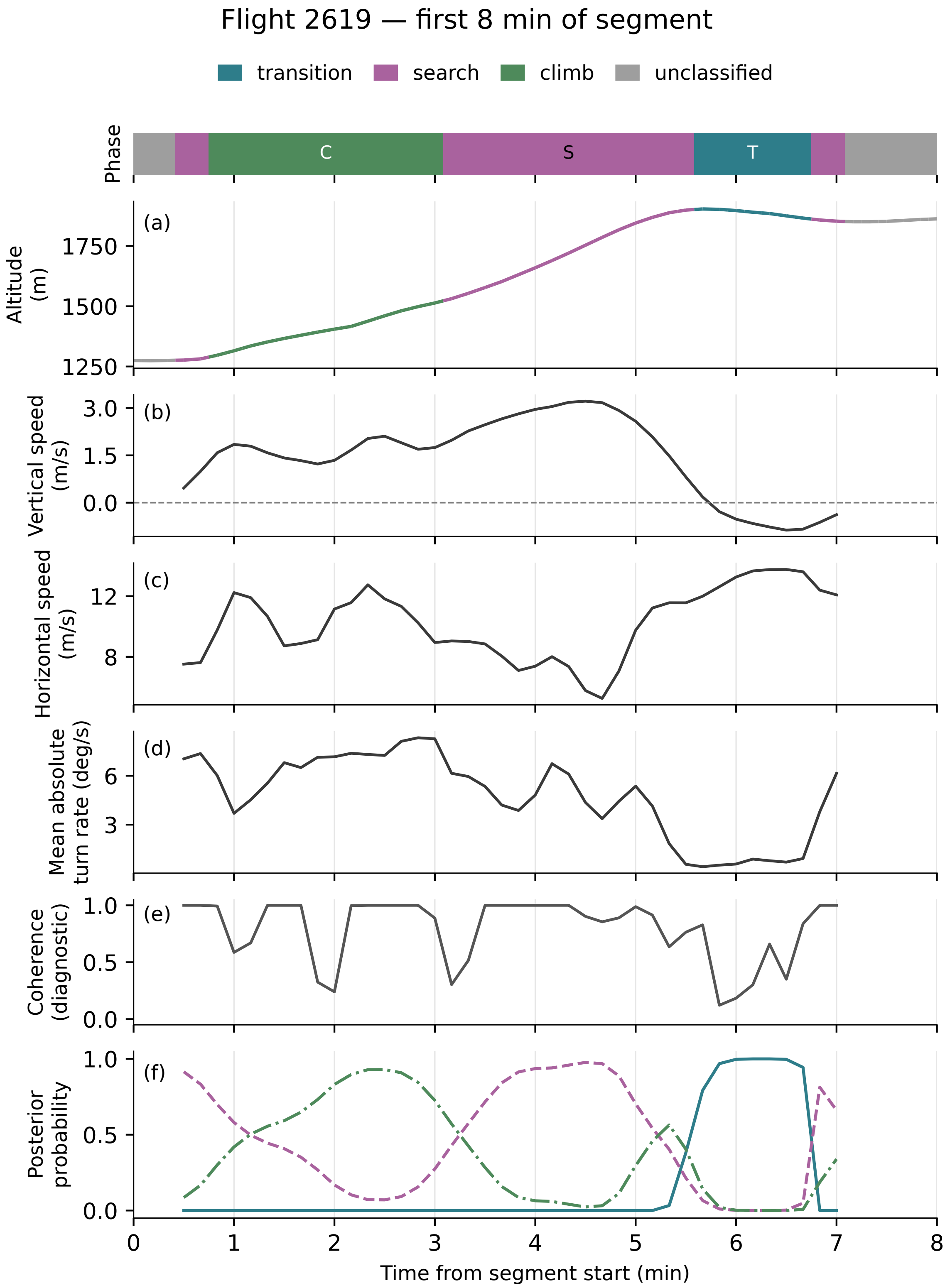


Figure 4.7. The corresponding eight-minute hang-glider validation excerpt. It uses its own fitted model and standardizer, with the same feature definitions and display convention as Fig. 4.6. A high posterior expresses certainty within the adopted model; it is not an independently measured probability of correct behaviour.

Figure 4.8. An eight-minute development excerpt from flight 20311250. The aligned phase strips compare the fitted four-dimensional reference, a change of sequence policy alone, and the current three-dimensional marginal with that policy. The altitude and vertical speed provide physical context. No human reference labels are present.

the cleaned archive used for evaluation; the regeneration procedure creates a new dated pack while preserving the previous pack and any labels already written.

Clearly interpretable intervals should be labelled across the whole candidate window; ambiguous boundaries can remain unlabelled. The fraction left unlabelled must accompany the scores, since evaluating only easy interiors would otherwise conceal the difficult part of segmentation. A second independent annotation of a subset will measure human agreement and boundary ambiguity. That agreement study has not yet been performed.

The deterministic 70/15/15% split operates at flight level, so no flight is shared between fitting and nominal holdout sets. This alone does not guarantee an untouched test: development flights 20275040 and 20279877 were inspected despite belonging to the nominal test partition. They, and hang-glider example flight 975, are excluded from the prepared annotation test pack and must remain outside final accuracy claims. Repeated pilots, days or sites can also make flights statistically related; generalization to a new region or season requires an additional grouped holdout, rather than being inferred from a random flight split.

Train annotations resolve state naming; validation annotations compare the original four-dimensional model, the marginal decoder with and without the sequence preference, and a refitted three-dimensional model. Model choices are then frozen before test scores are opened. For human label i and inferred label j, let n_{ij} count classified, annotated decisions. The core metrics are

$$\begin{aligned} \text{Precision}_i &= \frac{n_{ii}}{\sum_j n_{ji}}, & \text{Recall}_i &= \frac{n_{ii}}{\sum_j n_{ij}}, \\ F_{1,i} &= \frac{2n_{ii}}{\sum_j n_{ji} + \sum_j n_{ij}}, & F_{1,\text{macro}} &= \frac{1}{3}\sum_i F_{1,i}. \end{aligned} \tag{4.25}$$

If a precision, recall or F1 denominator is zero, the evaluator assigns that metric zero and retains the phase in the three-state macro average. The corresponding accuracy is $\text{Acc} = \sum_i n_{ii} / \sum_{ij} n_{ij}$ on the same annotated, classified decisions; the evaluator requires at least one annotated, classified decision. On that nonempty set all these classification scores lie in $[0, 1]$. The confusion matrix reveals whether search is systematically confused with climbing or transition; macro-F1 gives a rare phase the same weight as a frequent one. Accuracy alone can be dominated by the most occupied phase. Bootstrap intervals [21] obtained by resampling whole flights preserve within-flight dependence, although they do not remove dependence between related flights. The existing evaluator provides these classification metrics. Boundary errors, annotation coverage and between-observer agreement remain additional validation measurements to complete; no values are supplied before labels exist.

4.6 What the segmentation will let us measure

The next empirical chapter will characterize all three phases before selecting a stochastic transport model. For a phase occurrence $I = [a, b)$, useful primitive quantities are its duration $D = b - a$, horizontal path length L, net displacement R, and altitude change:

$$L = \int_a^b v_h(t)\,\mathrm{d}t, \qquad R = \|\mathbf{r}(b) - \mathbf{r}(a)\|, \qquad \Delta z = z(b) - z(a), \qquad \mathcal{P} = R/L \in [0,1] \quad (L > 0). \tag{4.26}$$

Here $\mathbf{r} = (E, N)$ and $v_h = \|\dot{\mathbf{r}}\|$ refer to the same horizontal curve; this consistency is needed for the triangle inequality $R \leq L$. The discrete implementation must use a matching position-based path length rather than assume that separately smoothed derivatives satisfy the inequality exactly. The ratio $\mathcal{P}$ distinguishes local circling from directed progress without assuming that either occurs in a particular phase. The ground displacement of a climb can be substantial in wind; search is not automatically an immobile waiting state. Distributions of $(D, L, R, \Delta z)$ and phase-conditioned velocity correlation will quantify those differences.

A phase-conditioned MSD requires an explicit conditioning convention. One useful choice keeps the *entire* displacement interval inside a single occurrence of phase i:

$$M_i(\tau) = \frac{\sum_I \int_{a_I}^{b_I - \tau} \|\mathbf{r}(t+\tau) - \mathbf{r}(t)\|^2\,\mathrm{d}t}{\sum_I (D_I - \tau)_+}, \qquad I \text{ an occurrence of phase } i. \tag{4.27}$$

Integrals for $D_I \leq \tau$ contribute zero, and the observable is defined only when its denominator is positive. This weights eligible time origins; long occurrences contribute more, and short occurrences disappear at large lags. A plot must therefore show the contributing occurrences and durations, and be checked against equal-occurrence or equal-flight weighting. Conditioning only the two endpoints would define a different observable because it could cross other phases.

Prior phase-resolved results [2] supply comparisons, not values that the segmenter must be tuned to recover. Agreement in phase MSDs or time budgets cannot independently validate a classifier designed around the same kinematic expectations. Differences in sampling, geography, wind and the phase definitions must be considered before attributing a disagreement to the dynamics.

4.6.1 Keeping model assumptions separate from measured physics

An HMM imposes a geometric latent dwell-time law. With decoding persistence $\widetilde{a}_{ii}$, its model prediction is

$$\Pr(D_i = m\Delta) = \widetilde{a}_{ii}^{\,m-1}(1 - \widetilde{a}_{ii}), \qquad \mathbb{E}[D_i] = \frac{\Delta}{1 - \widetilde{a}_{ii}}. \tag{4.28}$$

for $m = 1, 2, \ldots$ and $0 \leq \widetilde{a}_{ii} < 1$. A Viterbi run-length histogram is not a sample of observed physical durations independent of this assumption. It depends on the emissions and the decoder as well as on behaviour. Runs cut by a gap, quality mask or feature boundary are censored; merely discarding them can bias the surviving duration distribution. Testing non-geometric phase durations requires human reference intervals and a sensitivity analysis against a model with explicit duration distributions, such as a hidden semi-Markov model [37]. The imposed exit preferences likewise prevent interpreting the current decoded phase order as unqualified evidence of physical memory.

The later transport model must reproduce both these phase observables and the global constraints of Chapter 3. In particular, the duration of a moving leg and its displacement should be studied jointly. A constant-speed Lévy walk couples a leg's length to that *same leg's travel time*; dependence between its length and a *subsequent* climb or search is a separate question. A stereotyped phase cycle by itself also does not disprove renewal at the level of complete cycles. Renewal requires independence of successive cycle variables, which must be tested at the chosen event scale.

4.7 Chapter summary

The chapter has built the hidden-state model from its probability law, connected each feature to a measured kinematic quantity, and separated parameter fitting from sequence decoding. The current labels use a three-feature marginal of a fitted four-feature Gaussian HMM and an explicit sequence preference. Coverage, numerical diagnostics, short trajectory excerpts and a decoder comparison make those choices inspectable, without supplying an independent estimate of behavioural accuracy. The current archive leaves 8.24% of retained paraglider time and 5.88% of hang-glider time unclassified. The narrow coherence components and negative final likelihood changes remain numerical limitations, while the ascent labelled search in the hang-glider example exposes a behavioural ambiguity that posterior probabilities cannot settle.

The next step is the independent phase validation. Once its errors and boundary uncertainties are known, phase durations, displacements and correlations can be used alongside the global observables to test a physical transport model.

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Appendices

Appendices 2.A and 2.B supply the spectral and geodetic definitions used in Chapter 2. Appendix 3.A gives the squared-displacement distributions supporting Chapter 3.

Appendix 2.A

Power spectral density and Welch's method

The spectra in Chapter 2 compare fluctuations in recorded altitude. They contain both flight dynamics and measurement error. A spectral estimate does not separate those contributions without further assumptions.

2.A.1 From covariance to a spectrum

Let $x_n = z_n - \mathbb{E}[z_n]$ be a real, uniformly sampled, second-order stationary process, with sampling interval $\Delta t = 1/f_s$ and covariance $C_k = \mathbb{E}[x_n x_{n+k}]$. When the covariance is summable, its two-sided spectral density is

$$S_2(f) = \Delta t \sum_{k=-\infty}^{\infty} C_k e^{-2\pi i f k \Delta t}, \qquad |f| \leq f_s/2. \tag{2.A.1}$$

For real signals S_2 is even. The one-sided density doubles the positive-frequency contribution, giving

$$\int_0^{f_s/2} S_1(f)\, df = C_0 = \mathrm{Var}(x_n). \tag{2.A.2}$$

Its units are $\mathrm{m}^2\,\mathrm{Hz}^{-1}$ for an altitude signal. The Nyquist frequency $f_s/2$ limits distinguishable sampled frequencies: physical motion above that frequency may be aliased into the observed band.

A full flight altitude series need not be stationary. Equation (2.A.2) therefore gives the stationary reference interpretation; it is not a claim that the variance of an entire climbing and descending flight equals the integral of its detrended Welch estimate. Here the spectrum is a finite-record diagnostic, computed after detrending shorter windows.

2.A.2 The estimator used here

For window j of length L, subtract its least-squares linear trend and denote the residuals by $y_{j,n}$. With a periodic Hann taper w_n, define

$$I_j(f_k) = \frac{\left|\sum_{n=0}^{L-1} w_n y_{j,n} e^{-2\pi i k n/L}\right|^2}{f_s \sum_{n=0}^{L-1} w_n^2}, \qquad f_k = k f_s/L. \tag{2.A.3}$$

This is the two-sided normalisation. For a one-sided estimate, positive bins are doubled except for the Nyquist bin when L is even; the zero-frequency bin is also left unchanged. Welch's estimate averages the resulting periodograms over K overlapping windows [38]:

$$\widehat{S}_1(f_k) = \frac{1}{K}\sum_{j=1}^{K} I_{j,1}(f_k). \tag{2.A.4}$$

The implementation uses $L = 256$, half-window overlap, linear detrending and the arithmetic mean within each flight. The frequency spacing is f_s/L; tapering broadens the effective resolution. Details of the eligible intervals and the separate aggregation across flights are given in Choice of altitude channel.

An individual periodogram has substantial sampling variability. Under suitable stationarity and weak-dependence conditions, averaging independent windows reduces its variance approximately by K. Overlapping windows are correlated, so their effective number is smaller. Increasing L improves frequency resolution but leaves fewer windows to average; keeping L fixed also retains smoothing bias as the record grows. The mean Welch estimator is not, by itself, resistant to arbitrarily large outliers.

Linear detrending suppresses slow variation within each window; it does not remove all physical motion or isolate sensor noise. Likewise, a high-frequency plateau can be compatible with measurement noise without uniquely identifying its cause. Comparisons between GNSS and barometric spectra must account for missing-channel selection, sampling, interpolation and the response of any subsequent filter.

Appendix 2.B

Geodetic transforms: from (φ, λ, h) to ECEF to ENU

This appendix derives the two formulas that Sec. 2.7.5 uses as given: the geodetic-to-ECEF transform, Eq. (2.20), and the ECEF-to-ENU rotation matrix. It closes with the size of the tangent-plane error quoted there. Notation as in the body: WGS84 semi-major axis a, flattening f, $b = a(1-f)$ the semi-minor axis, $e^2 = f(2-f) = 1 - b^2/a^2$ the first eccentricity squared [15].

In the exact transform, h denotes ellipsoidal height. The GNSS channel in an arbitrary HPG record need not follow this convention (Sec. 2.7.1). For orthometric height H, the relation is $h = H + \mathcal{N}(\varphi, \lambda)$, where $\mathcal{N}$ is geoid undulation. The pipeline currently inserts the recorded GNSS value as a height proxy; it does not infer a datum or apply a geoid correction. A height error δh perturbs a horizontal displacement from the origin by approximately $|\delta h| d/R_\oplus$ on a short, level route. This is a sensitivity estimate, not an established error bound for every recorder. Moreover, a spatially varying geoid correction does not cancel exactly from altitude increments.

2.B.1 Geodetic coordinates to ECEF

The ECEF frame is Cartesian, centred at the Earth's centre O: the Z axis along the polar axis, the X axis through the intersection of the equator and the prime meridian, Y completing the right-handed triad. By symmetry, a point at longitude λ lies in the meridian plane obtained by rotating the X–Z plane by λ; writing p for the distance from the polar axis, its ECEF coordinates are

$$X = p\cos\lambda, \qquad Y = p\sin\lambda. \tag{2.B.1}$$

The problem therefore reduces to the meridian ellipse: find (p, Z) of a point at geodetic latitude φ and height h.

The surface point. The meridian section of the ellipsoid is the ellipse

$$\frac{p^2}{a^2} + \frac{Z^2}{b^2} = 1. \tag{2.B.2}$$

The geodetic latitude is defined through the surface *normal* (Figure 2.12). The gradient of the left-hand side of (2.B.2) is proportional to $(p/a^2,\ Z/b^2)$, so the outward normal at a surface point (p_s, Z_s) makes an angle φ with the equatorial plane given by

$$\tan\varphi = \frac{Z_s/b^2}{p_s/a^2} \qquad \Longrightarrow \qquad Z_s = \frac{b^2}{a^2}\, p_s \tan\varphi = (1-e^2)\, p_s \tan\varphi. \tag{2.B.3}$$

Substituting (2.B.3) into (2.B.2), with $b^2 = a^2(1 - e^2)$,

$$\frac{p_s^2}{a^2}\Big(1 + (1 - e^2)\tan^2\varphi\Big) = 1 \quad \Longrightarrow \quad p_s^2 = \frac{a^2\cos^2\varphi}{\cos^2\varphi + (1 - e^2)\sin^2\varphi} = \frac{a^2\cos^2\varphi}{1 - e^2\sin^2\varphi}, \tag{2.B.4}$$

that is,

$$p_s = N(\varphi)\cos\varphi, \qquad N(\varphi) \equiv \frac{a}{\sqrt{1 - e^2\sin^2\varphi}}, \tag{2.B.5}$$

and, back through (2.B.3),

$$Z_s = (1 - e^2)\, N(\varphi)\sin\varphi. \tag{2.B.6}$$

The quantity $N(\varphi)$ so defined is the *prime-vertical radius of curvature*; a computation shows that, away from the poles, it is the distance along the normal from the surface point to the polar axis (the segment Q–P of Figure 2.12): the normal direction in the meridian plane is $(\cos\varphi, \sin\varphi)$, and following it inward from (p_s, Z_s) a length N lands on the axis, $p_s - N\cos\varphi = 0$ by (2.B.5). At the poles the normal lies on the polar axis, so this intersection construction is degenerate; the formula for N remains defined by continuity.

Adding the height. The height h of a fix is measured along the same normal, whose unit vector in ECEF components is

$$\hat{\mathbf{u}} = (\cos\varphi\cos\lambda,\ \cos\varphi\sin\lambda,\ \sin\varphi). \tag{2.B.7}$$

Adding $h\,\hat{\mathbf{u}}$ to the surface point (2.B.5)–(2.B.6) in the meridian plane and restoring the longitude via (2.B.1):

$$X = (N + h)\cos\varphi\cos\lambda, \qquad Y = (N + h)\cos\varphi\sin\lambda, \qquad Z = \big(N(1 - e^2) + h\big)\sin\varphi, \tag{2.B.8}$$

which is Eq. (2.20). The asymmetry between the horizontal factor $(N + h)$ and the vertical one $\big(N(1 - e^2) + h\big)$ is the ellipsoid's flattening at work: on a sphere $(e = 0)$ both reduce to $R + h$ and the geodetic and geocentric latitudes coincide.

How far the two latitudes differ. The geocentric latitude ψ of a point on the surface is the angle its position vector makes with the equatorial plane, so from (2.B.5)–(2.B.6), at $h = 0$,

$$\tan\psi = \frac{Z_s}{p_s} = \frac{N(1 - e^2)\sin\varphi}{N\cos\varphi} = (1 - e^2)\tan\varphi, \tag{2.B.9}$$

the classic relation: the two agree only where $e = 0$, or at the equator and the poles, where tan is 0 or infinite. To see the size of the gap, write $\psi = \varphi - \Delta$ and expand for small e^2. Using $\tan(\varphi - \Delta) \simeq \tan\varphi - \Delta\,(1 + \tan^2\varphi)$ in (2.B.9),

$$\tan\varphi - \Delta\,(1 + \tan^2\varphi) \simeq (1 - e^2)\tan\varphi \quad \Longrightarrow \quad \Delta \simeq \frac{e^2\tan\varphi}{1 + \tan^2\varphi} = e^2\sin\varphi\cos\varphi, \tag{2.B.10}$$

that is

$$\varphi - \psi \simeq \frac{e^2}{2}\sin 2\varphi, \tag{2.B.11}$$

which is the expression quoted in Sec. 2.7.5. At this order it is largest at $\varphi = 45°$, where it equals $e^2/2 = 3.347 \times 10^{-3}$ radian, or $0.192° \approx 11.5'$. Mistaking one latitude for the other therefore displaces a fix along the meridian by $R_\oplus\,(e^2/2) \approx 21$ km at mid-latitudes—the $\sim$20 km quoted in the body. The error vanishes at the equator and at the poles; the leading-order approximation is symmetric about 45°.

2.B.2 ECEF to local ENU

The local frame at the origin fix $(\varphi_0, \lambda_0, h_0)$ is spanned by three unit vectors: East, tangent to the parallel; North, tangent to the meridian, pointing toward increasing latitude; Up, along the ellipsoid normal (2.B.7). Away from the poles, East is the direction in which the position (2.B.1) changes under λ alone,

$$\hat{\mathbf{e}}_E = \frac{\partial \mathbf{X}/\partial\lambda}{\|\partial \mathbf{X}/\partial\lambda\|} = (-\sin\lambda_0,\ \cos\lambda_0,\ 0), \tag{2.B.12}$$

At a pole, longitude fixes a conventional orientation of the horizontal axes; the displayed vector still defines that convention although the derivative vanishes. Up is $\hat{\mathbf{u}}$ of (2.B.7) evaluated at (φ_0, λ_0), and North completes the right-handed triad,

$$\hat{\mathbf{e}}_N = \hat{\mathbf{e}}_U \times \hat{\mathbf{e}}_E = (-\sin\varphi_0\cos\lambda_0,\ -\sin\varphi_0\sin\lambda_0,\ \cos\varphi_0). \tag{2.B.13}$$

The ENU components of a displacement $\Delta\mathbf{X}$ are its projections on the triad, $E = \hat{\mathbf{e}}_E \cdot \Delta\mathbf{X}$ and likewise for N and U; stacking the three row vectors gives exactly the matrix of Sec. 2.7.5,

$$\begin{pmatrix} E \\ N \\ U \end{pmatrix} = \underbrace{\begin{pmatrix} -\sin\lambda_0 & \cos\lambda_0 & 0 \\ -\sin\varphi_0\cos\lambda_0 & -\sin\varphi_0\sin\lambda_0 & \cos\varphi_0 \\ \cos\varphi_0\cos\lambda_0 & \cos\varphi_0\sin\lambda_0 & \sin\varphi_0 \end{pmatrix}}_{R} \Delta\mathbf{X}, \tag{2.B.14}$$

whose rows are orthonormal by construction, so R is a rotation ($R^{\mathsf{T}}R = 1$) [16]. Two remarks used in the body follow directly. First, the rows depend on the *geodetic* latitude because Up is the ellipsoid normal — the same φ that enters Eq. (2.20), which is why no latitude conversion appears anywhere in the pipeline. Second, the matrix is evaluated once per flight, at the origin fix: the frame is a fixed tangent frame, not one that travels with the wing.

2.B.3 Size of the tangent-plane error

Two different distances must be distinguished. On a sphere, an arc of length d subtends a three-dimensional chord

$$c = 2R_\oplus \sin\frac{d}{2R_\oplus} = d - \frac{d^3}{24\,R_\oplus^2} + O(d^5/R_\oplus^4). \tag{2.B.15}$$

The horizontal EN projection from a frame anchored at one endpoint is instead

$$d_{EN} = R_\oplus \sin(d/R_\oplus), \qquad d - d_{EN} = \frac{d^3}{6R_\oplus^2} + O(d^5/R_\oplus^4). \tag{2.B.16}$$

This deficit is 0.033 m at 20 km, 0.51 m at 50 km, and about 33 m at 200 km. These estimates assume surface points and an origin at one endpoint. For increments between two remote points in a fixed frame, projection distortion also depends on their distance and bearing from the frame origin. It cannot be bounded from increment length alone. The *vertical* coordinate of a distant surface point is a different matter: the surface falls below the tangent plane by the sagitta $\approx d^2/(2R_\oplus)$, which reaches tens of metres already at $d = 20$ km. This is why the analysis never takes its working altitude from the plane geometry: the vertical coordinate is the adopted altitude channel itself (Sec. 2.7.5, Sec. 2.7.1), and the U component serves only the geometric construction.

2.B.4 Size of the spherical-approximation error

Sec. 2.7.2 computes inter-fix distances by the haversine formula, Eq. (2.2), which treats the Earth as a sphere of fixed radius $R_\oplus = 6{,}371\,\mathrm{km}$ regardless of latitude or heading. This section bounds what that costs against the WGS84 ellipsoid.

Local metric. For a short step at geodetic latitude φ, the ellipsoid and the sphere give

$$ds^2_{\text{ell}} = M(\varphi)^2 d\varphi^2 + N(\varphi)^2 \cos^2\varphi\, d\lambda^2, \qquad ds^2_{\text{sph}} = R^2_\oplus (d\varphi^2 + \cos^2\varphi\, d\lambda^2). \tag{2.B.17}$$

Define $d\sigma = ds_{\text{sph}}/R_\oplus$ and the bearing α in this spherical metric by $d\varphi = d\sigma \cos\alpha$ and $\cos\varphi\, d\lambda = d\sigma \sin\alpha$. Then

$$R_{\text{eff}}(\varphi, \alpha) \equiv \frac{ds_{\text{ell}}}{d\sigma} = \sqrt{M(\varphi)^2 \cos^2\alpha + N(\varphi)^2 \sin^2\alpha}, \tag{2.B.18}$$

interpolating between the meridian value $M(\varphi)$ ($\alpha = 0$, north–south) and the prime-vertical value $N(\varphi)$ of Eq. (2.B.5) ($\alpha = 90°$, east–west) [15]. $N(\varphi)$ is already in hand; $M(\varphi)$ follows the same route, differentiating the surface point $(p_s(\varphi), Z_s(\varphi))$ of Eqs. (2.B.5)–(2.B.6) with respect to φ and using $ds = M\, d\varphi$ along the meridian:

$$\frac{dp_s}{d\varphi} = -\frac{N(\varphi)(1-e^2)\sin\varphi}{1-e^2\sin^2\varphi}, \qquad \frac{dZ_s}{d\varphi} = \frac{N(\varphi)(1-e^2)\cos\varphi}{1-e^2\sin^2\varphi}, \tag{2.B.19}$$

so that $M(\varphi)^2 = (dp_s/d\varphi)^2 + (dZ_s/d\varphi)^2$ collapses, via $\sin^2\varphi + \cos^2\varphi = 1$, to

$$M(\varphi) = \frac{N(\varphi)(1-e^2)}{1-e^2\sin^2\varphi} = \frac{a(1-e^2)}{(1-e^2\sin^2\varphi)^{3/2}}. \tag{2.B.20}$$

Comparison to the sphere. The reference radius used in Eq. (2.2), $R_\oplus = 6{,}371\,\mathrm{km}$, agrees with the ellipsoid's mean radius $R_m = (2a+b)/3 = a(1 - f/3) = 6{,}371.009\,\mathrm{km}$ to within $10\,\mathrm{m}$. The fractional difference, expressed as ellipsoidal length relative to spherical length, is approximately $[R_{\text{eff}}(\varphi, \alpha) - R_m]/R_m$, bounded between its values at $\alpha = 0$ and $\alpha = 90°$. Expanding Eqs. (2.B.5) and (2.B.20) to first order in the flattening f (using $e^2 \simeq 2f$),

$$\frac{N(\varphi)}{R_m} - 1 \simeq f\Big(\sin^2\varphi + \frac{1}{3}\Big), \qquad \frac{M(\varphi)}{R_m} - 1 \simeq f\Big(3\sin^2\varphi - \frac{5}{3}\Big), \tag{2.B.21}$$

both of order the flattening $f = 1/298.257223563 \approx 0.335\,\%$ at every latitude: from $-5f/3 \approx -0.56\,\%$ at the equator ($\varphi = 0$, meridian direction) to $+4f/3 \approx 0.45\,\%$ at the poles ($\varphi = 90°$, where $M = N$ by symmetry). Thus 0.3% is a regional estimate, not a global bound.

At the latitudes flown. Most of the archive launches from the Alps box $\varphi \in [43.8°, 46.6°]$N used to classify take-off sites (Sec. 2.9, Fig. 2.23). Evaluating Eq. (2.B.21) across that range moves the bound by under 0.02 percentage points, so its midpoint $\varphi = 45°$ is representative:

$$\frac{N(45°)}{R_m} - 1 \simeq \frac{5f}{6} \approx 0.28\,\%, \qquad \frac{M(45°)}{R_m} - 1 \simeq -\frac{f}{6} \approx -0.06\,\% \tag{2.B.22}$$

(numerically, $N(45°) = 6{,}388.84\,\mathrm{km}$ and $M(45°) = 6{,}367.38\,\mathrm{km}$ against $R_m = 6{,}371.01\,\mathrm{km}$). At 45°, the sphere underestimates an east–west ellipsoidal step by about 0.28% and overestimates a north–south step by about 0.06%. Across the latitude interval above the absolute discrepancy remains below 0.3%, or roughly 3 cm on a 10 m step. The bound only opens up toward the 0.45–0.56 % extremes for the small remainder of flights nearer the equator or at higher latitude (Sec. 2.9, "elsewhere"), corresponding to at most roughly 6 cm on a 10 m step. The uncertainty of a GNSS increment must be assessed separately, since errors at successive fixes can be correlated.

Appendix 3.A

Squared-displacement distributions

These figures express the test of Section 3.3.1 in squared metres, using the same observations, weights and six lags. Squaring the bin edges preserves probability masses, while the exponent doubles according to Equation (3.27). This representation adds no independent evidence for self-similarity.

The shape of a density also changes under this transformation. If a component magnitude has a finite nonzero density limit $p_Y(0^+) = c$, then $p_{Y^2}(s) \sim c/(2\sqrt{s})$ as $s \downarrow 0$. An approximately straight descending part of a squared-component density on logarithmic axes can therefore come from this Jacobian near zero; it is not by itself evidence of a heavy upper tail. Very small positive bins and exact zero atoms describe numerical observations, without establishing physical displacement resolution at those scales.

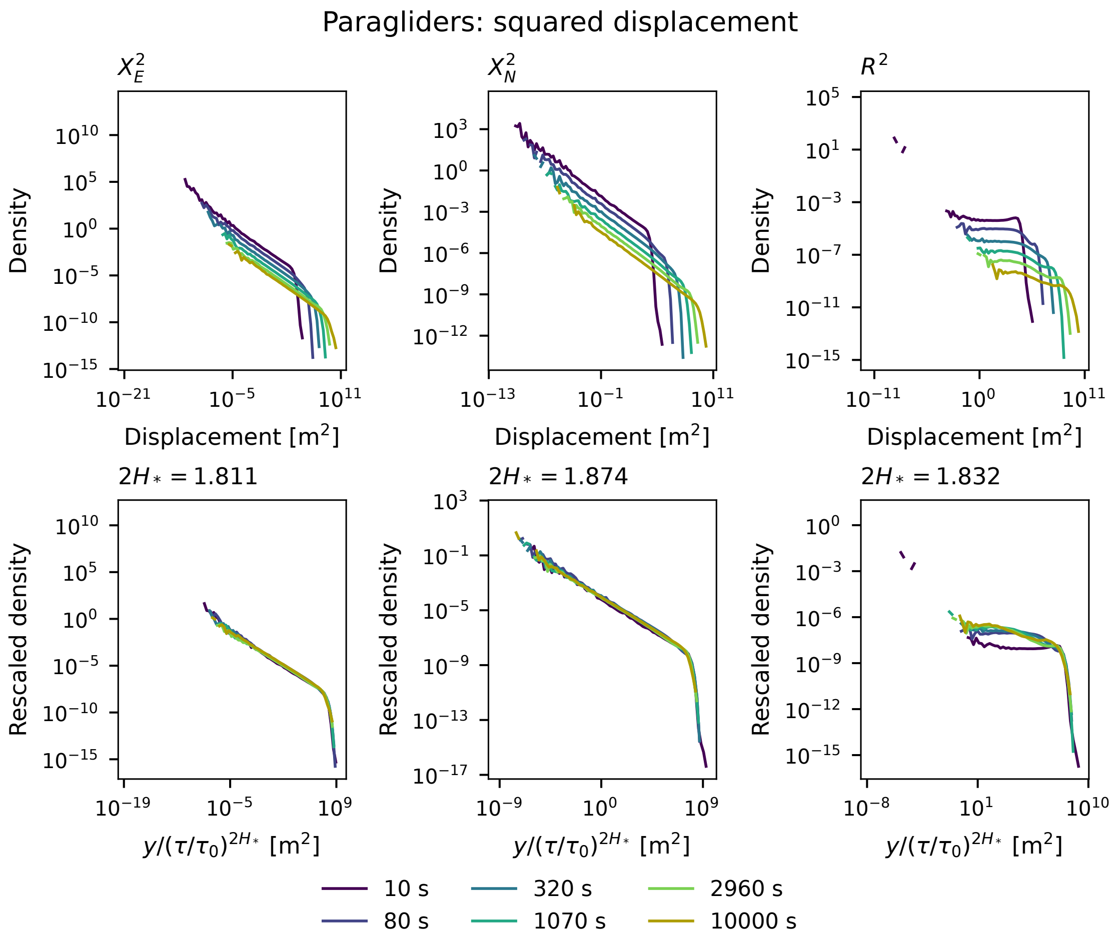

Figure 3.A.1. Paraglider distributions of X_E^2, X_N^2 and R^2. Top: raw densities; bottom: rescaling by $(\tau/\tau_0)^{2H_{j,*}}$, with $\tau_0 = 1{,}000\,\mathrm{s}$ and the doubled exponents from Table 3.5. Zero atoms are recorded separately.

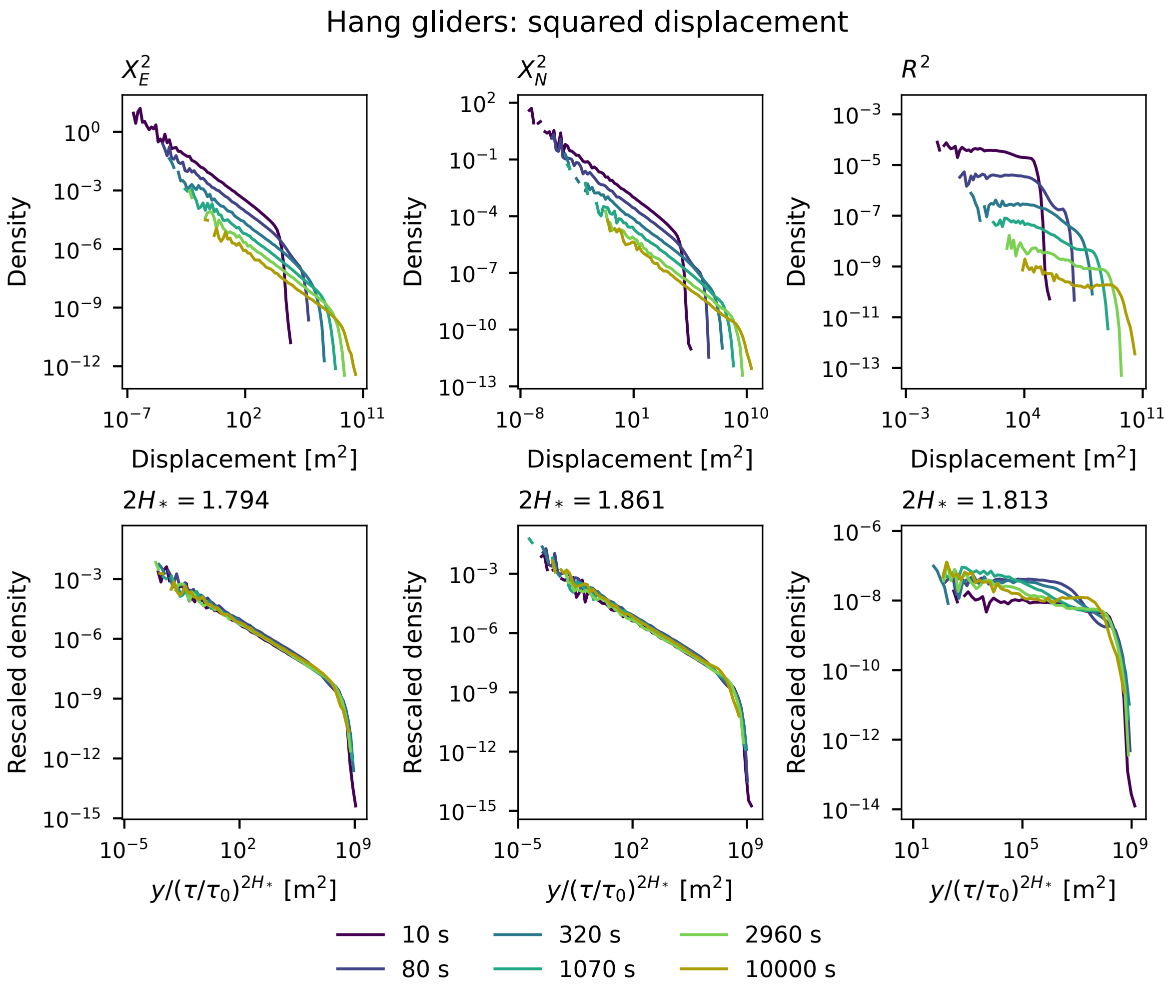

Figure 3.A.2. The same squared-displacement distributions and rescaling for hang gliders. Changes of shape seen for magnitudes remain under the monotone square transformation; the corresponding ECDF separation is exactly unchanged.